

Reconstructing Phonology: A Modest Proposal

John M. Anderson

PREFACE

What follows argues that research in the reconstruction of phonologies of the past, particularly that of pre-historical phonologies, is often based more on phantasy than argument from evidence. I offer a compact survey of common sources of argument appealed to, and their frequent misuse, and other abuses. My focus is on the earlier history of English, the language the study of whose history I am most familiar with, but which is not exclusively or, of course, comprehensively addressed here. I am not, unfortunately, an anglo-philologist.

The present survey highlights successive misinterpretations of the relation between phonology and phonetics and between phonology and writing, involving ‘litteræ’ and so-called “taxonomic” phonemes, or “systematic” phonemes’, and that between phonological structure and ‘metrical evidence’. Claims concerning alleged historical phenomena are typically complemented by appeals to alleged ‘fundamental’ or ‘uniformitarian’ principles of phonology. Different assumptions concerning these phenomena have led to competing formulations that cannot be respectively evaluated, and indeed to competing alleged solutions to issues that cannot, on the available evidence, be resolved. Modesty requires that researchers admit that they can gain only a very partial idea of the character of languages of the past, especially if the language is pre-historic.

The text concludes with two Appendices. The first of these supplements the main text in illustrating problems with the application of a range of recent phonological frameworks to phenomena discussed in the main text. The second deals with morphological innovations neglected in the main text.

Concerning my presentation, the following quotation may be helpful for the reader: ‘<w>hen I use the pronoun *we*, I mean the “inclusive *we*”, inviting the reader to consider a problem with me together and not the very impersonal “editorial *we*”, with its peculiar singular form *ourself*’ (Chao 1968: vi). My *we* in intention may embrace anyone interested in the world of (English) language users.

Writing, when properly managed (as you may be sure I think mine is) is but a different name for conversation. As no one, who knows he is about in good company, would venture to talk all; – so no author, who understands the just boundaries of decorum and good breeding, would presume to think all: The truest respect that you can pay to the reader’s understanding, is to halve this matter amicably, and leave him something to imagine, in his turn, as well as yourself.

Laurence Sterne, *The Life and Opinions of Tristram Shandy, gentleman*, Book II, Chapter XI, p. 99 (The World’s Classics 40. OUP).

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Chapter 1

Introduction

‘Reconstruction’ in relation to the study of language tends to be associated in many people’s minds (if present at all) with diachrony. And it is indeed obviously necessary to reconstruct languages no longer spoken, even those that have extant descendants. The existence of the latter can indeed help in the pursuit of the phonology of unrecorded ‘predecessors’. Reconstruction of the past manifestations of a language, particularly as concerns phonology, is helped, for instance, in the form of attested morphophonological alternations in a later manifestation of that language, such as the present-day English minor alternation in *mouse* vs. *mice*. This is commonly called ‘internal evidence’ for previous stages of the language concerned (in this case English), given that, unlike the above pair, members of the same paradigm commonly share the stem of words in that language. Something has changed, disturbing the expected morphophonology. Our understandings of past and present are mutually supported. Our main interest here is with the reconstruction of the past, however.

But that is only part of the story. What lies behind such an eccentric contrast in the stem of these and other number forms, as well as the absence of a suffix? As anticipated in their titles, Chaps. 4-6 below compare the alleged role of different conceptions of phonological structure, particularly those based on littera and phonemic vs. subsegmental and featural as well as segmental, and prosodic (suprasyllabic (in a Firthian [1947] sense) properties, in attempts to reconstruct (morpho)phonological

history. We shall here evaluate these different potential interpretations of the sources of evidence for reconstruction of the past of languages.

Very helpful, obviously, can also be (particularly alphabetic) ‘written evidence’ of any past manifestations of a language variety, though the helpfulness of this depends on the purpose of this writing and the transparency of the script and the alphabet. The early written history of the *mice* plural for *mouse* as *mys* vs. *mus* singular starts us on the evolution of this bizarre morphology. Indeed, in the absence of writing, reconstruction must be extremely schematic. The role of writing and its interpretation will occupy us throughout this volume, given their importance. However, even at this point, one should also not assume a simple correspondence between the letters of an alphabet and phonological elements. Particularly, there is more to phonology than a sequence of segments, often unacknowledged even if the spelling is augmented with diacritics, indicating (word or sentential) accent or some other phonological feature, segmental or suprasegmental or subsegmental, and including lexical tone (as in Swedish) vs. sentential/utterance-based. See already Chap. 2 on the role of writing and problems in its interpretation. And I also return in Chaps. 3 & 4 to a consideration of such often neglected aspects of phonological structure as just mentioned and their significance for reconstruction of earlier forms. In the first of these we look at confusions that may arise from misinterpretation of the status of phonology and phonetics and concerning the status of phonetics in language; and Chap. 4 explores further the neglected role of suprasegmental contrasts or prosodies. In Chaps. 7-8 I exhibit some of the alleged ‘universals’ and ‘structural preferences’ appealed to in traditional attempts at reconstruction.

Texts that are metrically structured present particular problems, but also potential access to the non-metrical rhythmical structure of phrases and sentences, depending on the nature of the metrics, which must often itself be a reconstruction. Chap. 5 comments on (mis)interpretations of the status of metre in Old English, though (aspects of) the metrical system may be preserved in later manifestation of the language(s) concerned, as an aspect of internal evidence.

Especially in the absence of written records of a language, but also in their presence, of course, the existence of apparently related languages, especially those with early written attestations, can offer potential reciprocal evidence, via formulation of a shared reconstructible ‘ancestor language’ of such putatively related reconstructions: this introduces ‘comparative evidence’ (Chap. 9). But of course the relatedness itself also involves a complex reconstruction, particularly given that the potentially ‘related’ languages may offer possibly different kinds of evidence from different periods. We have, for instance, the familiar dating discrepancy between the earliest written Frisian and written Old English, which reconstructed languages are usually regarded as ‘closely related’, i.e. in sharing (parts of) a close-to-immediate ‘ancestor’. Moreover, ‘similarities’, rather than reflecting shared ancestry, may instead

be the result of copying from one language to another or of shared innovations among languages that are not obviously related at the time envisaged for the innovation concerned. See further both Chaps. 9-10 on the reconstruction of relatedness.

As anticipated, Chaps. 2-4 compare the alleged role of different conceptions of phonological structure, particularly those based on *littera* and phonemic vs. subsegmental and featural as well as segmental, and prosodic (suprasegmental) properties, in attempts to reconstruct phonological history. We shall evaluate these different potential interpretations of the sources of evidence for reconstruction of the past of languages. And in Chaps. 7-8 we look at assumptions concerning putatively universal or at least common phonological changes, particularly changes said to be evident in the recent history of present-day languages. As indicated, Chap. 9 scrutinizes comparative reconstruction; and in Chap. 10, the last chapter before the Conclusion, suggests graphic dependency-based notations for expressing the suggested relative timing of relations among individual reconstructions of the past, whether or not they have a predecessor in common. Before that, dependency will be introduced, along with possible linearity, as a crucial part of the structuring of the products of components of the grammar, particularly in relation to phonology.

So much for historical reconstruction, for the moment. Reconstruction, however, is central to the activity of all students – or indeed users – of language. The aim of the grammarian is to reconstruct the linguistic abilities and language use of speakers of a language in terms of related representational systems devoted to this aim. Crucial here, in the case of an extant language, is access to native-user informants, who ideally are or are included among the investigator(s). Any account of a language involves reconstruction. When we learn a language, in my (post-Saussurean) view, we are enabled to associate meaningful units with (sequences of) perceived sounds in that language. Together these combinations are the poles of the signs of the language; the lexicon is a set of representations that relate one of these poles to the other and the syntactic valency of the expressed pole and the morphological structure (if any) of signs, as well as those aspects of the phonology that distinguish the expression of signs. Language is thus a mental phenomenon the structuring of whose domain is the responsibility of grammar in the wide sense, including lexicon, phonology, and syntax, and I assume it to be substantively based, specifically on stored and constructible mental structures, including the registering of contrastive sounds, sounds that collectively express poles that can differentiate signs with distinct meaningful content poles.

The information in the mental lexicon, including morphological structures, if any, as well as immediate contextual information, enables the construction of syntax (though encyclopaedic knowledge and context may be invoked, of course, in usage), and also the phonological structure expounding, as the other pole of the minimal sign, the word, the basic unit of syntax, and other sign-types, including lexical phrases as

well as compounds. In use, syntax is built crucially, as well as on awareness of context, on the categorial valencies or other colligational requirements, supplied to signs by the lexicon. And the syntactic categories of sentences and more and less extensive sequences are associated with the accommodating of the phonologies of the individual words in a sentence or other sets of words; and the phonologizing of sentences includes the accommodating to each other of words in sequence and the placement of the prominent/tone-bearing, or tonic, accent(s) and the selection of particular suprasegmental tones, completing the composite linguistic representation that is potentially implemented as (part of) an utterance. Anderson (2022b, particularly Part III) describes in detail such a building of the linguistic structures on which there depends English usage.

The means of transmission of the products of exercising the acquired language capacity are not part of grammar, however; to implement a linguistic representation, primary transmission utilizes our capacity to make and recognize a range of noises, by means of respectively articulatory activity in the tract from the lungs to the oro-nasal cavities and the auricular apparatus for hearing noises. Given the purely mental status of grammar and the absence of a vocabulary for its perceived sounds (beyond figures based on physical sensations – *soft*, *hard*, and the like), variation in these articulatory activities can, by default, provide us with labels for these associated internally perceived sounds – though what are perceived as very similar sounds can be produced in alternative ways, and vary among speakers. As well as articulatory descriptors, we can use the products of such instruments as spectrographs to observe differentiations in acoustic records that correlate with perceived sounds, thus providing an alternative or supplementary source of implementation-based labelling for our perceptions.

We cannot convey phonology, a mental construct, being the expressive poles of signs, directly, but reconstruct it from its implementation as audible articulation. The reception of these noises by activity in the ear channel is thus a complementary aspect of how I interpret implementation, specifically (successful) reception vs. expression. Similarly, understanding of the other, signified pole depends on our experience not merely of use of the significance of the sign, including non-grammatical uses, such as recognition, along with pronunciation, of the identity of an unseen speaker or their provenance, but also knowledge of the culture(s) it serves. We can identify those aspects of implementation that express contrast, that differentiate signs. We can on this basis reconstruct and identify the basic units of phonology deployed in particular languages. This sketch of the content of grammar and linguistic structure, together with their implementation, is again illustrated in more detail as manifested in English, in Anderson (2022a, b).

The need for reconstruction is more obvious, however, when we are dealing with languages no longer spoken. With their different roles in implementation, articulatory and auricular factors, as observable in current languages, may, indeed, contribute to

reconstruction of the changes in language that are traditionally referred to as ‘sound-changes’. Many changes can be interpreted as being motivated by the (sometimes contradictory) needs of aspects of implementation rather than (morpho)phonological or syntactic factors. But, as expressed above, the means of transmission are not part of phonology, not part of grammar. A corollary of this is that there is no unique linguistic level of ‘phonetics’, since differentiation of articulations and acoustic records can be varyingly detailed, and no particular non-contrastive degree of differentiation has a grammatical motivation.

Phonology, on the other hand, is concerned with contrastive differences in the sounds we perceive, internally or externally, i.e. without or with the presence of phonetic implementation, possibly limited to mental activity. Also, as anticipated, and argued in later sections, representations of phonology (and syntax) require a subsegmental notation, and the domain of some phonological features, may also be larger than the segment, the basic sequential unit of the phonology; they are suprasegmental, or prosodic. In what follows I adopt, as indicated, a post-Firthian sense of prosody as a contrastive feature that is attached to a suprasegmental node in linguistic structure, even to a morphological unit, or associated with a word form. Couper-Kuhlen (1986: especially ch. 1), for instance, provides a survey of different interpretations of ‘prosody’. And a prosodic structure as interpreted here is illustrated in example (7) below, in Chap. 4.

We can, of course, use graphic symbols to represent a sequence of the products of particular articulatory configurations of varying delicacy (using relatively ‘broad’ or ‘narrow’ transcriptions), but none of these represents a systematic linguistic level, unless contrastive. They have no grammatical status. Transcriptions and spectrograms can have various ‘instrumental’ functions (as in field-work with an undescribed language, for instance or, as noted above, in speaker identification as well as linguistic-variety affiliation, particularly in the absence of visual clues), but they are not part of grammar, i.e. (the characterization of) our linguistic capacity itself, including how it directs the elaboration of linguistic structures. Implementation (articulation and reception) is obviously part of the language phenomenon, but it is not part of grammar; it enables language to communicate to users other than the speaker, though ‘speakers’ can and do communicate with themselves, usually silently – the internal monologue that makes up a large part of thinking.

What we have been contemplating immediately above is direct, synchronic reconstruction. Reconstructing synchronic grammar, including syntax, morphology and phonology, has its particular difficulties: apart from anything else, these arise because we are dealing with unobserved mental capacities and their (co-)activations, which, of course, vary over time, situations, and users. Unsurprisingly, too, there is no widespread agreement concerning grammatical structure, including views in the sketch I have just offered. This is all the more reason for historians of language(s), in

particular, to be modest concerning what can be reconstructed about no-longer existing minds. See too Allen (1992) and Colman (2021), for example, on the difficulties in interpreting and assessing the ‘evidence’ taken for granted in many accounts of early English, as we return to.

All of this underlies the need for ‘modesty’ I am proposing, with particular reference to reconstructing languages of the past. Any reconstruction will be partial and contingent; and in particular we cannot reconstruct ‘real’ former, particularly pre-historic, languages. ‘Pre-Old English’ is not the name of a language or even a succession of languages, but designates a possibly temporally ranged set of those systems of properties or distinctions we can reconstruct with more or less confidence about the immediate predecessor(s) to recorded Old English. The reconstruction of the latter and its culture can be closer in explicitness to those of a currently spoken language. In particular, concerning pre-historical predecessor(s), we can reliably reconstruct very little of linguistic variation among the speakers whose speech is hypothesized to have possessed certain reconstructed properties. All the more necessarily, reconstructed prior-languages (i.e. ‘pre-pre-language’) such as ‘West Germanic’ and the rest of such ‘pre-pre-reconstructions’ based on correspondences among pre-languages or other prior languages, do not label languages rather than, as we move backwards in time, sets of increasingly contingent properties, or systemic contrasts, shared by ‘languages’ that are hypothesized to be ‘related’, via a common ancestor.

Symptomatic of lack of modesty in reconstructing past phonologies has been, on the one hand, the labelling of certain reconstructed correspondences between particular reconstructed prior-languages as ‘sound-laws’ (despite apparent exceptions), and, on the other, the premature assumption of certain ‘universals’ of phonological structure or ‘fundamental principles’ of reconstruction, both of which we shall return to in various contexts.

Unfortunately, recent (particularly traditionalist) reconstructors of non-current languages have continued to pay little attention to the cautionary observations of their predecessors in the mid-to-late twentieth century. Some of these cautions are brought together in Collinge’s (1965) short but subtle and allusive (let’s, please, avoid the fashionable pretentious partially-assimilated cliché ‘nuanced’) presentation of ‘some linguistic paradoxes’ arising in attempts at both synchronic and diachronic reconstruction. I do not attempt to resolve or dissolve all the paradoxes. But it still seems necessary, above all, to examine again the status of what can be reconstructed about languages that are no longer part of the knowledge of a set of speakers. Ritt (1994: preface), indeed, recognizes that the need for modesty doesn’t apply only to the traditionalist applications of philology; but it is recent manifestations of ‘historical linguistics’ that are especially lacking in modesty, as I shall try to illustrate.

Harris (2000: 98-99) is even more forceful, to the extent of subscribing, alas, to final absurdities concerning the accepted status of ‘sounds’ vs. ‘writing’ themselves, even if interpreted as applying only to historical reconstruction:

What would be a mistake ... would be to ascribe validity to the tables of sound changes that authors of textbooks on historical linguistics are fond of compiling ... The ‘sounds’ represented in such tables are abstractions reconstructed by linguists on the basis of certain hypotheses and the application of certain analytic methods. The supposed ‘evolution’ of these sounds is equally hypothetical, and metaphorical to boot. Yet the misleading impression is given – and accepted by many students – that the table of letter-form changes and the table of sound changes are somehow on an equal footing, and merely facets of the same linguistic story. ... *letters and figures are cultural artefacts*. Sounds are not (except in so far as their identification is a second-order activity, which itself depends on familiarity with writing).

Contrast with the above view Strang’s (1970: §9) informal statement of a more familiar view: ‘Any study of how a language functions, structurally at a given point in time, or historically through time, must make sense in relation to speech’ as a not merely second-order activity, unlike Harris’ ‘letters and figures’.

What follows, then, is my own plea for modesty in claimed reconstructions of phonologies of the past, including assessment of appeals to (the secondary status of) writing and its variable status as a possible source of evidence for the culturally and linguistically basic phonological structure, basic even in the ‘digital age’, in which, despite Harris, language users tend to illiteracy, not aphasia.

Reconstruction of syntax, however, requires very extensive original written material and detailed evidence of its situation in a society and its culture. As evidence, continuous prose texts are obviously more appropriately articulated than, say, glosses and (even more) glossaries, or lists of names or single names. The latter may provide valuable evidence for phonology, and sometimes morphology – and particularly lexical semantics in the case of glossaries, but little of relevance to syntax apart from identification of content poles, for instance, and parts of speech. Thus, as conveyed by my title, I confine myself here to the reconstruction of phonology, since our capacity to reconstruct syntax is even more problematic, given its extensive demands on evidence of meaning and colligability; and, as noted, it is my impression that in the representation of even present-day syntactic structure, researchers have still less in common than in studies of phonology. Appendix II is intended to illustrate the complexity of morpho-syntax, even in recent English. Appendix I, on the other hand, illustrates the failure of various recent phonological frameworks to illuminate

phonological histories in particular. But we now proceed to an initial consideration of the role of writing in the reconstruction of languages of the past.

Chapter 2

Written evidence: orthography, litteræ, and phonemics

Let us begin to look at the consequences for historians of language of the assumptions concerning evidence observed in the introductory chapter. Reconstruction of the ‘value’ of the letters in orthographically recorded earlier languages is perhaps the most commonly attempted activity, doubtless given its (variably) relative closeness to the means of transmission in alphabets and, less so, syllabaries (such as the linear B of Mycenæan Greek – whose deciphering is described in Chadwick [1958]). The presence of a substantial body of surviving orthographic representations renders the reconstructed ‘language’ historical rather than pre-historical and thereby more accessible to reconstruction, particularly if the goal(s) and context of making the original written record are accessible and understood.

However, despite being an important source of evidence for phonological reconstruction, the status of written records is the first reason for modesty in what can be claimed about the (no longer current) language so represented. For the goals of even alphabetic writing are not the same as the goals of the would-be reconstructor of phonology. Of course, normally the writer seeks to enable the reader to identify, in the first place, individual lexical items or sequences of them; but the writer can normally assume, with respect to the contemporaneous user, knowledge of the language and the writing system employed and the relationship between them, as well as relevant aspects of the culture of users. And use of parts of an established spelling system can be routinized, conventionalized; and, indeed, the same system may have different values in different speech communities. Not much of this may be available to the reconstructor, though other aspects of the culture may also be helpful in identifying such phenomena, depending on the range of texts and on other evidence of culture. However, given its importance, I reiterate that the usage of communities of users of writing may reveal intentions in doing so that are not necessarily close to the aims of the reconstructor.

But, also, the initiators, in particular, of deployment of an alphabetic writing system for a specific language may misinterpret or be misled by existing applications of the alphabet elsewhere. And, crucially, (re-)appliance of a writing system includes concern with economy of graphic symbols: what is generally seen as the ideal writing system includes the minimal set of graphs to distinguish those segment types associated with lexical items that are phonologically distinctive, or contrastive, at least in some linguistic context. But the contrasts at different positions in a lexical item typically differ, and this variable ‘value’ is something necessarily obscured, given the economical goals of orthography, even with otherwise apparently reliable alphabetic writing. And, of course, different languages appeal to different as well as like contrasts; the cross-linguistic value of graphs may be variable in order to cope with this. And

there may also for various reasons be recourse to more than one script, as in those Germanic languages that appeal to graphs from the runic alphabet as well as or instead of Latin-based graphs.

Much of what I have to say here and what follows is anticipated by Cecily Clark (1992: 123-124) in her discussion of what she dubs ‘The myth of the Anglo-Norman scribe’.

Spelling, as anyone conversant with Modern English must already appreciate, rests not upon phonetic transcription but upon some system of conventionalized representation clear enough to allow the items in question to be recognized by anyone familiar with the language involved. So a “spelling” need not – and often in Modern English (or Modern French) does not – correspond symbol by symbol with the equivalent sequence of phonemes; some languages have managed for centuries with ideograms unrelated to any vocal realization of the items represented. Literate people of the Middle Ages would have been attuned to the idea of conventionalized spelling through their training in Latin, which was everywhere set down in similar form, irrespective of local idiosyncrasies of pronunciation. Because of thus being conventionalized, spelling systems must ... be assessed in their own cultural and chronological terms.

‘Phonemes’, however, – as well as of course ‘ideograms’, for instance – are terms of their times which I reject here as part of phonological notations, for reasons made more explicit in the case of ‘phoneme’ as we proceed.

Apart from the typical absence from many writing systems of indication of suprasegmental aspects of phonology, what I have introduced as the discrepancy of goals between writers and reconstructor means that the latter cannot assume that the scribe anticipates their interests in how the graphs (and possibly diacritics) of the alphabetic system map on to phonological units. ‘Respecting the scribes’, often advocated, cannot be based on the assumption that the aim of a writer is to serve the interests of the later scholar, or even necessarily that the writers intend to try to represent their own phonological system. Moreover, as anticipated above, writing can have non-linguistic uses, so that, for instance, a common important function is decorative, as on coins and in certain types of document, and on stone monuments.

An instance of possible clashes of goals, even in relation to the linguistic uses of writing, comes from the failure of orthographic recognition of neutralization in a particular position of a contrast found elsewhere; see, for instance, Davidsen-Nielsen (1978: ch. 10) or, for a rather different view, Lass (1984: ch. 3). Recognition of neutralizations and of the consequential archiphonemes (or ‘neutralisees’) is disfavoured in alphabetic systems on grounds of symbol economy. Similarly, the

segment-based nature of alphabets means that a contrastive feature that spans a sequence of segments, i.e. any type of Firthian prosody (discussed below), is misleadingly attributed to whatever the individual graphs represent. The latter mismatch can be overcome, or at least acknowledged, and partially represented by diacritics, though this is not typically practised. And though suprasegmentals and neutralizations may be recognized by the reconstructor, the requirement that contrast is a property only of segments can lead to intended phonological representations and spellings that are misleading from the point of view of the reconstruction of where a phonological contrast lies and doesn't lie.

A failure to acknowledge neutralization is evidenced in present-day English orthography by the (perhaps over-)familiar spelling in *spit*, where what the *p* represents is a neutralized element in place of a contrast one finds exemplified in *pit* vs. *bit*, a neutralized element that in this case is, for the typical user, no more like the onset phonology of *pit* than that of *bit*; it lacks both the aspiration of the former and the voice of the latter. As again anticipated above, we need, indeed, to recognize that there are different systems of contrast at different positions in a word. Application of the 'once a phoneme always a phoneme' assumption, for instance, in the employment of alphabetic writing obscures both the distribution of contrasts and thus the different phonological values of their graphs in different positions. This demand for economy is perhaps symptomatic of the ever-increasing commoditization of every community or domain invaded by a market mentality. Swadesh (1934/1958: 35) already advertises 'the phonemic principle' in the claim that '<a> phonemic orthography provides the most adequate, economical, and effective method of writing a language'.

Actual writing systems often vary in representation of particular segments in what seems to be the same word form. There are different possible reasons for the occurrence of such graphic alternations. They may indeed be simply that, if, say, the source alphabet(s) offer(s) more symbols than the new application of an alphabet requires, or different source alphabets offer alternative graphs, whose joint use disguises a lack of phonological contrast. Or graphs may be used diacritically, to distinguish a different value from the usual to be attributed to an adjacent or co-syllabic graph, or socially significant variation, as in *ship* vs. *sip*, *spite* vs. *spit*, *shite* vs. *shit* – instances of the untidy usage of contemporary English. Or alternations of graphs in a particular environment may indeed result from the existence of a neutralization (as illustrated above and further below). But rather than an 'archiphoneme' being distinctly implemented from both of the neutralized segments, it may differ from only one of them, as well as from both. Thus, in a number of (e.g. some Germanic languages) some word-final plosives are neutralized as voiceless.

Such alternations have also been interpreted, however, as a sign of 'sound-change in progress' or 'spreading' – simply through time or through a population? But the latter is particularly unlikely if the alternation occurs in the same text produced by a

single scribe in the former interpretation or indeed to texts from a particular period and a shared location. And the ‘indecision’ interpretation is unlikely. Such claims of ‘sound-change in progress’ should be approached sceptically: it is not characteristic of the use of a shared spelling convention. Thus, to continue with familiar examples, the occurrence of both *þ* and *ð* in early Old English texts is not associated with a (change in a) phonological contrast, and the obvious difference in sound that they might be associated with, i.e. in voicing, as usually reconstructed elsewhere, is apparently not contrastive here. We have non-contrastive alternation, partly associated with position in a word form. Such a symbol difference could usefully indicate a (admittedly marginal) contrast in current English, particularly in marking morphophonological distinction: *thigh/thy*, *mouth* (noun/verb), *loath/loathe*: *mouth* has a single digraph in both forms, and the last pair are distinguished by a diacritic use of *e*. And neither of the Old English graphs *ð* or *þ* is appealed to. In Old English, adoption of the Greco-Roman and Runic alphabets creates unnecessary complexity in representing these segments.

Concerning diacritic status, recall that the phonological value of *c* in Italian is often determined by a following diacritical graph, as well as segmentally contrastive graphs: in both *chiesa* ‘church’, with diacritic *h*, and *casa* ‘home’ the onset is, in traditional terms, plosive, voiceless and velar. And, to return to earlier history, Daunt (1939) argues that various vowel digraphs in Old English that had been and continued, in developments of the ‘Luickian tradition’, to be interpreted as ‘short/long diphthongs’ are rather to be interpreted as implementing a (short/long) vowel and a diacritic that recognizes a partial, non-contrastive value, a colour, of an adjacent consonant: thus, in her view, the ‘backness’ of the lateral in *eald* ‘old’ is signalled in this way. One might interpret this phenomenon as involving ‘graphic extraction’ of a non-contrastive feature of the consonant. Daunt suggests that this is based on a practice of Celtic orthography probably familiar to the earliest scribes of English. Daunt’s presentation is somewhat confused, however, by invocation of ‘glides’ from vowel to consonant. However, at the very least, she is illustrating something of the range of possible relationships between spelling and phonological/phonetic phenomena.

On suggested inter-language influence see too Laker (2009) on the development and representation of (dental) fricatives as contrastive in early Old English, where it is argued that this also involves Celtic influence, though this is based on little non-phantastic evidence, as well as offering a tedious story. Unlike loan-words, loan-phonology/orthography is difficult to establish, given the internal variations in many languages that lead to changes that are attributable to internal factors, as well as the commonness in languages of certain changes. And the conclusiveness of Laker’s (2014, 2017) arguments based on investigations of Germanic dental fricatives is also vulnerable to the observed general markedness of such phones, as well as confusion concerning phonemicness. And Baldi & Page’s (2005) review of Vennemann (2003)

illustrates some of the difficulties in characterizing the source of similarities of any kind between languages, particularly if they are pre-historic: do particular instances involve a genetic relationship or the result of a contact loan or merely coincidence? The distinguishing of the source of shared properties is another controversial area to arise in this chapter, which is concerned specifically with the status of written evidence. We shall return below to the cluster of difficulties encountered here.

A different phonological situation where the drawing of graphic distinctions is not contrastive involves prosodies (as described above, in Chap. 1), of course. The contrast between the entire codas of *ant* and *amp* and *ink* involves a prosody: the ‘place of articulation’ belongs to the coda as a whole; segmentally, the representation of either the distinctive nasals or the distinctive plosives or of both is arbitrary, and signalling the distinction in both nasal and plosive is inaccurate; a single prosodic interpretation of the coda is appropriate; the feature implemented as a traditional ‘place of articulation’ is prosodic here.

Almost finally, as concerns the well-attested functions of letters, I have recognized in my own experience a not uncommon relation between spelling alternation and phonological neutralization. I illustrate this situation with the following personal anecdote.

In Greece I live near the village of Φοινικούντα in southern Messinia, and I have come across alternative romanizations in public orthographic representations of the name, notably *Finikounta*, *Finikounda*, and even *Finik(o)uda*. The distinction between the first two alternatives is rather obviously purely graphic; there is no phonological contrast: there is no voicing contrast in plosives after a nasal, the latter of which may itself be reduced to pre-nasalization of the plosive. Initially in a word the plosives are distinguished in Greek orthography as τ (voiceless, non-aspirated, very common) vs. ντ (voiced), for instance; but the latter is most familiar in loanwords (e.g. ντους ‘shower’): in phonemic terms, the (possibly pre-nasalized) voiced initial plosive is apparently to be taken to have a nasalized non-initial ‘allophone’, also represented as ντ in Greek spelling, again with no contrast with the simple voiced equivalent. Recognition of this neutralization, with varying pronunciation, is undone in the roman variations in spelling. This is, of course, a special case, in involving a largely intrusive (non-native) marginal initial contrast and a neutralization in the coda, but it illustrates some of the sorts of problem in interpreting orthography and, in particular, use of alternative orthographies – which can, however, also be revelatory.

As anticipated, we shall encounter still further examples of the suggested diversity of roles that might be attributed to particular graphs in what follows, including especially speculations about ‘sound-change-in-progress’. It is crucial, as anticipated above, that a reconstructor acknowledge the reasons for and purposes of the available different types of written material, from, in e.g. Old English, epigraphs, glosses, and glossaries, to the various document types labelled as ‘charters’, and chronicles, and to

verse and riddles, as well as recognition of the date of the copy of a document that is extant. Unsurprisingly, the reconstruction of the phonology of even written Old English remains controversial, if attempted at all (rather than reconstruction being simply a question of a ‘pronunciation’ being assigned!).

Campbell’s quaintly titled (but in accord with philological tradition) *Old English Grammar* of 1959 does not pay any attention to the phonological system of Old English, as well as virtually ignoring syntax and non-inflectional morphology – admittedly again not exceptional in ‘grammars of Old English’ (cf. e.g. Brunner’s [1942] revision and Cook’s [1885] translation of Sievers’ *Altenglische Grammatik* [1882/1986/1898]). And this tradition is continued by Hogg’s *Grammar* (Hogg [1992], Hogg & Fulk [2011]). Is this nomenclature associated in part with an obsession with etymology, and diachrony generally? But it is not the practice of modern grammars of even the classical languages, for instance, or of grammars of Modern Greek such as Τριανταφυλλίδης (1941). One English word for what is elsewhere a ‘grammar’ seems in traditional descriptions of Old English to be, among other things, ‘guide’, as in Mitchell & Robinson’s (1964) *A Guide to Old English*, which has some sections on syntax; cf. too Lass (1994) *Old English: A Historical Linguistic Companion*, with a short ‘chapter’ labelled as ‘Topics in OE Historical Syntax’, pp. 216-240. This imbalance in scope of coverage may also, of course, reflect the difficulties of reconstructing syntax (even in current languages). But, on the other hand, Mitchell’s (1985) *Old English Syntax* is not part of a ‘grammar’, though again built on an obscure traditional framework. Sturtevant’s (1917) is one of the few early studies of ‘language change’ that recognize the various traditional components of a grammar.

The concern with change in these other philological ‘grammars’, however, is unsurprising, not just because of the aim of establishing the mutually supporting connections in and between ‘language-families’, pervasive in the nineteenth century, but also because many ‘sound-changes’ are in themselves ‘diachronic’, unfolding over time. I have in mind here not just the social spread or contraction of ‘changes’. Many familiar ‘sound-changes’ necessarily involve a protracted phonetic change that eventuates in a phonological change, a change (such as e.g. loss of the context of the change) in the system of contrasts. But such staging may not be recognized as such, given the lack of concern with contrast. This lack is fundamental: phonology serves to identify minimal lexical items (words) or the components of phrasal lexical items.

There is, instead, in e.g. Campbell’s (1959) book a chapter on ‘writing, orthography and pronunciation’. The discussion of ‘pronunciation’ has to do with illustrating the Old English ‘approximate’ ‘phonetic system’ (which is, however, necessarily not reconstructible, unlike the system of contrasts). The alleged ‘phonetic system’ here is determined by the set of graphs, allowing for the redundant use of some pairs of graphs in not representing distinct ‘sounds’. The sounds and the graphs are

represented, as far as possible, in the same way, as italicized letters (unless the ‘sounds’ are in a table of the ‘system’).

This is, again unsurprisingly, reminiscent of the classical tradition of the ‘littera’ that survived until quite recently – and, indeed, still has its advocates, notably Lass (appendix to [2015]). And see too Lass & Laing (2012: 76, note 7), which exemplifies the assumptions of their sequence of papers on Middle English. The littera is a composite of graph, or written shape (*figura*), spoken value (*potestas*), and name (*nomen*); Anderson (2014a: §1, 2022c) offers sketches of the history of the concept and its deleterious impact on the study of phonology. The same graph-based notation is used by Campbell even for reconstructed sounds, a practice that pervades the study of languages of the past (another well-known instance in the area of Old English being Brunner [1942]), though the reconstructed pre-historical sounds/letters are sometimes distinguished as such by an immediately preceding asterisk (familiar elsewhere in works on language, however, as a marker of ungrammaticality – though perhaps after all not an unfortunate ‘coincidence’!).

Unfortunately, however, such practices persist (as in Ringe [2006]), undefended, perhaps because indefensible (cf. again Anderson [2014a: esp. §1]). Martinet’s description of (1955: 27) is still appropriate as concerns many researchers:

People in general and philologists in particular do not like being disturbed. Those who, from the time they were students, are used to certain ways of thinking, naturally resent the introduction to their domain of revolutionary theories. Many scholars, who would have been glad to accept new suggestions forthcoming in slow succession, reacted unfavourably when, some fifteen years ago, they were presented with the new and somewhat aggressive doctrine of phonology. As a rule they did not launch attacks against it, but simply carried on their work as if nothing had happened, waiting for the phonologists to justify their claims.

All that has changed since then, for the most part, is the adoption of the (now discredited) concepts of ‘phonemics’ by some philologists, and, more recently, the making of appeals to concepts from one tradition of sociolinguistics, perhaps the most inappropriate area of linguistics for the purposes of reconstruction of pre-historical languages, given the primitive state of our understanding of language variation and, particularly, the paucity of evidence for the social situation of speakers of these historical and pre-historical languages, precisely because they lack current users. But most of the rich ‘structuralist’ tradition of studies in phonology and its history that is alluded to in Stepanovičius (1987: ch. 1), for instance, is ignored – as it is in some of his own analyses of the ‘phonological history’ of English. More positively, the core tenets of generative phonology have otherwise been widely spurned by philologists

(but the generativist tradition involved is represented by contributions to a number of collections such as Anderson & Jones [1974]). A much later defence of the lexical-phonological variant of generative phonology is offered by McMahon (2006, 2007), based on its capacity to allow for two synchronic vowel shifts in present-day English operating in morphophonologically opposed directions – not possible in ‘optimality theory’ (whose ontology is uncertain for other reasons and it is irrelevant here). McMahon’s claims, typical for generativists, mistake morphophonological alternations and their history for phonological rules (see Anderson [2014: 36-37]).

Fortunately, too, invocation of sociolinguistics, ‘inspired’ particularly by Toon (1983), and thereby ignoring much earlier work, largely reduces in such historical work to the frequent attribution to alternations of graphs for apparently the same segment in the same word-form the status of evidence of ‘sound-change in progress’. However, established spelling systems are slow to recognize the results of sound-change, let alone reflecting their being ‘in progress’ (whatever that might mean). Familiar spelling systems of the recent past or present do not typically ‘work’ in this way, as marking ‘sound-change in progress’, and there is no likelihood that even non-standardized orthographic traditions involve such recognition of ‘sound-change in progress’.

This assumption concerning graphic interchange is bad enough: see again Colman (2021). But Toon (1983: 98-110) argues indeed that alternation between <a> and <o> before nasals involves replacement of the former by the latter as a consequence of the alleged dominance of one dialect (Mercian) over others. And Ringe (2014) simply assumes the appropriateness of Toon’s analysis: ‘<f> or the linguistic consequences of Mercian dominance see especially Toon 1983’ (2014: 184, fn. 6). But King (1992) provides a devastating critique of Toon’s arguments, particularly the weakness of its reliance on charters.

As suggested above, and as acknowledged by King (1992: 35), another, not unusual, situation is that these alternations indicate neutralization associated with a particular environment. Davidsen-Nielsen (1978: 204-6), for instance, in discussing the early West Saxon system of phonological contrasts of short monophthongs, offers an analysis of the alternations in early West Saxon between the graphs *a* and *o* and between *æ* and *e* before nasals as indicating neutralization of phonological contrasts in a particular context; this would be not uncommon as an attested happening in these circumstances, given that the following nasal provides a ‘noisy’ environment, widely recognized as potentially encouraging reduction in a preceding vowel-system.

Campbell’s position regarding the phonetic values of the graphs representing the ‘sounds’ of the Old English ‘phonetic system’ collapses at the observation that, as I have described, there can be no such thing as a ‘phonetic system’: phonetic representations can be as detailed as you like, consistent with audibility, but no set of non-contrastive sound differences, as such, constitutes a system. Systemicity becomes

involved only when sign-differentiating contrast is invoked. This untenable ‘phonetic system’ position is embedded in the pervasive quasi-litteral notation of Campbell (1959).

But this is made worse by Hogg’s (1992) espousal of ‘phonemics’ – in his view (p. viii) ‘so that issues are not muddled by theoretical squabbles’, despite, or because of, its merely partial recognition of contrast. Hogg alludes to participants in the ‘digraph controversy’ (see §6 below), but doesn’t devote much attention to earlier ‘phonemic’ accounts such as the well-known ‘*Diachronic Grammar*’ of Welna (1978), as well as invoking only rather sparsely the rich American phonemic tradition initiated by Sapir (1925/1957) and Bloomfield (1926/1957), with both diachronic and synchronic orientation – as anthologized in Joos (1957) and Lass (1979), for instance. And Swadesh (1934/1957) and Twaddell (1935/1957), for example, both repeated in Joos (1957), offer rather different conceptions (‘squabbles concerning?’) of the ‘phoneme’. Twaddell (1935/1957), unusually, allows for neutralization as introducing a distinct phoneme. And Yuen-Ren Chao (1934/1957) surveys a range of issues concerning the phoneme (see too Chao [1968: especially ch. 3]). Thus, it is not helpful for Hogg to announce simply that ‘I take as my foundation phonemic theory’ (Hogg [1992: viii]). Which phonemic theory? Not, apparently, that presented by Twaddell (1935/1957), for instance. And how is it a ‘foundation’, and for what?

Moreover, the assumptions adopted by most phonemicists are incompatible with an adequate characterization of contrastivity, as I shall spell out below. Swadeshian ‘phonemics’ is crypto-litteral, partly motivated by orthographic requirements; and the phoneme is also segment-based, non-featural, in its most familiar, post-Bloomfieldian form. This is particularly transparent in relation to the reconstructing of historic languages in this tradition. And Hogg’s (1992) substitution of Campbell’s (1959) chapter on ‘Writing, Orthography, and Pronunciation’ with one on ‘Orthography and Phonology’ is doubly misleading: the distinction between writing and orthography is neglected; and the difference between the quasi-littera-based ‘phonemics’ apparently adopted by Hogg and post-Saussurean contrast-based phonology is unacknowledged. Thus the apparent preference for Swadesh’s ‘phonemics’ is without motivation.

Swadesh (1934/1957: 35), as we have seen, overtly relates the goals of orthography and phonemics. Compare (as always) Pike’s familiar sub-title to his *Phonemics: A Technique for Reducing Languages to Writing* (1947). This means that neither a script nor a set of structuralist phonemes is the basis for an adequate reconstruction of a phonological representation, as shown already on the basis of the English *sp/st/sk* set by Twaddell (1935). He recognizes the polysystemicity of phonological representation, and thus the presence of neutralization. Hockett (1942: note 11) misconstrues the consequence of Twaddell’s argument, while citing the basis for it; and Hockett’s appeal to ‘economy’, which is clearly orthographically motivated, is dressed up as ‘a corollary of the general scientific principle that the simplest

description which accounts adequately and accurately for all the facts is to be preferred' (§6.12 [6])!

Twaddell's example of the neutralization in Modern English of plosives in the initial clusters spelled *sp/st/sk* is already present in Old English. The sibilant also realizes a neutralization: it is the only segment type that can initiate such an onset; it contrasts only with its absence. Moreover, the combined neutralised status of both the sibilant and the plosives in these clusters can be said to be reflected in the alliterative patterns of Old English metre. Wrenn (1953: 78), for instance, comments under item (7) of his metrical 'rules': '<t>he sounds represented by *sc*, *st*, and *sp* ... can alliterate only with a word of the same two initial letters, and never with a single *s*'. (Here I ignore the complication that *sc* in historic Old English is usually interpreted as representing a single segment in certain environments – a further development in the realization of the neutralizations.) For more recent discussion of these neutralizations in relation to Modern English phonology see e.g. Anderson (2014a: §2).

The plosive neutralizations in Modern English illustrate rather blatantly that a contrast is missing. Its realization typically corresponds to neither of the contrastive segments in initial position: it is plosive but neither voiced (*bit*) nor aspirated (*pit*), as cited above in this chapter. This is the most severe type of neutralization: elsewhere the implementation of a neutralization may coincide with that of one of the contrastive segments. The implementation of the initial segment in these clusters does occur (rather widely) elsewhere; but in these initial clusters it realizes a neutralization of every consonant. In accordance with the spelling in both Old and more recent English, the sibilant in such clusters apparently contrasts elsewhere with other voiceless fricatives, and like them is also in codas subject to voicing 'agreement' in obstruent clusters (as in *ask*/**asg*/**azg*/**asg*) and phenomena associated with 'alveolars' in general. The 'agreement' phenomenon suggests a prosodic treatment here too: the shared property belongs to the cluster not to a particular segment.

It has been declared (by Davidsen-Nielsen [1978:7]) that the 'plosives after foot-initial-*s*' phenomenon is 'a very poor illustration' of neutralization, 'since voice remains distinctive in *asbestos* as opposed to *asperity*'. This is, on the contrary, a very poor example of counterevidence. [zb] or [sb] is not a viable syllable onset (or coda) in English: a syllable (and indeed foot) boundary falls between the two relevant segments in *asbestos/asperity* but the first cluster in *aspect* is not foot-initial. (He also cites a foreign name borrowed into American). The neutralizations that occur in onsets spelled *sp* represent a radically neutralizing segment that is elsewhere a prosodic or segmentally contrastive voiceless fricative, and a neutralized set of plosives. Syllable structure is thus also crucial in reconstruction, synchronic or diachronic. But this is to stray a little from our present concern, with neutralization and the evaluation of phonemics (as is the observation that *ptarmigan*, *psittacosis* and the like are exotica, unlike *#sp* etc.).

The above plosive neutralization in Modern English is realized by a segment type which is unlike either of the other articulatory types (voiced, aspirated) of plosives that occur elsewhere in onsets. But in many neutralizations there appears only one of the segments that contrast elsewhere, as with the word-final devoicing of German (though misconstrued in Lass [1984: 56-57, specifically in example (4.4)]). And also in the case of (e.g. some Scottish) speakers of English who do not aspirate voiceless initial stops (Wells 1982: 409), the neutralization after the initial sibilant is manifested as one of the elsewhere-contrasting segment types. Another kind of neutralization involves combinatory variety of reduction in the contrasts, as do the restrictions in English illustrated by the initial clusters [**sl-*] (but *shrimp*), [**ʃr-*] (but *slump*). And this is common with other position-restricted contrasts that are prevalent in present-day English, for instance (see e.g. Anderson [2022a: chs. 11-13]). On types of neutralization see the succinct discussion in Lass (1984: ch. 3), and, more fully, that in Davidsen-Nielsen's (1978) critical historiography.

We've been looking at positional system reductions. It seems to be appropriate to recognize also 'system expansions'. Thus, in present-day English a contrasting velar nasal typically appears only when closing a closed syllable, ignoring a following inflection (thus no plosive in *ring(s)*); and it is segmentally non-contrastive before a (voiceless) plosive that shares with it a 'place of articulation' prosody (*link*, as well as *limp* and *lint*); and it is absent in onsets. Traditional spelling, as in *ring*, suggests it is non-contrastive but prosodic: place of articulation agreement in the coda. And in some current varieties of English it is generally not contrastive. The velar nasal also fails to occur, except as prosodically determined, before the comparative suffix (cf. *longer* vs. *long*) and medially in a few lexical items mainly again ending in *-er* (*linger*, *finger*) or before a lateral (*tingle*, *angle*, *bungle*). The main nasal system I take to consist of a bilabial and an alveolar, and the velar is a variant of the nasal neutralization in codas before plosives; and it has been added to the contrastive system only as a syllable closer (*bum/bun/bung*). The neutralization value before voiceless plosives is determined by the coda prosody (*bump/punt/bunk*); but we find only a voiced alveolar prosody (*fund*). We have increased contrastivisation in this environment rather than neutralization. But this is also a kind of neutralization.

Despite demonstrations of the polysystemicity of phonological representations (see e.g. Fudge [1967] on English and other languages) – so that typically different sets of contrasts are shown to be associated with different positions in onset or syllable or foot structure and under different accents – and, crucially, of the related commonness of neutralizations, writers on language history nevertheless persist in invoking Swadeshian 'phonemic theory'. As noted, in the preface to his *A Grammar of Old English*, Hogg (1992: viii) even explains concerning this choice of 'theory' that 'I prefer to take as my foundation phonemic theory, so that issues are not muddled by theoretical squabbles'! Espousal for phonological representations of 'phonemic

theory’ and its orthographic aims is not just not squabble-free, but its crypto-literal basis is in itself a barrier to the establishment of the phonological systems involved.

A suggested phonological representation that takes the form of such a (tellingly labelled) ‘transcription’ (Hogg [1992: §2.8]) is inadequate in various ways, one of which is, as we have seen, the failure to recognize neutralization. Phonemics cannot accommodate neutralizations without adding to the set of phonemes further symbols that represent segments that participate in reduced subsystems of contrasts. So that we might represent Old English *scrud* as /Skru:d/: /S/ contrasts only with its absence, and the following plosive /k/ neutralizes whatever differentiates /k/ and /g/ elsewhere. As I have again noted, in Modern English initial /k/ and /g/ typically differ as aspirated voiceless vs. unaspirated voiced, but after the initial sibilant the neutralization is neither aspirated nor voiced – or at least there is no contrast between voiced and voiceless plosives after a foot-initial voiceless sibilant.

The neutralization segment /κ/ (or ‘archiphoneme’ [despite Davidsen-Nielsen: 1978: ch. 10]) is distinct from both contrastive velar plosives, but in other neutralizations the archiphoneme may be implemented as one of the neutralized segments. /Σ/ in *spit*, however, as again already observed, is apparently realized/implemented and spelled in the same way as voiceless *s* in most other environments, including initially, as in *sittan* ‘sit’ (cf. e.g. *hittan* ‘hit upon’). The use of ‘Σ’ thus violates what has been called the ‘biuniqueness’ requirement of phonemics, imposed on the relation between phonemes and their realizations. This is conveniently formulated as follows in Lass’ textbook (1984: 24): ‘**Biuniqueness.** Any phone in a given environment must be an allophone of one and only one phoneme – to prevent ambiguity and to secure unique read-off’. This requirement is important in constructing a simple, economical alphabetic spelling system for the language involved, but it obscures the systemic location of contrastivity, whose establishment I take (perhaps immodestly?) to be a fundamental goal of phonology. We have subsystems of contrast, the contrastive relevance of which is generally recognized in relation, say, to ‘long’ vs. ‘short’ vowels in Modern English and other languages.

But Campbell (1959) is not interested in the phonological system(s) of Old English: in his book, as I have observed, there are a few initial sections on ‘Writing, Orthography, and Pronunciation’, but most of the phonic concern of the book is with particular pre-Old English ‘sound changes’ and the ‘sound changes’ occurring in the historical period of Old English, and the relative chronology of these. Hogg (1992) is also mainly devoted to these concerns, with the disadvantage noted above of overtly invoking phonemics. Even more misleading are Lass & Anderson (1975), though they do concede the polychronality of their approach. And, in relation to present-day English, Chomsky & Halle (1968) impute to synchronic phonology, formulated as ‘rules’, the diachronic mutations associated with ‘sound-changes’. However, the effects of these ‘sound changes’ are reflected synchronically, if at all, in morphological

alternations or neutralizations or de-neutralizations, and the ‘changes’ themselves are not a synchronic phonological phenomenon. See again Anderson (2014a: 3). Saussure (1915: Part 3) distinguishes carefully between ‘les changements phonétiques’ (Chap. II) and ‘conséquences grammaticales de l’évolution phonétique (Chap. III)’.

I conclude this initial look at written evidence from the past and present, as illustrated in the relation of this specifically to early English, with a formulation of perhaps its major limitation as evidence for the language of the time. The formal and/or specialized language(s) of these documentary sources – religious, legal, historical, literary – do not for many centuries give us access to the language(s) of the illiterate majority of speakers of early English.

Subsequent chapters introduce mainly further limitations in the practice of reconstruction to those encountered in the present chapter.

One factor that I shall be unable to pursue here in any depth is the diversity of written sources invoked in studies of what can be deduced from particularly early Old English. These range from ‘charters’, a term that, as indicated, includes a variety of document types, to glosses and glossaries, and inscriptions on stone crosses and coins. With this last pair, in particular, inscriptions are typically decorative as well as language-bearing, and the language on early charters is Latin, with only names in Anglo-Saxon, though these may be given Latin inflections.

Stanley (1988: 317) illustrates the pitfalls of interpreting name forms in Latin charters as evidence for Old English sound change, while ignoring the ‘possibility of the influence of Latin orthography’.

Campbell (1938: 150) discusses such forms as *bernhaerdi* and *bernhardi* as if they were evidence for the operation of sound-changes in proto-Kentish, and he has no reference to the fact that neither the Hlotharius nor the Uuhtredus charter in which they occur (Sweet 1885: 427-8 Nos 4 and 5 [S.8 and S.21 respectively] ...), both of them Latin and both of them early, show not a single unlatinate diphthong; and Ström (1939: 92) considers the absence of breaking <ea> before <r> + consonant without considering the possibility of the influence of Latin orthography.

This illustrates the kind of text-based controversy that can arise in interpreting the past.

Sadly, absurd notations have been encouraged even by attempts to represent transcriptions of corpora of recent spoken languages. Advocates of this, such as Durand and Pukli (2004), are often critical of use of the IPA here, as subjective, which is indeed the case. But Durand and Pukli prefer a notation that is at least equally subjective, a computer-readable notation called ‘SAMPa’ (‘SPEECH ASSESSMENT METHODS PHONETIC ALPHABET’). However, this too is subject to the accuracy of the user or particularly the initial investigator. The representation offered by these

authors of a Scots native speaker's (mine, possibly ours) utterance which I might transcribe as [hò:dʲarh wí:ft] is /*hod j@r wiSt*/ (Durand & Pukli [2004: 44]), which may then undergo various 'codings' (see too Carr & Durand & Pukli [2004]). Durand & Pukli conclude with (p. 45) 'Our simple claim is that there are no good theories without good data'. My claim with respect to a present-day language is that good data are in the head of a competent user of the language (see e.g. Anderson [2022a: Preface]) – allowing for variation. The linguistic knowledge of users of the past is obviously much less accessible, and only indirectly manifested. But computer corpora are simply not 'good data', particularly when tagged, of course, so that the analysis is prejudged for users of the corpus (see e.g. Anderson [2022a: 559], in addition to other questions concerning the ontology of corpora in general).

But perhaps the most dramatic development in the recent study of phonology and its representation is the adoption and abandonment by generativists of the phoneme after adopting two levels of its application, and their confusion with history. This is one of the two death blows administered by the generativists themselves. The other death-blow was the adoption of the transformation and all the difficulties it brought with it. Collectively, they destroyed generative linguistics. But the latter is not our concern here.

Chapter 3

Linguistics, grammar, and phonetics

My starting point for what is concluded in this chapter can be seen as the assumption of Saussure as described in Joseph (2012: 236, 237), which begins to unpack the above title.

In the *Mémoire*, the phoneme is not conceived of as a sound as such, but a unit within a system. Phonemes will find their material substance in sound, but that is not their essence. The essential thing is how they function relative to the other units in the same system. Whatever their exact sound happens to be is accidental, contingent. ... The reality of language, Saussure would insist, lay not in sound, not in the muscular movements needed to produce sound, not in the vibratory acoustics of their transmission and perception – but in form, understood as mental patterns, ultimately cerebral traces, socially shared, that make it possible to produce and recognize substance as meaningful language.

(see Saussure [1879]). I want to contrast vocal implementation and auricular extrapolation of phonological representation of content-expounding contrasts with the complex mental status of grammatical structure and what it manipulates, and its determinate stratification into levels. Stratification arises principally in the differentiation of the content and expression planes, but it also arises from the order in the interfaces in which representational properties such as hierarchization (e.g. by dependency relations) and sequencing are introduced. There is, however, as Kaye (1995) affirms (vainly, in its effect on many researchers), no phonetic level of grammatical representation. Unfortunately, for him and others, such a recognition leads, perversely, to the regarding of many phonetic events as part of phonological ‘derivation/change’.

The content of what implements phonological elements in sound can be variously detailed, as indicated in traditional alternative ‘transcriptions’, a variety partially concealed by the traditional distinguishing of ‘broad’ and ‘narrow’ transcriptions, which embody phonologically arbitrary conventions, not necessarily of systematic status; spectrograms and other instruments also may be of varying definition. Likewise, the non-determinateness of articulatory implementation is complemented by the reception by hearers of sound of varied definition and detail, from which contrasts may or not be extracted.

Determinate grammatical levels and particularly planes arise only when contrast is expressed by the poles of signs and the extra-lexical and lexical structures of syntax and phonology based on the information in lexical entries. Signs are learned conceptual entities with colligational specifications; but, as observed, their

implementation involves quite different capacities, motor control (articulation) and hearing (audition). David Abercrombie wisely entitled his 1965 collection of papers *Studies in Phonetics and Linguistics*. But I would have had to reassure him that my intention here is not to demean what he makes the former discipline in his title; my aim is merely to remind us (possibly unnecessarily) of a crucial difference in what these two disciplines, particularly if distinguished as grammar vs. phonetics (regarding both as linguistic disciplines), are concerned with, and a consequent limit on the domain of grammar.

I am assuming, then, that grammar, in an inclusive sense, gives an account of linguistic knowledge, which includes the ability to assemble linguistic structures on the basis of what is stored in or can be provided by redundancy specifications in the lexicon. I do not distinguish between ‘competence’ and ‘performance’, which distinction introduces a mode of evasion of evaluation of claims about language; but I recognize mis-performance as something for researchers to be aware and wary of. The major distinction I would make is between the information that the language user can extract from the lexicon and the creation of non-mutational syntactic (content) and phonological (expression) structures that, with the help of awareness of context, are assigned at the interfaces between these three components: the lexicon, the content plane, and the phonological plane.

I avoid here irrelevant grammatical particularities as much as possible. Nevertheless, in what follows I shall assume a ‘substance-based’ grammar: both syntax and phonology are based on mental substance, conceptual and perceptual, and they give it a particular structure based on dependency relations whose nodes are characterized by bundles of substantive features. This is not just because I would advocate such a grammar – though I do (cf. e.g. my contribution to Andor [2018]) – but because this sharpens the distinction between grammar and the implementation of grammatical structures. And, as we proceed, beginning with this chapter, I shall introduce some, I hope useful, notation drawn from such a framework – as illustrated in Anderson (2011c: particularly chs. 2-4) and (2022a, b).

In terms of such a substance-based view of grammar, I contend that the study of how linguistic knowledge and the linguistic structures that are its artefacts do not include the study of how language structures are made audible, via vocal articulation; and, indeed, the extraction of linguistic structure by hearing via the auricular mechanism is also not part of grammar. Nor are ‘transcriptions’ or acoustic records part of grammar. A critic of this view might conceive of this theoretical position as ‘paradoxical’: a grammar based on substance does not include the study of the substance by which linguistic representations are implemented or recognized. I suggest that this ‘paradox’ is not such; and the clue to that is in a comparison of the substances. This position is not paradoxical because the substance of grammar is conceptual and perceptual, mental, while that of what is described by phonetics is operational; it is an

interface that involves motor control of the vocal tract that provides audible implementation of phonological structure or the recognition of linguistic structure in the sounds that are registered auricularly via the ear tube. These are quite distinct capacities from knowledge and use of grammar as such. Phonology represents, in the first place, contrasts in the internalization of perceived sound that differentiate among signs, more specifically those contrasts belonging to the expression pole of the sign, particularly the minimal sign, or word.

As well as with lexical contrasts, phonology is also concerned with extralexical contrasts, particularly in tone and its placement, that signal (conceptual) content distinctions, such as emphasis and questioning, as well as the adjustments to minimal signs in sequence to do with the placement of sentential accentuation and of word-boundary phenomena such as attachment of an unstressed initial syllable to an accented syllable ending the preceding sign. In *Move around the corner?*, *a-* and *the* form part of the feet governed by the accented *move* and *-round* respectively and the latter are part of the tone group headed by the primary/tonic accent on *corn-*, which bears the tone that (in this case) signals interrogation. This is no place to elaborate the phonological representations I would suggest (drawn from e.g. Anderson's [2022a, b] *Grammar of English*), but perhaps this brief description will have conveyed something that will be helpful. However, Appendix II here provides a detailed account of a (diachronic) change in grammatical structure, particularly the (morpho-) syntactic.

Phonology represents sound images, such that there is typically an arbitrary relation between the expression and content poles of the sign, though there have been various different kinds of attempt at recognizing 'sound symbolism' (see e.g. Colman [2014: §5.4.2] and the references given there). And the content pole and syntax represent conceptualisations, but are often also perceptual, at least figuratively, particularly metaphorically or metonymically, in dealing with non-concrete, particularly mental domains. As a familiar example, thoughts may be represented as part of the content of the brain or even of the head, as in *I just can't get that out of my head* – or, perhaps more picturesquely, Dashiell Hammett's *I woke next morning with an idea in my skull*, at the beginning of ch. IX of his *Red Harvest*. These are suppletive, not merely supplementary, figures; they are instrumental in the expression of thought in speech. Indeed, *mind* and *brain* can be seen as offering different perspectives on the same entity, though the interaction of the perspectives is still to be clarified.

In this way language is indeed the principal 'language', or carrier, of thought, though recourse to it seems to vary with individual language users. Artificial, or philosophical 'languages', or 'syntaxes', are simplifications of 'real' language (often naively believed to be 'improvements' on the latter). Much language use is thus not implemented; much 'communication' and interaction is with our selves, often silently, without even requiring, to ensure this, the intervention of a strict librarian. Given the prevalence in language of envisualisation and figurativeness, including iconicity,

much content as well as expression is ultimately perceptually-based, unless conventionalized in usage. Extra-lexical content and expression structures are erected on the basis of the contents of the lexicon, i.e. as anticipated, at the interfaces that feed hierarchized and sequential content and expression structures.

Language users select from the lexicon signs that are equipped with valencies that allow them to be assembled into potentially self-standing meaningful units. But at the same time, such units may be self-standing only by virtue of encyclopaedic knowledge and/or awareness of the context, both situational and textual. For an (obvious) instance, many answers – e.g. *Tomorrow*. – can be understood only in context. Language, however, helps contextualization by supplying deictic and referential signs, which may often be supplemented by ‘literal’, gestural deixis. Semantically-based syntax and perceptually-based phonology are substantively based, as is phonetics, but differ in focus, in the substance represented; I thus am not in agreement with Hale and Reiss (2000) and others who would deny substance to phonology. That substance is typically implemented in sound, but there are other media available, involving graphic and gestural, in many cases parasitic upon phonetics. We also need to recognize that syntactic structures can be lexicalized and acquire a distinctive sense, sometimes distinguished as ‘idioms’, sometimes confusingly labelled as ‘constructions’, presumably in the latter case because they are not simple signs.

As I have observed, studies of the history of English and other languages are particularly susceptible to confusion of phonology and phonetics, and concerning the location of contrast. Thus, for instance, Hogg’s (1992) description of various ‘sound-changes’ reconstructed as having occurred in pre-Old English or historical Old English confuses the nature of the change. Hogg adopts what he terms a ‘phonemic theory’ (again [1992; preface, p. viii]), and his notation distinguishes between contrastive forms/segments and phonetic in terms of inclusion within ‘/ /’ rather than within ‘[]’, though he does sometimes relapse into an indeterminate italic notation, as in representation of the *i*-umlaut ‘sound-change’ in (his p. 124) ‘**gaduling* > *gædeling*’, ‘companion’, with the pre-historic (so not testified orthographically) form distinguished by the asterisk. *I*-umlaut is introduced in more detail on his p. 121 as involving historical sequences such as */*gaduling*/ > */*gadyling*/ > */*gædyling*/ > /*gædeling*/ = *gædeling* ‘companion’, where the different stages separated by ‘>’ are apparently interpreted, by the use of ‘/.../’, as phonemically different, involving changes in the system of contrast at particular vocalic positions. But all that the reconstructed successive first three (prehistoric) representations that he suggests is a suprasegmental expansion of the phonetic implementation of the typically vocalic secondary feature that I label (following Anderson [2022a: ch. 12]) as **i** (a secondary feature implemented in the area of ‘high front unrounded’, in articulatory terms), which combines with **u**, and to give {**i,u**}, and then with **a**, to give {**i,a**}. What is more

plausible is that these are implementational (phonetic) changes, one of them finally lost. A final mystery is that these phonemic forms seem to end up as a spelling!

Overall in the above history, we have a change from *i* as associated with a vocalic segment, as represented by the primary and secondary feature combination $\{V\{i\}\}$, a primary vowel feature *V* combined with the secondary *i*, together the final vowel represented throughout Hogg's sequence. The *i* in this morphologically derived word stem 'spreads' through the derived noun, to association with the node governing the whole derived form, and an unbracketed final form (spelling?) and with eventually a weakening of the medial vowel in the post-initial sequence. But there is no evidence that the changes are independently contrast-creating, (/ /), or of how these assumed changes are ordered, unless we assume that this kind of 'harmonic' change, or 'spreading', must take place segment by segment. However, the medial vowel reduction may precede or follow the 'harmony', since a possibly reduced segment need not express or block a 'spreading' of the *i* vowel quality. And the 'spread' of the feature need not appeal to the intermediate stages in Hogg's sequence; indeed, there is no evidence for these intermediates. Either way we have the development of a prosody, a change from intrasegmental segment to suprasegmental status for *i*: a change in location of the contrast.

I shall look more formally at what is happening in *i*-umlaut in §4. But firstly I shall look at a related but perhaps more spectacular synchronic phenomenon in various languages, prompted by Hogg's rather vague comment on *i*-umlaut (p. 121, again) that 'There is, therefore, no reason to suppose that we are dealing with anything other than a type of vowel harmony'.

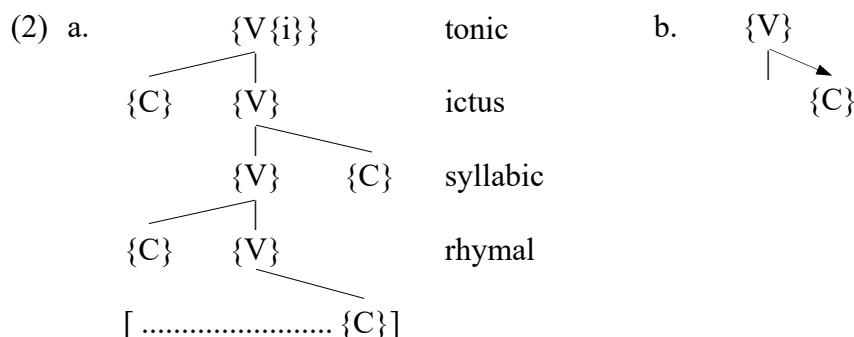
Phonology vs. phonetics is an important distinction which, however, is often blurred by students of language. 'Phonological spreading', for instance is often attributed to the 'vowel-harmony' systems discussed in relation to a number of languages, but I shall suggest that this 'spreading' is associated not with phonological spreading from segment to segment but with articulatory implementation. I shall illustrate this with respect to a simple hypothetical but typical 'vowel-harmony' system.

I take 'vowel-harmony' to involve a prosody in a variant of the Firthian sense (1948) – the 'word prosodies' of Lass (1984: §10.2.3). A certain feature (or feature value, if you prefer) characteristic of vowels is manifested throughout a word form in my (hypothetical) example, but certain segments, especially non-vowels, may render it more or less opaque at that point in the sequence. Phonologically there is a single contrast, the presence vs. absence of the prosody; its 'spreading' – better, implementation in 'real' time – over the stretch of sound subordinate to the prosody-bearing tonic (root) node of the word is manifested in articulation, but there remains only one contrast, that between the prosody and its absence. Implementation does not change this; it is not a grammatical phenomenon.

We might store such an expression form in the lexicon as in (1), where the square brackets mark the boundary of the expression pole of a minimal, or word-stem, a sign consisting of a variably large set of segments, where only the relative order of the syllables and in some cases their eventually shared provisional content need be specified (though this is not relevant here).

(1) [{i} {{C}} {V} {C} {V} {C} V {C} ...]

{i} is the prosodic phonological feature, representative of common word-prosodic features. And it is extrasegmental: it lies outside the set of segments, represented here by unspecified capitals, apart from the C/V distinction, whatever they might be. In the interface between lexicon and lexical phonology these segments are linearized and project a dependency structure in which {i} is attached to the tonic, the head of the word form, stem or full word, depending on the language.



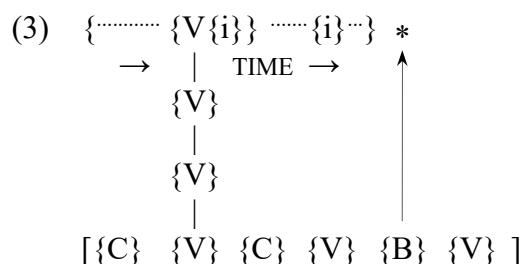
Let's say, for simplicity of illustration, that we take the second segment here at its possible face value in (1) – 'V' – as a vowel, and that this vowel supports that word-form phonological head, so that we have, whatever else, the subjoined (unsequenced) dependency structure of vowels in (2a) above the V of (1). From the top we have a tonic (intonation-carrying) head {V}, an ictus (foot), a syllabic, and a rhymal {V} (which are all potential harmony-carriers, though here we are concerned with a harmony-bearing tonic and its implementation). Each line in (2) is a directed arc: direction embodies dependency of one dependency-terminating node on another, the upper one, the head; the lower one is dependent. The dependent {C}s in (2) are adjoined, here as {V} → {C}, as redundantly spelled out in (2b); they are dependent on and linearly distinct from a {V}. {V}s except the tonic are subjoined – they occur at the same point in time as their heads: the ultimate head (tonic) governs a construction. Two dimensions are thus involved, sequence in time and dependency/subordination.

{V} in (2) is a major category, one of those features that alone or in combination represent the major categories that determine the basic distribution of segments; {i} is a minor feature, often subclassifying a {V}. The rest of the structure need not

concern us, so far. The present harmony is a feature of the tonic, the node that carries ‘primary accent’ and bears a tone when the word is pronounced in isolation. A non-intonational minor feature of the word head and its articulation are expressed (other thing being equal) throughout the word-form, though some segments may be more or less opaque to it, with the typically most transparent being vowels, as suggested by the traditional nomenclature.

However, it is likely that this whole configuration in (2) will be stored, or at least the set of segments, their syllabic structure, and the value of tonic-prosody. (1) provides the non-redundant set of contrasts, one of which is extra-segmental with respect to the word structure, without specifying the redundant tonic selection, which is erected in the interface between lexicon and lexical phonology, in which accentuation and sub-syntactic sequence are assigned. Figures 1 and 2 simplify the lexical representations, of course, by ignoring any indication of syllable sequence and presence of segments within onset or rhyme, which, unlike sequence within onset and rhyme, are not redundant (see Anderson [2011c: part III], and Anderson [2022a: chs. 6, 11-13]). But these absences are not pertinent to our concern with the ‘spreading’ phenomenon and its status.

There is a number of ‘harmony’ systems. But generally implementation of the harmonic feature goes from left to right, i.e. in accordance with time within the domain of harmony. In some systems there may be a segment type, whose nature varies in different systems, that is not just (relatively) opaque, but ‘blocks’ the harmony from being manifested in the following sequence of segments, i.e. in segments that are articulated after it. And appeal to articulation is the point of my illustration. For ‘blocking’, as with ‘spreading’ itself, applies to the implementation of the harmony, which is in real time. It is not a phonological phenomenon: what is phonological here is the presence of the harmony, implemented phonetically within the word form, which can contrast with its absence in distinguishing lexical items, however far its implementation extends within an item. The possibility of ‘blocking’ is a by-product of implementation of the prosody, in the form of a particular articulation, which, as occurring in time, can manifest the ‘blocking’ of the prosody in the course of its implementation, as diagrammaticized in a simple way in (3), where {B}, whatever its value, is a ‘blocker’.



Once more, in the hierarchy of $\{V\}$ s, each is subjoined in dependency to the one above, and i is representative of the secondary features that may sub-classify vowels, whatever else. Here the tonic vowel is the root of the dependency ‘tree’, other aspects of whose structure are not specified here. The upper dotted horizontal marks the path of implementation of the prosody $\{i\}$, starting at the beginning of the phonological, or expressive, pole of the lexical item; and the upward arrow indicates the implementation of the ‘blocker’, i.e. $\{B\}$. The ‘blocking’ belongs to phonetics; it does not introduce a contrast, which is a property of phonology, specifically in this case of the tonic node; here we have combination of a prosodic contrast and (unspecified here) segmental contrasts and their phonetic implementation, involving ‘blocking’ of the implementation of the prosody in the present case. If readers prefer, or demand, ‘real time’ to be matched with ‘real examples’, they may consult the extensive survey in van der Hulst (2018), which also provides further complications in implementation. These include interaction between the implementations of different harmonies, or the possibility of more minor differences, such as the location of the tonic node within the lexical item – though the ‘real-time spreading’ is still from the beginning of the first accessible segment of the relevant construction, comprising the head and its subordinates.

There is also a brief discussion along the present lines of two familiar ‘harmony’ systems in Anderson (1992a; 2011c: §§4.2.1-4.2.2). But that account does not insist on the implementational nature of the ‘spreading’, and, ‘retro-perversely’(!), uses a different terminology from the present description. But we can extract from that and other accounts of the Finnish and Turkish systems ‘real’ analogies for the simple hypothetical situation just described, as well as complications such as ‘blocking’ or ‘opacity’, or many complexities found elsewhere, as again illustrated in van der Hulst (2018).

The often cited Finnish word *pöytä* ‘table’ can be said to ‘contain’ the prosodic feature $\{i\}$, and it contrasts in this respect with *pouta* ‘fine weather’, which lacks $\{i\}$. The former item differs from the latter in the presence of an i attached to the word tonic. In the lexicon this could be stored in the manner of (1) (though sequence of syllables is stored, and within syllables is redundant, apart from whether they belong to onset or coda), with the prosodic feature given outside the set of segments, as in (4).

(4) {i { {C{ {u}}}, { {V{a.u}}, {V{u}}, {C}, {V{v}} } } }

Lexically, *pouta* would lack the **i**, and so fail to show ‘harmony’. In the lexical phonology structure of *pöytä*, but not *pouta*, {i} will be attached to the lexical tonic at the interface to lexical phonological structure, as in the artificial example (2a-b) suggested above.

What we have been looking at here is the reconstruction of suprasegmental phenomena which, as well as involving phonological features as the basic entities rather than the segment are themselves implementational spreadings of these. We take up similar phenomena in the following chapter in the form of prosodies, and there and in the chapters that follow I shall elaborate on the character of grammatical structure, phonological, lexical, and (particularly in Appendix II) syntactical and morphological.

Chapter 4

Umlaut as prosody

We have been looking at word prosodies. There are also, of course, prosodies associated not just with word-dominating phonological nodes but also with lexical-derivational or inflectional structure, and sometimes a diachronic segmental ‘trigger’ of the prosody can be identified and may remain as such, for a period at least. There is evidence for the diachrony of at least some prosodies that involves a rather more intricate picture than the synchronic presence of a word ‘harmony’ (with its own varieties) perhaps suggests.

For instance, there was introduced in Chap. 3, a suffix-induced prosody in pre-Old and Old English and other languages that I shall again represent as {i} (implemented as ‘high front unrounded’ in articulatory terms), which is introduced by the presence of certain suffixes both derivational and inflectional, though Campbell (1959: §192) too suggests that it, traditional i-umlaut, ‘may be a mere vowel harmony’ – whatever he may imply by that, particularly the ‘mere’.

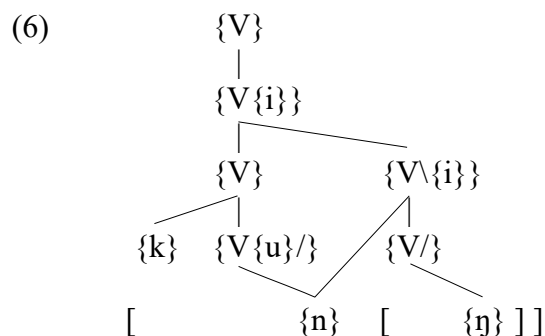
The development of the umlaut variety of prosody described by Barbara Strang (1970: 387-8) is manifested in the history of the derived form in Old English spelled *cyning*, ‘king’. One might, in the notation of Anderson (2022a: ch. 6), represent the development of the pre-prosodic suffixed form lexically, again crudely, as in (5).

- (5) a. [{k} {V{u}/} {n} [{V{i}/}{η}]] →
 b. [{k} {V{u}/} {n} [\{i} [{V/} {η}]]]

In (5a), {V{u}/} is transitive (or ‘checked’ or ‘short’ or ‘lax’, if the reader prefers), so taking a following complementary consonant rather than an optional adjunct; its transitivity is marked notationally by an immediately following ‘/’, as shown in the representation. This first (transitive) vowel segment has a secondary feature {u} (high back rounded, in articulatory terms). The characters of the other (non-vocalic) symbols (such as {η} – segment or cluster?) are immaterial, as simply ‘place-holders’, at this point. The outer square brackets in (5a) again enclose the word form/stem, and the suffix has its own brackets included within the outer brackets. The two vowels belong to distinct local contrastive systems.

The first step in the change towards a prosody, in the representation in (5b), is the assimilation of the first, accented vowel to the quality of the suffix, creating the possibility of a prosody, with an outer and an inner bracketing of the suffix. Simple {i} is excluded from the innermost bracketing in the suffix. This and the immediately preceding ‘\’ mark {i} as inviting a prosody: it is not simply part of a vowel segment within the suffix; and the ‘\{i}’ notation specifically ‘asks for’ a governing {i}. This is the diachronic mechanism of umlaut.

This results in a morphophonological prosody, triggered by the presence of the (in this case, derivational) suffix with the potential for projecting a prosody. A segmental feature has come to be implemented as prosodic: this status is represented more transparently in the partial dependency tree in (6), where the suffixal vowel is attached to the preceding ictus.



At the interface with the lexico-phonological interface a suprasegmental (lexical) phonological dependency-based structure is erected, and the prosodic feature is attached to the nearest governing $\{V\}$, the ictus that in this instance governs the vowel of the base, as indicated in the more extended dependency structure of lexico-phonological representation of (6), which displays both adjunction (the undergoing of sequenced government) to the ictus and adjunction to the rhymal $\{V\}$, as satisfying the rhymal requirement. And there is also mid-foot ambisyllabicity: the mid-foot consonant satisfies the transitivity of the first vowel and the (unmarked) initial-maximalism of the following syllable.

At this point it may be helpful to describe further aspects of ‘dependency’ as assumed here, particularly since, unfortunately, the term ‘dependency’ is promising to become as variable in usage as ‘grammaticalization’, often attributed to lexicalization/idiomatisation. The relevant view of dependency invoked here was inspired by the work of Hayes (1964) and others in that decade and is introduced in Anderson (1968, 1969, 1971) and developed in later works of Anderson and Böhm; and it is presented most recently in a ‘conversation’ published as Andor (2018), and in Böhm (2018) and Anderson (2022a, b). This dependency relation was introduced in phonology by Anderson & Jones (1974), which also mentions shorter presentations of the time. Developments are presented in Anderson (1986a, 2011c) and Anderson & Durand (1988b). Anderson & Ewen (1987), despite its title, is an amalgam of ‘dependency phonology’ and phonological/phonetic ‘geometry’. The progressive integration of syntax, phonology, and lexicon is attempted in Anderson (1992b, 2004, 2011b, c, 2022a, b), and elsewhere.

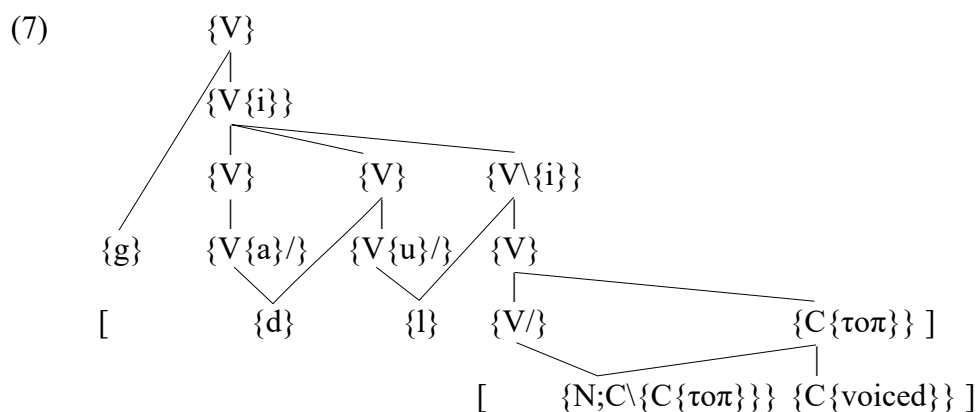
As noted in Ch. 3, the solid lines joining category symbols in (6) represent dependency relations; they are ‘arcs’ (‘directed’ lines) and may involve either

adjunction (slanting) or subjunction (vertical), i.e. dependency accompanied or not by a distinction in sequence (time) between head and its dependent(s). The dependents of dependents are said to be subordinate to a head; together they constitute a construction; and a head that is not also a dependent is the head of an independent construction, a potentially independent sentence in syntax or tone group in phonology.

The capitals in (6) designate primary substantive phonological features and the lower case secondary. The ‘/’ symbol signals a transitive category and ‘\’ triggers a particular prosody head. The two ‘*n*’s are ‘place-holding’ letters. The first of these is ambisyllabic, and the second is adjoined to the head of the coda. The inner and lower square brackets in (6) enclose a suffix, here derivational. The representation of phonological structure is not ‘flat’ but two-dimensional, with vertical dependencies (straight or slanted) and simple linear sequencing; and phonologically there is both a lexical structure and pre-utterance structure, and optionally metrical structure, as in the top-line of each of ‘(124)’ from Anderson (2022b: ch. 34) quoted below in Chap. 5, on appeals to metrical evidence (though we should acknowledge the variety of accentual and metric structures found elsewhere). The structure in (6) ignores other aspects of the morphology of the form.

The spelling of the first vowel in *cyning* represents the combination of {*u*} and {*i*}: the value *u* is perceived as a step closer to *i* on the front-back articulatory dimension: the combination {*u,i*} is high-front-rounded in articulatory terms. The prosody is implemented on any compatible segment between the source vowel and the target vowel of the assimilation: a so-called ‘harmony’, again an implementational phenomenon. See too Anderson (2011c: §4.4.1), for instance. But for a more comprehensive account of *i*-umlaut in the context of other developments in Old English see Colman (2005).

In trisyllabic feet where the first two syllabics lack an inherent {*i*} feature the whole foot is still subject to such a prosody triggered by the last syllabic. We thus find our old friend with the spelling *gædeling*, ‘companion’, with the remains of ‘double umlaut’ (Campbell [1959: §203]), where the first two syllabics are usually reconstructed as having {*V{a}*} and {*V{u}*} as their pre-prosodic ancestors. At this prehistoric stage the prosody can be reconstructed as in (7), where again non-active consonants are abbreviated as a simple traditional transcription, except that I have opened out the final cluster, headed by the less sonorous consonant (thus, qualitatively more differentiated from the syllabic), to which we shall return below.



The first vowel is heard as {a;i} (**a** dominates over **i**, giving a ‘fronted’ {a}, spelled *æ*), since there is a contrast with {i;a}, spelled *e*; and the second vowel again is perceived as the simple combination {u,i}.

There are also two ambisyllabities following the tonic accent in (7). And the medial syllable is subsequently reduced, reflected in the spelling; this is not unexpected, given that its ‘transitive’ vowel shares its complement with the onset of the following, final, syllable as well as sharing its onset with the complement of the first vowel, thereby fulfilling onset maximization; the vowel thus occupies a classic position for diachronic ‘weakening’, or reduction, of a vowel to occur. A reduced vowel is plausibly regarded as opaque to the prosody, and, anyway, the reduction of this vowel apparently post-dates the triggering of the prosody. In the coda of the last syllable, as normally in any consonant cluster, the more sonorous consonant, a nasal, depends on the less, the more plosive-like and more differentiated from **V**.

I note too in relation to (7) the ‘further’ (consonantal, or coda) prosody involving the final cluster. The content of the final coda prosody and its trigger, where {τοπ} is a variable over (articulated) ‘place-agreement’ and {V;C{c}} is a place-neutral nasal, and the coda is interpreted as consisting of two rather than one consonant – unlike in the stop-gap representation in (6). If the third syllable in (7) is argued to have a non-primary accent (whose presence is contentious, particularly since the ‘evidence’ for this is metrical and the latter is based on pre-utterance prominence, not lexical accentual structure), the placement of the vocalic prosody and its trigger would need to be adjusted. I observe in passing that reduced vowels, represented as {V</>}, may or may not have a satisfied transitivity, depending on whether or not a suitable following coda consonant is available.

When/if the {i}-suffix involved is lost, as it is in *bec*, the nominative/accusative plural (and genitive and dative singular) of *boc*, ‘book’, one might expect the prosody to disappear. But the manifestation of the prosody in the implementationally transparent formerly intervocalic consonant as well as the vowel quality is retained: thus *bec* ‘books’ is usually interpreted as phonologically ending in a palatal consonant

of some sort; Campbell (1959: §428) and others indicate this with a dot over the *c* graph. The rhyme segments are collectively contrastive: indeed, the rhyme apparently bears a prosodic {i}, and it is inflectionally contrastive. We have a paradigmatic prosody associated with the rhyme of forms of this item. *Boc* ‘book’, consistently in nominative/accusative singular forms, is interpreted as lacking ‘high unrounded frontness’, or (better) {i}, throughout its rhyme. The prosody is ‘triggered’ by particular inflectional values; it too is morphophonological, but inflectionally, not derivationally. In forms like *king* the prosody has subsequently been segmentalized and installed as an instance of a member of the major transitive-vowel subsystem.

Sometimes the prosody is apparently neither derivational nor paradigmatic, but a simple rhyme prosody, without a ‘trigger’, as in *bæc*, ‘back’, or *pic*, ‘pitch’ (Campbell [1959: §428]); but historical *bæc* has lost any prosody it might have had (Campbell [1959: §434]), as has the plural of *book* – though in this case apparently as the result of morphological analogy.

Adze in modern English exhibits a coda prosody of voicing, attached to the head of the coda, realized as the plosive. But this is taking us further off on a side-issue: the prevalence of prosodies in phonology and thus of persistent articulations in the implementation of phonology. Implementation does not involve simply the identifying of so-called ‘allophonies’ – i.e. polytopical contrastive units, involving contrasts that occur in different positions – as well as the filling in of redundancies. The ‘filling-in’ of these, however, takes us back to the main points here, the non-grammaticality of phonetics, and thus the lack of a principled point at which to stop filling in non-contrastive implementational details as part of grammar, i.e. the lack of a non-arbitrary phonetic level.

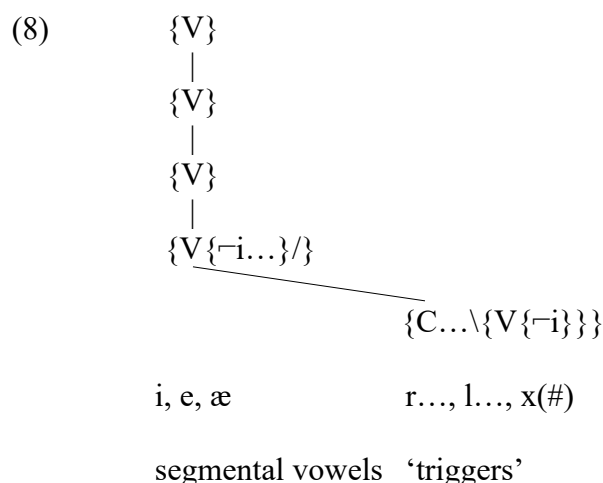
Another prosody, whose implementation is more contentious, involves the (initiation of the) coda or a glide to it (?) as a consequence of the Old English ‘sound-change’ that is usually rather prematurely labelled as ‘breaking’, though the only thing that is uncontroversially ‘broken’ is the spelling of the syllabic concerned (with two successive vowel graphs): hence the continuing ‘digraph controversy’ in studies of Old English. We are concerned in this case with forms such as (in these examples) a transitive (short) monosyllabic *feoh* ‘cattle’ and *bearn* ‘child’ and with similar intransitive (long/tensed) vowels. The vocalic phonetic implementations are contentious, given doubt concerning the precise scope of the ‘backness’ rhyme prosody that seems to be involved and the phonological or phonetic interpretation of the digraphs: is the ‘backness’ projected by the coda initiation realised as the second component of a diphthong, or as a feature of the rhyme or even a diacritic of the backness of the following consonant.

Hogg (1992: ch. 5, section II) defends the traditional ‘diphthongal’ interpretation, with his usual confusion about bracketing: as in ‘The short vowel /e/ is regularly broken to /ëu/’ (p. 88), even though the proposed diphthongization is conditioned by

the context. The tradition is also defended by Kuhn & Quirk (1953, 1955) and Kuhn (1961), in response, particularly, to Stockwell and Barritt (1951, 1955, 1961). Other alternative views have been offered by Daunt (1939) and Hockett (1959).

A similar dilemma is associated with the later ‘back umlaut’ phenomenon said to be exemplified by *heofon* ‘heaven’. The ‘backness’ prosody here is projected by the vowel of the following syllable and it affects only transitive (‘short’) vowels as well as being dialectally more restricted. Perhaps again the positing of a prosody is as far as we can safely conclude here on the basis of the available evidence, unless the spelling is simply diacritic. What we lack in this case too is clear evidence of how the front rhyme vowel interacts with the back prosodic vowel it is assigned in accordance with an equivalent of the structure for traditional breaking given in (5) below. Compare Lass’s passing suggestion of ‘breaking before back vowels’ (1994: 51) as a title for the immediately above phenomenon, given the shared problems of interpretation with ‘breaking’.

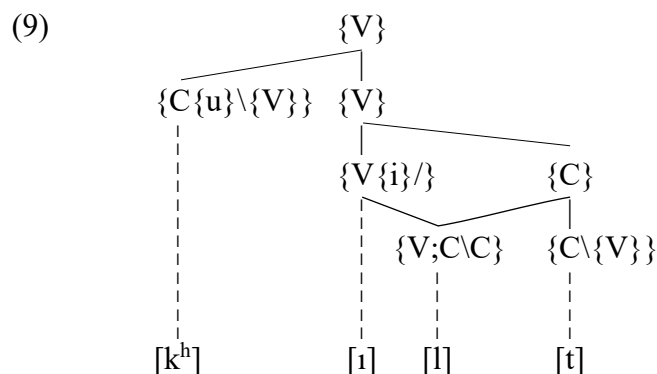
We can sketch out the assignment of the prosodic value in traditional breaking of transitive vowels as in (8), perhaps, where a non-front feature ($\{-i\}$, with ‘ $-$ ’ as an absence marker) is added to the rhyme that the coda depends on, a representation that ignores the head of the coda in the case of the liquids (as well as the onset).



With ‘back umlaut’ the trigger is the back vowel of a syllabic adjoined to the rhyme head, as in the *i*-umlaut of (6); but again the implementation of the affected vowel is apparently uncertain. This is another context where a reconstructional notation must be capable of admitting incompleteness in our interpretation: in such instances different interpretations are difficult to evaluate; the value of the affected vowel is similarly contentious. And again a prosodic development replaces segmental implementation. But ‘breaking before back vowels’, rather than the structure in (8),

however, is a simple parallel to the fuller representation in (6), but involving $\{-a\}$ rather than $\{i\}$, would be appropriate.

In outlining these prosodic histories in this and the proceeding section, I have simplified the structural representations. Compare a full phonological syllable in Modern English in (74a) of Anderson (2022a: 85), given here as (9), with a rough transcription to help the reader, where $[k^h]$ is in contrast with $[g]$ (as in *guilt*), and they are neutralized as $[k]$ after initial $[S]$ (as in *skill*).



The uppermost $\{V\}$ is the syllabic and the lowest is the rhyme, complemented by the immediately preceding consonant in the coda; and the intermediate $\{V\}$ is introduced by the plosive modifying the rhyme head. The higher consonant in the coda is the less sonorous, which the lateral is therefore subordinate to. There is no prosody, but there is a distinct system of contrasts at each position.

The claimed non-grammaticality of phonetics does not entail a denial that differences in implementation are one kind of indication of variation among and within social and/or localized groups of speakers. Phonetic phenomena are also important among factors promoting language change. And the evolution of the implementational capacities involved facilitated the innovation of language itself, having rendered our ancestors ‘language-ready’ (see e.g. Hurford [2014], extending the tradition of Darwin and Saussure). But they are not part of grammar, and there is no determinate phonetic level of representation. Even the lower limit of the capacity to discriminate perception of sound varies among language users. On the other hand, much use of language, as the primary medium of thought, is silent, unimplemented phonetically – or, indeed, visibly, graphically or gesturally – though, not uncommonly, (intended) solitary reading is accompanied by lip movements, at least. An extreme manifestation of non-implementation is instanced by the affliction of tinnitus.

Chapter 5

Accent and metre in reconstruction

I noted above a final omission from many applications of alphabetic systems: the absence of any representation of suprasegmental structure, apart from ‘mood’ markers (such as ?, !) and clause delimiters (;, :, ,, .), though the comma also has an enumerating function, and the dash (–) makes a loose connection; and various signals of status may be signalled by font variation. Particularly evident even in many modern orthographies is the absence of any indication of variable accent placement within the word, whether tonal or dynamic or quantitative. Obviously, this is important for reconstruction when accent-placement in the word is not on the same syllable-position, relative to the end or beginning, in polysyllabic forms. Present-day Greek, for instance, very generally marks the (dynamic) accent, even though there are some (particularly morphophonological) regularities in selection of placing. But in Italian, indeed, only non-penultimate accents need be marked. In Old English, as in present-day English spelling, there is no indication of the variable accent placing, if we ignore the acute accents occurring in Old English texts sporadically and with various, often unclear, functions, though in both stages of English there are generalizations concerning accent-placement, briefly acknowledged at this point.

There are (dynamic) accentual regularities in words that are shared by Old and Modern English, associated particularly with word class and morphology, but often they differ, especially as a result of intervening native morphological simplification and lexical innovation, as well as the acquisition of ‘loans’ from other languages; and both English accentual systems offer some controversies. The modern system, which has a complicated recorded history, involves a number of non-native accent-determining factors, including the role of some affixes and loan types (see e.g. Anderson [2022b: chs. 27-8]).

The survival of some versions of Old English metrical works provides an awkward addition to the range of texts containing representations of the language of the period. In the first place, this last phrase must be qualified by admission that the surviving text of, say, *Beowulf* has often been regarded as from a much later period than that at which it is estimated the original was composed (see Wrenn [1953: Introduction, §4], Chase [1981], Klaeber [2008⁴: Introduction, §II] – though the 3rd edition of this last is more carefully executed). This conclusion concerning dating is partly based on apparent discrepancies in use of the reconstructed metre and phonological reconstruction of the spellings of certain forms. And this brings us to the nub of the awkwardness in invoking metre: even if there were no time discrepancy in textual versions, a (misapplied) reconstruction (i.e. placement of word accents) is based on an abstract reconstruction (metre): see too McCully (1992), Colman (1994: §3.6), and further below). Such dynamic metres impose additional patterns on the accentual and other

characteristics of sequences of word forms; and there is little litteral evidence (except for the pattern of alliteration and its correlation with verse lines) for the placement of utterance and particularly word accent or the role, if any, of syllable weight. (On dating evidence in general see Amos [1980]).

Both later English and languages thought to be closely related to English – a source of evidence we shall pursue below – are reconstructed as showing isochronic dynamic accent, and, crudely, a hierarchical distinction between the accent that bears tonal features (‘primary stress’) and that which doesn’t necessarily (‘secondary stress’) and unaccented segments and reduced: on accent in English, cf. e.g. Halliday (1967), Gimson, revised Ramsaran, (1989⁴; §10.5), Cruttenden (1997). It is thus perhaps not too immodest to suggest a similar suprasegmental character for (Pre-)Old English. These distinctions between tonic accent, ictus accent, unaccented, and reduced are relevant to phonology at both the lexical level and the syntacticized utterance-ready level, as is also the occurrence of ‘weak’, or ‘light’ forms, of words or affixes in certain circumstances. And, most pertinently, metre is built on pre-utterance (not lexical) structure, where ‘pre-utterance’ is the structure resulting from the building of a grammatical sentence, which is ready to be implemented phonetically, or uttered. The metre doesn’t reflect lexical accentual patterns directly, but only indirectly via sentential accent placement.

Metrically, dynamic verse typically manifests alternating dips and lifts; it may be that a metrical line has a lift followed by a possibly disyllabic dip or vice versa: double dips mean that neither syllable is more prominent than its adjacent syllables or line boundary. Anderson (2022b: 215) cites one suggested Modern English metrification of a familiar passage from Shakespeare:

‘(124)’ a. D L D L D L D & L D L
 | | | | | | | | |
 V V₃ V V₃ V V₄ V V₃ V V₄
 | | | | | | | | |
 You call me mis-be-liev-er cut-throat dog

Here the top line expresses versification (dip vs. lift) and the second sentential accentuation, if any, comprising primary, V₄, or secondary, V₃ (a notation to be explained below in §8), and ‘&’ marks the caesura (boundary of half-lines).

Anderson’s (124c) contains a disyllabic dip, in contrast with his (124b):

‘(124)’ b. D L D L D L D L D L D (E)
 | | | | | | | | | | |
 V V₄ V₃ V₄ V V₃ V V₃ V V₄ V
 | | | | | | | | | | |
 O thou foul thief where hast thou stow’d my daughter

c.	D	L	D	L	D	L	D	L	D	L	D (E)
	V	V ₄	V ₃	V ₄	V ₃	V	V	V ₃	V	V ₄	V
	O	thou	foul	thief	where	hast	thou	stow'd	my	daughter	

Both versions of the line have an extrametrical final syllable, and the caesurae are not indicated. These are not suggested as a template for Old English, but merely to illustrate something of the distance there can be between word stress and metrical structure in languages with dynamic stress. This is also the case with Strang's (1970: §181) proposal concerning Old English.

The distribution of lifts is presumably determined by the sequence of pre-utterance accentuations available; but this allows various possibilities, provided by rhetorical and discourse requirements and the distribution of alliteration. In metres that have been widely used in Modern English a lift displays local prominence (Anderson 2014b, 2022b: 215-17): as illustrated above, it is more prominent than its immediate neighbours, by virtue of accent or syllable-weight or even non-reduction.

Metrical structure in Old English has usually been reconstructed (following Sievers [1893]) as imposing on pieces of text (extracted by scholars from continuous, though often 'punctuated', texts) four lifts (verse accent/prominence), with a caesura between the second and third of these; and there are also patterns of alliteration linking the halves of the line together, typically alliteration of the third lift in a line with one or both of the first two. I again use the 'lift' and 'dip' terminology to avoid equating, improperly, the metrical units with pre-utterance accent, or – even more improperly – with lexical accent. Much remains for examination here, too much to deal with in detail within the limits imposed by our wider focus here.

Moreover, more recently (for instance, by Klaeber [1950³: 274-5]) and Colman [1988], there has been much criticism of the metrical system of Sievers. But there has been much complacency. Even the former cited critic is placatory.

Especially, Sievers's formulations of 1885 and 1893 came to be regarded as practically authoritative. These views, involving a very definite linguistic interpretation and criticism of the text, can no longer be fully maintained, and accordingly, the value of the following remarks has become somewhat problematic. Still, in the present state of uncertainty in these matters, it seemed best to retain these observations with such changes, however, as are demanded by a more liberal attitude in the treatment of the text.

Klaeber (1950³: 274)

We return below to Colman's (1988) criticism of the traditional (Sieversian) view of metre below. She (1988: 150) comments on the treatment of claimed exceptions to the traditional views of the correlation between metre and lexical accent, and she goes on to discuss the proposed contractability of disyllabic sequences to diphthongal.

However, I shall for the present leave the reader to pursue Colman's deployment of 'templates' in accounting for contractability. And having indicated the indirectness of correlations between metrical requirements and lexical accentuation, I do not pursue controversies concerning the metre of Old English (see more recently e.g. Stanley [1984], Russom [1987, 1998]), except to highlight that there is no direct link between metrical prominence and word stress. The related area where modesty is most obviously desirable is with respect to 'explanations' of changes in the spelling as instances of 'sound-change' involving accentual conformity to (controversial) 'metrical requirements', sometimes involving invocation of an elusive, not to say admittedly fictional, 'tertiary stress' in instances of lexical items.

In Fulk's (1992:177-8) discussion of alleged 'tertiary stress' in Old English it is assumed that long monophthongs and (even 'short') diphthongs (on the assumption that certain digraphs express such diphthongs) are allowed only in stressed syllables in Old English. But, the story continues, 'reduced' second components of compounds retain (secondary) stress on the root of the second component only if they are inflected. However, according to the story, uninflected forms may have stress 'extended to' them, 'by analogy'. But stress on the root of the second component of compounds is 'tertiary stress' rather than anything more 'emphatic', however, because in the second syllable of an example 'like *laðlice*' 'hateful(ly)' (Fulk [1992: 177]), quoted there out of context, it must be 'not enough for the alliteration of its initial consonant to disrupt the metrical pattern of the verse'. Again there is a failure to recognize that metrical structure is not based on intra-lexical structure. Such failure is widespread: consider, for instance, the review of Fulk (1992) by Minkova & Stockwell (1995), as well as the original.

Further, though the spelling alternations in the vowels of the second components of etymological compounds, such as *-dom/-dam* and *-full/-foll*, that initiate this discussion of Fulk's invite a possible analysis involving morphological change, compound component to suffix, say – entailing loss of (secondary) accent, as is common in present-day English; but this is rejected a priori by Fulk. He declares that 'it is a fundamental principle of historical phonology that initially sound changes are governed solely by phonological conditioning, with morphological factors taking effect only subsequently'. Compare Giegerich (1999; sub-title: *Morphological causes, phonological effects*). I suggest in what follows that there are no such 'fundamental principles'. To claim that there are such is an abuse of proper modesty, given the poverty of our evidence for languages no longer spoken and the limitations of our knowledge concerning even the dwindling set of present-day languages. Moreover,

conditioned ‘sound-changes’ are implementational, phonetic, until the conditioning is lost, if it is – and unless they are interpreted as part of a prosodic feature: see further Chaps. 4 above & 7 below.

Moreover, it is a common observation that we need to recognize conversions (see e.g. Colman [2018]), changes in word class not necessarily expressed morphologically, except indirectly by inflectional behaviour. Recall the familiar role of word class in determining phonological regularities in Modern English, for instance. This is illustrated by such pervasive pairs as *permit*, verb/noun, and *pervert*, verb/noun, with placement of lexical accent determined by the word class. Historical conversion to noun of the Latin-based verb *permit* is associated with adoption of the initial accent that one would expect of such an English noun; and, similarly, on conversion from the obsolete resultative-state adjective based on the verb, the noun *pervert* adopted the same initial placement of accent.

Word class can determine accent placement. And, throughout the history of English, the role of (change in) part of speech in (changes in) lexical accent placement is salient, as far as we can tell, and increasingly so with the adoption of non-Germanic forms. But, already in Old English, word-initial de-adverbial forms attached to verbs do not bear the accent, and are apparently weak, or reduced in weight, unlike those attached to nouns and adjectives. The complex noun is a compound, the complex verb is prefixed. Compare, in Campbellian notation, compound noun *ándsaca* ‘denier/apostate’ with prefixed verb *onsácan* ‘deny’ (see e.g. Campbell [1959: §§72-3]), whatever the morphological status of the final syllables. A differentiation in accent-placement has already occurred in the precursor of the historical language; and this is a ‘sound-change’ that is based on both syntactic word class and morphology, and such changes litter the later history of English. Of course, sometimes in Old English the primary accent of the source of a conversion is retained, as in nouns *bebód* ‘command’ and *forgífnes*, derived prefixed nouns (cf. verbs *bebódan*, *bebéodan* and *forgífan*), or compound verb *ándswàrian* ‘answer’ (cf. noun *ándswàru*). Our modern examples above show that the results of accentual change and its word-class/morphological conditioning cast a long shadow.

Fulk, however, appeals to metrical factors as accentually crucial for ‘word forms’: i.e. phenomena that are not even part of the core of linguistic knowledge, unlike syntax, phonology, and lexicon, including morphology. Verse is abnormal language, and its requirements are superimposed on the normal: these (abnormal vs. normal) phenomena should not be confused with each other; but the differences may indeed be instructive. As I have indicated, verse is served by the accentual pattern of words in sequence not that of individual lexical items. On this distinction see e.g. Anderson (1986), though recognition elsewhere of the distinction is, of course, far from unfamiliar; but it is typically confused in the tradition of studies of Germanic metre, in support of historical fantasies.

Particularly questionable in proposed reconstructions of Old English is the morphological status of alternations in the phonological structure of forms. Thus, Campbell (1959: ch. 2, B) claims that the metrical evidence shows that, as well as the second element of compositionally preserved compounds (*gældrèor* ‘blood of slaughter’, again in Campbell’s stress-marking notation), post-accentual ‘long’ (i.e. with a complex coda) syllables, suffixal (or not), that follow another such may bear ‘half-stress’; if the former syllable is not itself ‘long’ but is followed by another syllable (typically an inflection (*hláfordes*, genitive) it may still bear ‘half’ stress (contrast *hláford*, nominative). On this account, the stem of the overtly inflected form behaves like the second component of a compound, but not the nominative and other members of the paradigm with no overt inflection. We thus have a paradigm whose stem form changes accentual status – and thus apparent morphological status: suffix vs. compound component!

Also, sometimes the second syllable of the stem of such a disyllabic nominative may show the same spelling as an inflected form, as in nominative *andweard* ‘present’, compared with overtly inflected *andweardan*. Does this suggest secondary accent on the second syllable of the nominative? In a tradition that includes Hogg (1992: §6.7, n.1), this variant is regarded as the result of ‘analogical transfer’ of whatever the ‘digraph’ of the inflected form represents to the form without overt inflection. Without significatory motivation, we have a transfer of part of the stem of one form to the stem of another form in the same otherwise non-suppletive paradigm! And is the transfer a matter of spelling only?

A more likely analysis of Old English morphology and accentuation would come with an acknowledgement that Old English lexical accentuation, despite the intervening influx of non-native, and particularly non-Germanic, forms and the depletion of inflections, is not unlike that in Modern English, particularly in its association with word class and morphological status. Not only are verbal prefixes accentually extrametrical in both languages, as with *permít* (vs. noun *pérmit*), and *begán* (vs. noun *bígenga*), but also some modern verbal compounds have main stress on the second component, as in *ùndertáke* (if this is such); and this is even more common in Old English, as in *òfergán*; but this can be overridden if the verb is based on a noun, just as noun compounds based on verbs can retain the second-component accent of the verbs (Campbell [1959: §77]).

It is also unnecessary to appeal, with Hogg, to ‘analogy’ to account for the nominative form *andweard* rather than *andwerd*: the final unstressed syllable, with a short vowel, may or may not be reduced, while the genitive inflection in *andweardes* is regularly reduced. In Modern English, *answer* is not morphologically complex but, quite normally, retains the initial stress and the second syllable is typically reduced, whereas in *towárd(s)/towéard(s)* the first syllable has lost its stress to the following syllable and is a prefix with a possible weak (reduced) form. But the most important

task in reconstructing here is the disentangling of placement of the lexical accent from ‘metrical evidence’, and the reconstruction of pre-utterance (sentential) accentuation.

In general, then, in Old English lexical items the ‘primary accent’, i.e. the ‘tonic’ of Anderson (1987: §2, 2011c: esp. 294-8, 2022b: ch. 27), fell on the root-initial syllable (Strang 1970: 411), no matter what its ‘weight’ in terms of sonority, its syllabic complexity, or vowel-length. Compounds have two roots, one bearing tonic accent followed by an ictus-bearer. But already in Old English, word class is apparently relevant to accent placement, in forms with two roots, compounds. We can illustrate this with Old English nouns *ándswàru* ‘answer’ and *wīpersàca* ‘apostate’ vs. the verbs *àndswárian* ‘answer’ and *wīpersácian* ‘blaspheme’. But, as already observed, derived compounds may retain the accent of the derivational source. Often these adverbial components weaken to (accent-less) affixes or to morphological opacity. ‘Reduced’ *ge-*, in origin thought to be a deadverbilised formative, however, is extrametrical in Old English; it precedes the root of the word-form.

In present-day English we have noun *overcoat*, with tonic on the first component, but verb *overtake*, with final tonic and final ictus (on such compounds see Anderson [2022b: ch. 30]); and this distinction thus seems already to be applicable in Old English, specifically to the ‘first-position’ adverbs mentioned above in this section, where the category of the compound determines the placement of the primary accent, or tonic, on initial or second component: with nouns and adjectives the first component of nouns and adjectives normally bears the tonic, but in verbal compounds the sequence is ictus + tonic, secondary before primary accent.

Colman (1988: 150) adduces some problematic lines from Klaeber’s (1950³) *Beowulf* on which she comments in the light of the traditional metrical framework of Sievers, repeated here, along with the illustrations listed here:

Such claims might say something about what ought to be if Wrenn’s interpretation of the Sievers’ types, as well as the Sievers’ types themselves, are right: they suggest what ought to be in order for a half-line to conform to a particular Sievers’ type. But what if these lines never did conform? – or if there are “gaps” in Sievers’ formulations? It seems back-to-front to use Sievers’ abstractions to test the evidence of Old English texts rather than the texts to test these abstractions. And what are we to make of the forms of the 3sg. Pres. Subjunctive ‘be’ in the following lines of *Beowulf* (Klaeber [1950³]):

ic þæt þonne forhieve swa me Hugelāc sīe (435)
 bēagas ond brād gold. Sīe sīo bāer gearo (3105)
 rand gehēawe, þeac ðe hē rōf sie (682)
 ides to afnanne, þēah ðe hīo ænlicu sȳ (1941)

Geata dryhten, þēah ðe hē geong sȳ (1831)
 Helpan hildfrumen, þenden hyt sȳ (2649)?

The editorial hatchmarking of supposed disyllabic forms claims that both *ie* and *y* can represent both monosyllabic and disyllabic forms. The “metre” here gives no unambiguous evidence as to the presence of hiatus in this word-form: especially none as to supposed contraction in West Saxon. If anything, it would support the late West Saxon graphic alternation of *ie*, *y* and *i* cited above.

Below I suggest metrical structures for Colman’s lines that are cited as a problem for the Sievers tradition. Here the vocalic segments are associated with a L (lift), marking prominence, or a D (dip), but some Ds are assigned to two successive syllabics. Lifts are adjacent to at least one non-prominent: dip, a caesura (separating the two half lines), or line beginning or end.

ic þæt þonne for**h**iege swa me **H**ugelāc sīe (435)

┌──┐ ┌──┐ ┌┐ ┌┐ ┌┐ ┌──┐ ┌┐ ┌┐ ┌┐
 D D L L D & D L DL D

bēagas ond brād gold. Sīe sīo bær gearo (3105)

┌┐ ┌┐ ┌┐ ┌┐ ┌┐ ┌──┐ ┌┐ ┌┐ ┌┐
 L D D L D & D L L D

rand gehēawe, þeac ðe hē rōf sie (682)

┌┐ ┌┐ ┌┐ ┌┐ ┌──┐ ┌┐ ┌┐ ┌┐
 L D L D & D D L L

ides to a**f**nanne, þēah ðe hīo ænlicu sȳ (1941)

┌┐ ┌──┐ ┌┐ ┌──┐ ┌──┐ ┌──┐ ┌┐ ┌┐ ┌┐
 L D L D & D D L L D

geata dryhten, þēah ðe hē geong sȳ (1831)

┌┐ ┌┐ ┌┐ ┌┐ ┌──┐ ┌┐ ┌┐ ┌┐
 L D L D & D D L D

helpan hildfrumen, þenden hȳt sȳ (2649)?

┌┐ ┌┐ ┌┐ ┌┐ ┌┐ ┌┐ ┌┐ ┌┐
 L D L D D & D D L L

Functional syntactic categories, in Anderson’s (2022a) terms (copula, auxiliary, pronoun, article, preposition, conjunction, and affix), are typically (part of) dips but functionals may be utterance-prominent if demanded by the structure of the utterance. Emboldening of a graph marks an alliterating syllable-initial, necessarily present in the first lift of the second half-line (if there are two) and two or one in the first. The

metrical structure does not affect lexical phonological structure but it may be reflected in implementation of the utterance. In the fourth line the syllable-initial vowels alliterate, just as do initial *Sk*, the neutralized sequence in early English and Greek.

More generally, there is no apparent role for the Sievers types themselves in the study of metre in Old English. Lines and half-lines of the same types may incidentally manifest very different utterance structures. There is no explanatory role for this arbitrary taxonomy, no non-arbitrary relation between sound and sense.

Chapter 6

Morphology and internal reconstruction

Let us return to a more modest role in reconstruction that we can even more plausibly attribute to morphophonological alternations, as anticipated in the Introduction. For languages no longer in normal use but with a later recorded or even spoken descendant or descendants, there are often available clues supporting what is usually called ‘internal reconstruction’, in terms of ‘anomalies’ or ‘discrepancies’ of related forms in the descendant(s). Their diachronic interpretation often invokes a relatively inoffensive ‘naturalness’ assumption.

Typical ‘anomalies’ are phonological alternations between the formative in common of morphologically related words or, indeed, between members of an inflectional paradigm. Both types are suggested by the well-known alternations in spelling in many Old English texts, as in respectively *dust* ‘dust’ vs. *dystig* ‘dusty’ and *mus* ‘mouse’, e.g. nominative singular, vs. *mys* ‘mouse’, e.g. nominative plural. The commonness of such alternations between apparently contrastive vowel-bearing graphs – or different ‘sounds’, in Campbellian terms – in the same place in related word-forms suggests that these forms may have shared a common ‘sound’ in this place at an earlier stage in the development of the language.

This assumption concerning apparent anomalies in linguistic structure, specifically morphological structure, and its non-anomalous source, can be regarded as ‘natural’, or likely to occur, on grounds of a preference for transparency in morphophonological relatedness, in the absence of other preferences such as ease of implementation: the assumption is that such derivationally related base/stem components had previously a common phonological shape. And this provokes, and indeed requires, search for a possible motivation for the discrepancy. Most obviously in the present case, on the basis of supporting evidence (comparative and etymological), for instance, this is specifically assumed to involve presence vs. absence of a following high front segment in the ‘*i*-umlaut’-based alternations cited above. The segment’s continued presence is apparently evidenced in *dystig*. At this stage, the {i} (secondary) feature of the suffix is phonologically prosodic as a consequence of previous phonetic assimilation of the stem vowel to that of the suffix; as a prosody, {i} is attached to the ictus node of the first syllable, and is implemented in the sequence forming the subordinates of this node (see particularly Strang [1970: 387-8]). And the system of vowel contrasts can be further changed by loss of the prosodic ‘{i}-trigger’, as in *bec*, various oblique and plural forms of *boc* ‘book’. This is a viewpoint which we shall return to even more explicitly later in this section, particularly as embodied there in diagram (6) from above.

However, we should also note there are instances in particular systems where the mutation itself is the major or only signal of a category, as with particularly long-

established mutations such as verbal ablaut – or gradation – classes, in Old English (whatever its status elsewhere). For a compact presentation of one traditional interpretation of the Indo-European picture and problems with the history of Old English strong verb classes VI and VII, see e.g. Lass (1994: ch. 5 & §7.2.2) respectively. In the conjugations of Old English the ‘strong’ verb classes differ in the sequence of stem changes that occur in different forms; and the accented vowel of ‘present’ forms of ‘strong’ verbs of each conjugational sub-class is distinguished by vowel quality and/or length from its two ‘pasts’ (differing in number and person) and its ‘past-passive’ participle; this is illustrated by relevant stems of the verb ‘bear’: in a ‘sound/graph’ or ‘litteral’ notation, there are thus stems distinguished as in *ber*+INFLECTION – *bær* – *bǣr*+INFLECTION – *bor*+INFLECTION, where also the set of inflections attached to each grade is different. There are also further variations in particular contexts, such as the present non-first-persons in *-i-* in *bir(e)st/birð*, and in those roots associated with the presence of a post-vocalic nasal: *helpan* and *bindan* both belonging to class III; and the ‘past participles’ also have a different history. But none of these gradation vowels has morphophonological priority in Pre-Old English and indeed its immediate ancestors, though the infinitival vowel is the conventional citation form; the phonological variation is apparently largely morphologically motivated in Old English. This is also the case with other Indo-European languages, wherein, however, ablaut alternations may be more or at least differently distributed.

The above might be seen as an unsatisfactory situation for believers in the priority of phonic segment factors in sound-change, unless we can motivate an earlier stage with non-morphologically contrastive vowels in the related forms. In the view of Lehmann (1952), however, in the earliest stage of Pre-Indo-European, vowel contrasts are absent, though there is what one might call a ‘neutral syllabic element’, a syllabic V-node lacking secondary features in terms of the suprasegmental phonological dependency structure. Vowels are then developed phonetically, on my interpretation of Lehmann’s view, under the influence of ‘distinctive’ stress and then pitch accents (§15.6) and the successive loss of the latter and morphologisation of vowel variation. The question of on what forms such ‘distinctive’, or contrastive, accents are distinguished arises: lexical items vs. syntactic (sub)categories (or both)? Given the subsequent developments (as displayed in Old English strong verbs at least), the latter at least seems to be a likely possibility, without considering manners of manifestation of ablaut elsewhere in the reconstructed family members.

Now, this part of Lehmann’s (1952) history is speculative and controversial: see e.g. Hoenigswald (1954), already, and the objections of many subsequent Indo-Europeanists, which typically are concerned with interpretations of Saussure’s (1879) hypothesis concerning sources, later designated ‘laryngeals’. Certainly, too, Lehmann works within a phonemic framework, as did other, now rather neglected, American contributions to historical linguistics in the mid-20th-century, such as Hoenigswald

(1946), Hall (1950), or Hockett (1959), as well as other inclusions in Joos's (1957) and Lass's (1969) collections. But at least these phonemicists attempted to invoke global systems of phonological contrast. More recently, Lass (1994: ch. 5) offers a more articulated but still overwhelmingly segment-based alternative view as well as references. Unfortunately, more recent accounts of Proto-Indo-European 'phonology' like Ringe's (2006: §2.2) revert to quasi-literal notations that obscure the contrastive status of what is being reconstructed. But there is no reason not to revert to the view that at an early stage the vowels of Indo-European gradation implement morphosyntactic distinctions, distinctions that survive various 'sound-changes', perhaps those of the diachronic sequence suggested by Lehmann.

The morphosyntactic basis of the ablaut variants continues to be manifested in their being expressed by the stem forms of the strong-verb classes of Old English and morphosyntactic distinctions in other Indo-European languages. In Ancient Greek, for instance, vowel gradation is manifested in both nominal and verbal inflexional morphology and in derivational morphology (see Sommerstein [1973: §2.2.2], ignoring the 'generativist' trappings). The Old English strong-verb classes are differentiated in terms of their syllabic structure, and the synchronic grades of the Old English strong verbs of classes I-V, at least, are all triggered directly by the morphosyntactic categories of different stems, so that the values of these vowels are redundant, morphologically determined; and they need not be specified in the lexicon (Anderson [1970, 1988], Lass & Anderson [1975: ch. I], though 'generativised' in the last). These gradation alternatives thus have equal phonological status synchronically, and evidence of conditioning factors is obscure, except for the obvious morphosyntactic categories the stem vowels expound and the syllable-structure motivations for the different classes. This is not to deny the phonological contrastiveness of these vowels, of course: they are lexically predictable but are part of a prosodic implementation in use.

The earliest stages in Lehmann's (1952) diachronic account involve attribution to them of no lexically distinctive vowels, simply an undifferentiated nucleus/syllabic node whose implementation is determined suprasegmentally or morphologically; i.e. they are parts of prosodies comprising the nucleus and part of its environment. Is such a situation manifested in any current language? And how is it to be reconciled with a commonly adopted 'uniformitarian' assumption to the effect that there are some aspects of language that do not permit innovation? In particular, in the present context, is there a minimal set of vowel contrasts that can be found in any language: such requirements are often taken to be constants in language – though there is some lack of agreement on how these constants are to be formulated and on their attribution to certain languages, as we shall see. In the present instance there is absence of inter-vowel contrast only if the accentual or morphophonological or prosodic determination is ignored.

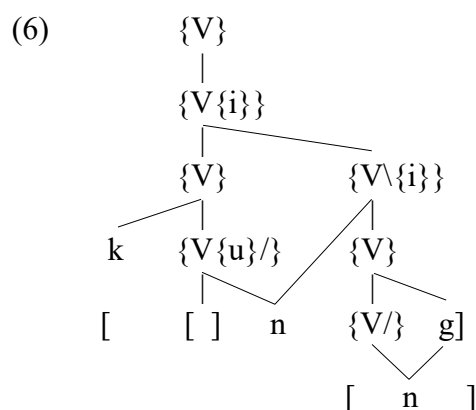
Lass (1994: 114, n. 11) objects that ‘it is part of the definition of “natural language” that such an object has at least three vowels in its system’ – if these vowels are in contrast, I presume? But this is another manifestation of lack of modesty: we do not know that what he asserts is the case, given the lack of competent and comparable analyses of many languages as well as the prevalence of doubts concerning the adequacy of many attempted analyses. And there is no principled account of why this alleged requirement would be necessary.

He declares later in the same note: ‘with either no exceptions or only some dubious ones, a language has to have at least one high vowel’. What then, precisely, is/are the principled constraint(s) that enable(s) us ‘to know when you’re talking rubbish’, as Lass (1994: 114, n. 11) again puts it? Such pronouncements are particularly strange in coming from the author of *Explaining Language Change* and *Historical Linguistics and Language Change*. Arbitrary restrictions inhibit research.

Lass (1994) provides no evidence for or against the requirement that a language can even manifest only one vowel. It would presumably be ‘high’, except that the presence of only one vowel has no substantive contrast, but is in contrast with its absence, and no particular contrastive vowel. What is the motivation for this empirically untested test?

A set of consonantal systems can be sufficient for an adequate range of contrasts for distinguishing signs. What is necessary phonologically (and phonetically) is the presence of a syllabic node, essential to the phonological structure that reflects part of the mechanism of implementation, the syllabics.

This is as illustrated again in (6) from Chap. 4, with tonic, ictus, syllabic, and rhyme {V}s, and where the square bracketing indicates morphological structure, with the affix within the word.



It is certainly not difficult to countenance in principle and perhaps even to envisage language systems unarguably lacking contrastive vowels, or, at least, fewer than three – just as a language may lack a categorial distinction between noun and verb

(and, of course, adjective), though their functions may be distinguished in other ways. Notably, their functional and substantive syntax is expressed by colligation with functional categories – see e.g. Bloomfield (1933: §6) or Swadesh (1936-8: 78). These are discussed in Anderson (1997: §2.1.4), who cites Swadesh on Nootka: ‘Normal words (vs. particles – *JMA*) do not fall into classes like noun, verb, adjective, preposition, but all sorts of ideas find their expression in the same general type of word, which is predicative or non-predicative according to its paradigmatic ending’, exemplified as his (2.19):

- ‘(2.19) a. Mamo’kma qo’?as?i
 He-is-working the man
 b. Qo’?asma mamok
 He-is-a-man the working’

The signalling of ‘nominal’ vs. ‘verbal’ here is conducted by subordination to different functional categories. Again, however, there is controversy concerning, in this case, the minimal complement of syntactic categories that can be attributed to a language, though there is a persistent tradition that accepts the minimal traditional 8-word-type found in traditional grammars, as well the minimalising position just discussed.

A digression here to consider the minimality of syntactic categorisation may throw some light on this area and how it compares with the phonological situation. Anderson (2022a) recognizes the simple syntactic categories of present-day English shown in Table 1 (from Anderson 2022a: 140, 205, 257, *passim*).

Functional		Contentive	
Operative	{P/}	Verb	{P;N}
Comparator	{P.N/}	Adjective	{P:N}
Determiner	{N/}	Noun	{N;P}
Functor	{ / }	Name { < _A > }	Pronoun { } Neither

Table 1: Syntactic Categories

The subscript <_A> stands for identification of individuals typically assigned by a naming ceremony, and adverbs are analysed as necessarily a combination of parts of speech (Anderson [2022a: ch. 7]). As well as adjectiveless languages, there are apparently languages that lack all the contentives (sometimes distinguished, rather misleadingly, as ‘lexical’) distinctions in this Table, as just illustrated: what is necessary in syntax, in Anderson’s terms, are the distinctions between {P/} (operative/finiteness) and {N/} (determination) and { / } (functor) and name { }, along with {P,N} for contentive. Such slanted lines aim to indicate transitivity, i.e. designate a category requiring a dependent, adjoined or subjoined (Anderson [2022a: ch. 5]).

Compare these syntactic types with the major phonological categories of Anderson's display in Table 2, quoted from p. 323 of the same volume, and extended in the light of what is looked at here, especially the 'semi' categories:

{V} vowels	{V;C} sonorants	{C;V} fricatives	{C} plosive s
{j} semi-vowels	{w}	{h} semi-consonants	{ʔ}

Table 2: Phonological Categories

Anything containing a **C** feature is a prime consonant, lacking in the representation of vowels. The lower line of categories have only secondary features, but occur in consonantal positions, and possibly in combination, as with the onsets in *huge* and *where* in English, but also with regular consonants, as in English initial *dwell*. There may be further semi-categories, such as the compound {ks}, as in *sex*, *exit*.

Some of these consonant types may be absent in particular languages. What can't be absent is the tonic {V}; its presence is necessary, structurally and phonetically (as will be again illustrated by my (6), repeated and explicated just below). Necessary too is either a coda or an onset or both, though this last is allowed only with intransitives ('long' vowels), of course. But particular sub-categories of **V** can, of course, be absent, as is {V{v;u}/} in varieties of English, the representation for a 'short low-mid-back' vowel ([ʊ]); unlike the syllabic node identifier {V}, (numbers of) particular vowels are not structurally necessary. Though the categories verb and noun may not be distinguished in syntax, in phonology it is only subcategories of **V** that can be universally absent in a language. But what emerges generally is the featural basis of both of the poles of the sign: both words and segments are feature-based (even if we ignore morphology, where features can include both types of feature).

Apparently supporting Lehmann's assumptions in the present context, on the other hand, are arguments like Kuipers' (1960) that currently Kabardian is (also) lexically a contrastively vowel-less language, where the implementation of particular syllabic features is determined by their environment, in this case phonological. Halle's (1970) tunnel-visioned 'generativist' negative response to this part of Kuipers' analysis is in turn rejected by Anderson (1991), though it is conceded there that parts of the evidence are in doubt, and complicated by notational divergences among descriptions of the language. But at most two contrastive vowel segments /a/ and /a:/ or /a/ and /ə/ might be motivated; and /a:/, the only form-initial vowel, has been analysed as /ha/, leaving /a/ as a default vowel quality, so non-contrastive. On either analysis, both of Lass's proposed restrictions are falsified. However, vowels and their 'conditioning' consonants together form a prosody 'triggered' by the consonant and headed by the

rhyme or syllabic node. We thus arrive at a possible ‘refinement’ of Kuipers’ argument.

If there are present-day languages that pervasively, or even just predominantly, behave as Kuipers suggests for Kabardian, then Lehmann’s reconstruction of pre-Indo-European ablaut conforms to some version of a uniformitarian requirement: among vowels only a syllabic node is necessary in language. Absence of independent lexical contrasts among vowels is then perhaps a persistent possibility. This is a more striking assumption than that of the universality of particular vowel contrasts, particularly in its distinguishing syllabicity from (contrastive) vowel feature. Compare Lehmann’s proposed historical narrative, such that merely phonological distribution determines vowel values only belatedly. An account of Proto-European vowel phonology involving prosodic developments based on adjacent segments and suprasegmental tones has some appeal. We take up the suggested general uniformitarian requirement in Ch. 8.

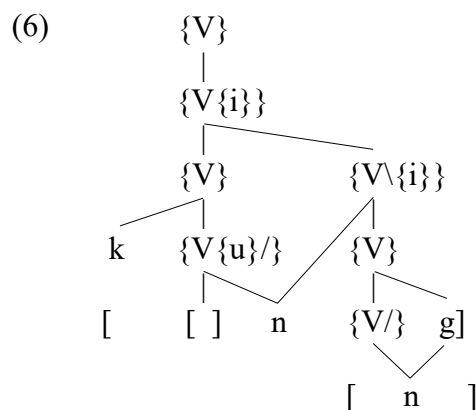
Tentativeness in reconstruction of pre-historic languages by comparison with hypothetically related languages increases with distance from historic descendants, unsurprisingly. Reconstruction of Proto-Indo-European and beyond has depended on the imaginative leaps that link aspects of a diverse range of descendants and the chance discoveries of further ‘relations’. A well-known instance of such a leap was, indeed, the positing in de Saussure’s (1879) *Mémoire* of hypothetical elements only indirectly evidenced in the Indo-European languages that were recognized at that time. Saussure himself showed due modesty in not immediately ‘spelling out’ a presumed character of the structurally necessary elements that he hypothesized, given the evidence available to him.

And the destination of the following out of his insight is still controversial, as illustrated by the continuation of contrastive attitudes such as are adopted in Lehmann (1952) vs. Hoenigswald (1954) and others, despite, or perhaps because of, the discovery of overt evidence of segments relating to Saussure’s speculations in subsequently discovered languages that have been constructed as related to (pre-) Indo-European. We return in Ch. 8 to the role in reconstruction of comparative evidence such as is appealed to here, as well as much more generally.

Given the limitations of internal reconstruction, particularly the frequent loss of the conditioning factors in such alternations as we have looked at, as in *mys*, plural, more can be reconstructed if the language is indeed historical, and particularly if in other cases the ‘conditioning’ segment is retained, as in *dystig*, or *cyning*. Also in such a situation appeal might plausibly be made to the naturalness of the conditioning. In the present case it is plausible to posit a ‘high front’ vocalic feature, as has been widely accepted. The transmission of this element is formalized as a morphologically-conditioned prosody projected by the exponent of the (reconstructed) suffix. The head

of the prosody is not independently contrastive unless the ‘trigger’ is lost or itself mutated.

The synchronic dependency-based representation repeated here as the structure of the form *cyning* in (6), showing its morphology and the prosody associated with the ictus of the first vowel, whose implementation stretches from this ictus to its subordinates to the right, including the second syllable – though intervening consonants may be variable in their capacity for accommodating the prosodic feature.



The implementation is manifested wherever a segment is compatible with palatalization. Via this lexical redundancy we have a wider implementation of the contrast previously associated with the implementation of the suffixal vowel {V{i}} alone.

The capitals in Table 2 are again primary features whose presence, combination, or lack of either distinguishes primary phonological categories that determine relative distribution within the pole in accordance with their relative sonority; recall {V} = vocalic, {V;C} = sonorants, {C;V} = fricative, {C} = plosive, where the semi-colon indicates the preponderance of the first feature. The sequence of subjoined {V}s are, from the top, tonic, ictus, syllabic, rhyme. But the structure of consonants is ignored in (6), the letters are convenient ‘placeholders’.

The lower-case secondary features make distinctions among the members of a particular category or categories, but some of them can occur independently, as indicated by the lower set in Table 2. And the basic and lowest rhyme vowel in both syllables is marked as transitive ('short'), indicated again by '/': a distinct dependent is necessary. The first 'n' is an ambisyllabic segment, shared by, dependent in, both medial coda and onset; the latter dependency follows from the preference for onset maximization. The vertical lines indicate dependency by subjunction (no sequencing), the slanting by adjunction (with sequenced dependent). The least sonorous node of the coda or onset is the head of these sequences; thus the velar plosive governs the nasal. The consonants here are indicated by familiar phonetic 'place-holders' for simplicity,

rather than a proper phonological representation. The square brackets enclose morphological suffixes: a base (with brackets include the form as a whole) and an enclosed suffix.

The ‘trees’ in (6) and below are not ‘proper’ in, for instance, Marcus’ [1967: 205]) terms; he offers ‘Theorem 1’:

Theorem 1. Let G be a finite graph of order greater than 1. G is a proper tree of center α if and only if the following three conditions are fulfilled:

- (1) Every vertex other than α is the terminal vertex of an arc, but of a single arc;
- (2) No arc has its terminal vertex in α ;
- (3) G has no circuit.

The segments represented by the two *ns* in example (6) above are ambisyllabic, thus violating (1) of the theorem, but necessarily: ambisyllabicity is a necessary property of phonological structure (Anderson [2022a: ch. 11]). And the vertical lines introduce several vertices to which a terminal vertex is subordinate (the transitive closure of dependency). The relationship between dependency and linearization is more independent than in a Marcusian ‘proper tree’, where all vertices are serialized: their exponents are not a ‘simple string’ (Marcus’ [1967: 206] ‘Theorem 2’). Marcus (1967:204) cites a number of earlier contributions, including Tesnière (1957), whom he introduces as the origin of condition (3); and he includes ‘Bibliographical Remarks’ (1967: 240-243). On Tesnière (1957) and more recent views of the ‘dependency’ relation see Böhm (2018). Phonological structure is nevertheless ‘projective’, unlike syntax, which allows defined violations of projectivity (Anderson [2022a: ch. 14] on ‘tangling’). On ‘projectivity’/‘non-tangling’ see Marcus (1967: ch. VI, particularly §10), with annotated bibliography in §11; and he offers an illustration (Figure 7, from French) of non-projectivity/‘tangling’ on his p. 247.

The prosody in my example (6) is required by the ‘\’{i}’ of the second, suffixal vowel to be attached to the ictus node of the first syllable that it modifies – actively, in that it requires the presence of the {i} on its governor; and this {i} is implemented in any compatible part(s) of the construction – here principally vowels – that is/are subordinate to that node. On such a view of ‘umlaut’ see too Colman (2005) and Anderson (2011c: §4.4.1) for more extensive discussion of these developments. I introduce, in Chap. 9, a further prosody, represented as in (8).

Already, we can observe that a prosody such as that in (6) is not mutative; it is the product of additions to the implementation of the forms concerned. Indeed, phonology in general is not mutative, especially synchronically. The Chomsky & Halle (1968) ‘vowel-shift’, for instance, is impossible as a synchronic phonological phenomenon. This statement may appear to be lacking in modesty, but it embodies a necessary part

of the distinction between phonology and its implementation. The only exception to the immutability of synchronic phonology involves the accommodation in the interface of a sequence of lexical phonologies to the pre-utterance phonology that they collectively belong to. We shall return to this area, but at this point our concern is with *i*-umlaut and the evidence for its implementation that leads to the addition of a prosody to the phonology.

If we also have sources of knowledge of the alphabet and the conventional auditory value in that alphabet of the graphs concerned, and particularly if the introduction of the graphic system based on that alphabet in representing the language concerned is relatively recent in relation to the surviving written records involved, and so not fossilized, we can hypothesize values for the graphs concerned in the '*i*-umlaut' alternations cited above. In the case of Old English, we have principally a Greek-influenced Latin-based alphabet already in use in other languages at the time, including mediaeval Latin. This has a role in understanding 'umlaut'.

The morphologically more marked formative of each pair of the above '*i*-umlaut' forms (derived or oblique or plural) has, consistent with the alphabetic tradition, a (first) vowel-graph that differs from the less marked in having as its value a front rounded rather than back rounded vowel, both hypothesized to be 'short' on other grounds. Similar in involving a back/front alternation are the apparent (long) low vowels in *hal* 'in good health' vs. *hælan* 'heal', though the value of both graphs would normally be interpreted as unrounded. The long vowel spelled *o*, interpretable as rounded and mid, alternates with *e*, as with *foṭ* 'foot' vs. *feṭ* 'feet'. But a few early spellings, retained in late Anglian (Northumbrian) Old English, suggest that the front vowel may have initially retained the roundedness of the back; such is *oeḏel* 'home'.

Another aspect of the interest of these last spellings is that the sequence of graphs can be interpreted as not necessarily a sequence of vowels but as the unsequenced features, of a single vowel, with *o* representing 'roundedness' and *e* 'frontness', and with the graphs sharing 'height' value. This is also exemplified by early spellings of words where we might expect *y* that instead involve *ui* – as in *suindriȝ* 'separate', with again separate expression of roundedness and frontness and sharing of highness. But we also find with the mid vowels, name elements such as *Oiþel*-, compared with *oeḏel*. The former in particular raises another possible, sequential, phonological interpretation, at some point, of the sequence of graphs. Again, any conclusion concerning the value of graphs must be tentative (and see further in the following chapter).

Let us look now specifically at some other common appeals to 'naturalness' posited in relation to 'sound-changes' in general.

Chapter 7

Implementational ‘Naturalness’

In all of the above ‘*i*-umlaut’ pairs the implemented value of one member, the morphologically more marked one, is closer to ‘high front’ than the other. And in some Old English forms there is a following graph interpreted as representing a ‘high front’ vowel or semi-vowel in the derived form, as further illustrated by *ænlic* ‘only’ vs. *an* ‘one’, or *ærlīc* ‘early’ or *ærist* ‘first’ vs. *ar* ‘before’. It is appropriate to invoke a rather modest naturalness assumption, compatible with what is taken to be the unmarked value of the Latin letters involved, to suggest that the alternation in these last derived forms is associated with the presence vs. the absence of the {i}-prosody demanded by the vowel in the following subordinate formative – hence the traditional labels, ‘*i*-umlaut’ or ‘*i*-mutation’.

This account invokes not structural, or contrastive, but implementational naturalness, specifically the diachronic naturalness of anticipatory assimilation in simplifying articulation and making the presence of a feature concerned more salient: we have naturalness in production. In such a case a feature of the value of a vowel is anticipated in articulation: the implementation of the feature spreads to preceding segments, presumably including, where possible, consonants intervening between the two vowels concerned, though this may very well not be indicated in the spelling. This diachronic formulation suggests a consequent **prosodic** synchronic phonological account, as suggested in (6) in Ch. 6 above, with the prosody, a high front unrounded feature attached, in the lexico-phonological interface, to the rhyme-head of the first syllable by the presence of particular suffixes. Compare Strang (1970: 387): ‘<p>erhaps the most plausible suggestion is that distinctions formally conveyed segmentally were transferred to a prosody of frontness, operating right across the stressed syllable.’

In a traditional non-prosodic, or rather pre-prosodic account, as long as the source of the assimilation is present, the segments showing assimilation are morphophonologically non-contrastive, though the orthography suggests that the phonological value of umlauted *æ* falls together with existing ‘long’ values of *æ*, as in *dæd* ‘deed’: the phonological distribution of the vowel is extended. This assimilatory account, however, also returns us to the question concerning the variety of spellings of the assimilation we have been looking at.

Do the spellings in *ui* and *oi* noted above represent the same results of assimilation as apparently does *oe*? Or is there at least an early stage at which diphthongs are a product of this assimilation, i.e. there is anticipation of a whole new segment or the end-point of a diphthong rather than a simple feature? Or do these spellings involve at some point diacritic marking of the following consonants as palatalized, participants in the implementation of the prosody or at least a non-contrastive indication of the

transition from the following palatalized consonants? Concerning the latter interpretations, Colman (2005: 36), citing Campbell (1959: §237), points out that there is no such consonant in forms such as *dry* ‘druid’ or inflected *gæþ* ‘goes’, however. What account or combination of accounts is more ‘natural’ here? And how decisive in development is relative ‘naturalness’? The results of assimilation are not determinable, in the absence of further evidence.

Moreover, events that are in some sense ‘unnatural’ seem to be attested. The reductive development of Scots phonology seems to have no consistent motivation. A discussion of this and the vocalism of Midi French is presented, with references (including Scobbie et al. [1999] and Aurnague et al. [2004]), in Anderson (2011c: §2.4). Or is loss of rhoticity in some other Englishes ‘natural’? [r] is a very vowel-ly consonant, but there is no obvious implementational advantage, for instance, in its (partial) loss (as described in Wells [1982: §§3.2, 4.1] – though that account is now dated as well as being expressed in transcriptions). And, in the case of English at least, ‘*r*-loss’ doesn’t seem to be communicatively desirable, given ultimate consequences in terms of loss of contrasts, with no obvious easing of articulation/auriculation. But, on the other hand, language users cope with homophony, though usually this again is lacking in obvious benefits (except for lovers of paronomasia).

As well as involving a complex and variable articulation, more difficult in a weak, post vocalic position, rhotics may constitute a ‘noisy’ background, as do post-vocalic nasals, promoting perception-enhancing reductions in vowel contrasts in such environments. But, as with post-vocalic nasals, a consequence can be not simply a distributional loss in the preceding vowel system: an [r] in typical RP English can be retained or even inserted intervocalically as an alternative to or in combination with a preceding reduced vocalic ([ə]) that is often interpreted as the termination of a new set of ‘diphthongs’ whose status is uncertain (see again Wells [1982: §§3.2.1, 4.1] for details). And we find a comparable development in some rhotic varieties of English, as perhaps in Scottish [biər], ‘beer’.

In many languages we have conflicts among a range of ‘naturalnesses’ – preservation of contrast vs. perceptual distinctiveness vs. articulatory ease vs. clarity of morphological relations etc. Some of these kinds of naturalness are primarily implementational: often ‘natural’ developments reflect implementation demands, not grammatical; and even implementational forces can appeal to opposing desirables, particularly identificational salience vs. articulationary facilitation in relation to phonology.

But ‘preservation of contrast’, for instance, is also often taken to be a structural desirability that is opposed not just to simplification in ‘noisy’ environments, but also to the alleged structural desirability of unified and balanced systems of contrast. Indeed, appeals to ‘naturalness’ invoke a greater range of different, often conflicting, factors, grammatical/structural as well as implementational. And I shall return in Ch.

8 to the invocation of structural ‘balance’ or ‘symmetry’ as allegedly preferred in phonological systems.

Notice finally concerning ‘i-mutation’, that alongside *ænlic* and *ærlic* in Old English we find *anlic* and *arlic*. The presence of the initial back vowel may reflect compound status for the latter forms, rather than weakening of the second component to suffix, in view of the failure elsewhere of the presence of this vowel to ‘trigger’ the change in the first component of a compound when it is in the second component. This at least would be structurally ‘natural’ in some sense, given the generally morphologically more ‘island’-like status of words, including compound components, compared with combinations with the morphological basic units of base and formative, where distinguished.

And it is a relatively modest assumption to suggest the presence of an earlier following high front vowel in the case of instances where there is no written evidence of this vowel but where the affected vowel is in an otherwise appropriate morphological environment – in the absence of further, contrary evidence. We are thereby positing a reconstruction of a pre-recorded stage where such a ‘triggering’ segment was present in *bec*, an inflectional variant of the ‘book’ word in Old English, as well as in all the other types of instance we’ve looked at. The change has at least two stages: firstly we have an assimilation, a phonetic change of the unlauted vowel; secondly, there may be a phonological change if the assimilatory environment is lost. We have either an increase in the distribution of the ‘triggering’ phonological vowel feature and its loss or a prosodic morphophonological status.

Thus, the relationship between the spellings and the mechanism of *i*-mutation is not completely transparent. Moreover, the interaction of the change with changes that created other alternations or other ‘anomalies’ involves the positing of other reconstructions that preceded or followed temporally that we have been concerned with. The changed vowel of *i*-mutation may be or become identical with an already existing contrastive vowel, as we’ve observed in the case of the first vowel in West Saxon *hælan*: *dæd* is also usually assumed to contain a long vowel, but it is also thought to be the product of another history and apparently doesn’t participate in the alternation we’ve been looking at here. This is an interacting part of the pre-history of Old English.

Such phenomena as just considered underlie the obsession of traditional accounts of Old English ‘phonology’, and so Old English ‘grammars’, with the relative chronology of ‘sound-changes’, at the expense of ignoring intermediate systems except, to some extent, when two pre-languages are argued to descend from a common ancestor. The prevalence of this kind of gap in concern for systemicity is exacerbated by the prevalence of littera-type notations in such works. The formulation of a synchronic phonological system involves the identification of contrasts and this requires featural representations that recognize both prosodies and positionally

distinctive subsystems and thus neutralizations in some of them. At any one time, also, there may exist partially similar systems associated with different groups of speakers, identifiable in some way, however vaguely, but particularly accessible in historical periods; but this too is a synchronic phenomenon, though of course mutable. The study of ‘sound-change’ should involve comparison of successive and coeval systems and their shared vocabulary, to the extent that this is possible. As a label for an extensive period, ‘Old English’ obviously doesn’t have a consistent or constant phonological system and similar traditional labels.

It is striking too in this context, that there is little acknowledgement in investigations of language histories of the often persistence in the phonological, morphological, and syntactic systems, as well as lexical content through the history of a ‘language’. Recall that Strang (1970: 419) reminds us of the importance of awareness of linguistic continuity: ‘<c>ontinuity can be looked at not only from the viewpoint of where our words come from, but also in reverse, from the viewpoint, of how much survives from the past; ...’. And she points to the persistence through the history of English of ‘stress patterns’ as well as lexical items and the core of syntax ‘except for object-placement’. And she concludes: ‘<i>n our end is our beginning; the nature and role of change, the interplay of change and stability, are exactly the same through the centuries as they are across the community at any one moment.’ Of course, the relative weights of change and stability vary in different communities and in particular times.

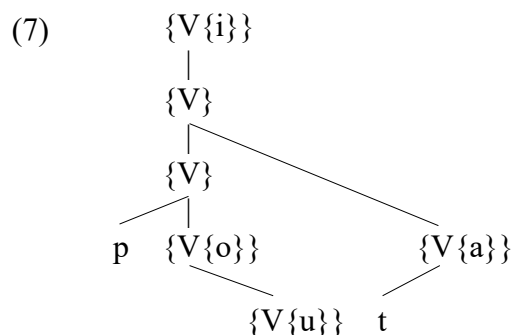
Even laying aside, for the moment, the non-systemicity and notational inadequacy of literal accounts of the implied phonic systems of the period, other evidence for the provenance of instances of, for example, the vowels concerned in the above discussion is welcome. And in general, since not all mutations result in alternations, reconstructions of unrecorded ‘languages’, pre-languages, need to be filled out, if possible, with reference to other sources of evidence; and we turned in Chap. 6 to one potential such traditional source anticipated above: this depends on shared dependence, preceding divergence. But let us acknowledge at this point systems that manifest a more extensive prosody than those associated with the umlaut type illustrated here.

I have in mind especially phenomena often labelled as ‘harmony, as noted above’; we come most commonly upon accounts of so-called ‘vowel harmony’, as introduced above. ‘Vowel harmony’, I shall argue, is an even more obvious manifestation of the non-segmental status of some occurrences of features. The ‘harmony’ terminology alludes to the agreement in presence of a feature value in the sequence of vowels, in particular in a word form. As remarked earlier, such phenomena are perhaps best known from the Uralic and Turkic languages. Here I shall briefly illustrate the phenomenon as manifested in Finnish rather than in a more complex manifestation such as is observable in, say, Turkish: the latter has two harmonies that are potentially found in stems at least, as well as harmonies ‘triggered’ by affixes. I am going to

suggest that the term ‘harmony’ is rather inappropriate as a description of the phonological status of what is involved; what I shall suggest is phonologically another manifestation of a prosody. The languages concerned have a vowel feature is an independent suprasegmental lexical property of word forms. Most striking are those prosodies where a vocalic feature is a property of the entire expression pole of a sign. I shall illustrate this phenomenon from Finnish, which, as acknowledged, is less complex than, say, Turkish: Even so, my look at Finnish is merely a sketch illustrating this type of prosodic behaviour. (For some references to work on Finnish see Anderson [2011c: §4.2.1], where the representations are, however, minimally specified and the intervening suprasegmental structure omitted, neither of which is our concern here. (On further complexities in systems of harmony see van der Hulst’s [2018]), possibly exhaustive, survey. Anderson [1986] illustrates another variety of prosody, illustrated from Modern English, which we shall return to.)

As in *i*-umlaut, the prosodic feature in Finnish ‘harmony’ is plausibly differentiated as the feature **i**. In traditional spelling we can differentiate the two signs *pöytä* ‘table’ and *pouta* ‘fine weather’ as respectively bearing the ‘harmony feature’ throughout the vowels and lacking it. But the ‘harmony’ feature is associated with the head of the expressive pole of basic signs, as in (7).

It is not a segmental feature or even ‘inserted’ by the presence of the feature in a dependent (morpho)phonological node, as in (6) above: its extended manifestation is as represented in (7), but this is an implementational rather than a phonological phenomenon.



(7) is much simplified; the most important point, as introduced above, is that the prosodic {i} attaches to the head of the expressive pole of the sign; it is both in a form's first syllable and bearer of primary accent. The 'harmony' is the crucial element in manifestation of the fusing, or co-implementation, of the segmental feature of vowel segments with the prosodic feature, giving, in the traditional spelling, from left to right in *pöytä*. And *pouta* 'fine weather', of course, simply lacks the prosody, and there is no fusion with {i} of {o}, {u}, {a}. Implementation of the prosody in Finnish is also not blocked by an affix boundary, as illustrated in *[[tyhmä]stä]* 'stupid + illative'; compare this with the prosody-free *[[tuhma]sta]* 'naughty' + illative. (On the representation of inflectional affixes in the present notation, see again Anderson [2022b: ch. 29], and compare the derivational affix of *[kyn[ing]]* in (6) above.)

However, in Finnish there are exceptions to the above regularities – especially, and unsurprisingly, with loan-words. And more complexity is associated, in native Finnish words, with segments that contain the *i* feature, however. There are two segment types that contain an inherent *i*: these are {i} and {i,a}. These can appear in forms that otherwise lack evidence of the prosodic *i*, as in {tuoli-lla} 'on the chair', where the morphology is given minimal recognition in the presence of the hyphen. Here we have a prosody-lacking form that contains a segmental {i}. In {lyö-dä-kse-ni-kö} 'for me to hit', we have both inherent vowels {i} and {i,a} and prosodic {y ö ä ö} – or {ü ö ä ö}, if you prefer, to trace the implementation of the prosody graphically. In prosody-free forms the {''} is lacking with vowel representations; and the presence of segmental {i} and {i,a} is compatible with either absence or presence of the prosody. But stems that contain *i* require a prosodic affix. I take segmental {ä} to be roughly [æ], while {i,a} is [e]. But the main interest here is the presence of a suprasegmental feature – a not uncommon phenomenon in language – and in this and other instances a minimal-sign (word-form) prosody, as shown in (7). Identifying segments that share a value is not enough to establish the place of contrastivity.

While 'harmony' illustrates the wide scope of some prosodies, there are other prosodies that are even more local than the umlaut-based type (see Firth [1948: 124], and contributions to Firth [ed.] [1957]). Consider, for instance, the behaviour of /h/ in

English. What indicates that the positioning of this segment type is a prosodic phenomenon is its determinate placement. In brief, /h/ appears only word- or syllable-initially. Thus, we have *hulk* and *hand* and *hideous* and *halibut*, where the only /h/ in both the monosyllabics and the polysyllabics is word-initial and accented; and the final occurrence the graph *bah* is graphic only in his respect, or a diacritic of intransitivity. In *behind* and many other word-forms the [h] is not word-initial but typically initiates an accented syllable; but a word-initial instance need not be accented. So we have *hilarious* and *hysterical*. And with two occurrences of [h] there is the notorious ‘borrowing’ *jojoba*, whose alien spelling is preserved.

This area is discussed more fully in Anderson (2001), and particularly (1986) and (2011c: §4.3.1) – though terminologically the term ‘prosody’ in the prologue to that volume denotes suprasegmental phonology, and the phonological and phonetic aspects of the present phenomenon are not adequately distinguished. Phonologically, the {h} segment is not linearized in accord with sonority or assigned to onset or coda, unlike the other segments in the word-form. In the interface to implementation, a {h} is inserted after linearization at an onset that is empty other than with a semi-vowel, as in *huge* or (for some speakers) *where*, or in two empty onsets if there are two. The completed linearization of the word-form is a preparation for phonetic implementation. Thus, schematically, the lexically stored {h}{a{n,d}}, where the coda is stored unlinearised, outside the rest of the word-form, is implemented as {{h}a{nd}}}. With *jojoba* there are stored two instances of {h}. {h} is a segmental prosody of a word form. The {j} and {w} semi-segments also invite a distinctive treatment, but not necessarily that offered in Anderson (2001.)

Chapter 8

Comparative reconstruction and pre-/pro-languages

If the language stage being reconstructed has recorded ‘sisters’, or ‘nieces’, or ‘aunts’, though such terms pre-judge such statuses (as well as being sexist – ‘yawn!’), evidence for any internally-based reconstruction may be extended by comparison among these; the evidence elicited can be said to be provided by comparative reconstruction. Thus, corresponding to Old English *cynn* ‘tribe, race, type’ there is also recorded Gothic cognate usually transcribed *kuni*, which apparently preserves an equivalent of the posited second, high front, vowel in the ancestor of Old English *cynn*. And other comparisons enable us or, at least, in combination with other kinds of evidence, help us to reconstruct other aspects of Pre-Old English.

Nevertheless, what can be reconstructed is far less than what can be achieved by synchronic reconstruction of an extant language, and even the latter is necessarily incomplete, as with any reconstruction of a variable, open-ended, indirectly accessible phenomenon. And reconstruction of unrecorded languages, such as Pre-Old English, is fragmentary, indeed typically constituting a set of individual or reconstructions of contrastive sets, as with *i*-umlaut and its consequences. Often, in the predominant tradition, these phonetic or phonological changes constitute a further temporal sequence rather than a general system. We need not go so far as, for instance, Pisani & Pokorny (1953: 79-80) or Pulgram (1959) in rejecting the reality of the reconstructions of language historians. But if they are actually devoid of some kind of potential reality, there is indeed no point in pursuing such reconstructions. Reluctant scepticism is preferable to both lack of modesty and complete dismissal.

Certainly, we cannot treat pre-historical reconstructions as we would those of present-day languages. The former reconstructions consist of a set of properties the acquisition of whose members may not even be coeval, and whose sequence may be difficult to establish. These fragments do not form a system. And they do not fully exhibit the extent of internal variation among users of the language that adheres to ‘real’ languages. This failure is particularly familiar in the case of pre-historical Indo-European ‘languages’, where the linguistic reconstructions are the main source of our knowledge of culture and localization(s) of the users of the ‘languages’. A **pre-language** is fundamentally different from a ‘real’ language. But the main point of the preceding is to emphasize the need for modesty, not despair.

‘Pre-Old English’ is not, then, a ‘real’ language, but, as typically presented, a set of reconstructed changes partially sequenced in relative time. We cannot reliably date the pre-historic differences whose consequences are manifested among historic varieties of Old English. In general, the further back in time we go, the less ‘real’ the reconstruction, though there may be fluctuations in the quality of the evidence in attested descendants or clarifications from newly discovered ‘cousins’. This declining

‘reality’ is particularly obvious in the case of pre-pre-historic reconstructions such as the reconstructed ancestors of Pre-Old English. Sociolinguistic distinctions cannot be extracted and even dialectal boundaries cannot be drawn with any confidence, given particularly a migratory status of speakers.

‘External evidence’ is also difficult to evaluate, particularly in differentiating between the likelihood of relationship vs. borrowing vs. typological commonness. We lack external evidence concerning location and contact. For instance, the distribution of pre-historic languages does not necessarily coincide with that of recovered artefacts from the past. Most obviously, the location of artefacts may reflect trading or imitation rather than manufacture of a distinctive artefact on that particular spot. The presence of forms with shared geographical, especially locatable ‘meanings’ may support the positing of potentially related languages. But Anthony & Ringe’s (2015) title *The Indo-European Homeland from Linguistic and Archaeological Perspectives* begs the question ‘did they have a homeland?’, whatever a ‘homeland’ might be, particularly for nomads.

Similarly, reliance on markedly retrospective chronicles can take us into immodesty in our reconstructions. On the content of the Anglo-Saxon Chronicle, Yorke (1990: 4) comments: ‘<t>he accounts as they survive show how later Anglo-Saxons wanted to see the foundation of their kingdoms, rather than what actually occurred’. Specifically, in the words of Sims-Williams (1983: 27), ‘<t>he *Chronicle* gives us the origin legends of Kent, Sussex and Wessex, cast in an annalistic framework’, and it is ‘a southern English view of the English conquest of southern Britain’, and (1983b: 29): ‘any attempt to reconcile the *Chronicle* with archaeological evidence is premature until it has been shown that the quality of the traditions the *Chronicle* preserves justifies the exercise’. Benskin (2002): dispels a striking belief, concerning which he concludes (p.95): ‘<w>hatever archaeology may show or come to show, and whatever credence it may be decided can be given to Procopius’s account, the *Historia ecclesiastica* affords no evidence whatsoever for Frisian participation in the fifth-century settlements’. This is just one aspect of the difficulties in establishing sociolinguistic phenomena in Old English.

Also misleading, of course, in historical times, is the assumption that scribes (or even coin die-cutters – e.g. Naismith [2012: 77-9]) are necessarily intending to represent their own pronunciation. If evidence for the language being reconstructed is preserved in written texts, the internal/comparative-based reconstructions of the language’s phonology can be complemented by literal reconstruction, especially if the writing system is a syllabary or, even better, alphabetic, as described above. This involves reconstructing both the writing system, including what differences in its apparent members are phonologically significant, and the phonological system that such writing seeks, in part, to record. However, as I have already discussed, the goal of an ideal writing system is not to represent the phonological system but only to

indicate distinctions in sound that can discriminate among distinct lexical items: a letter, like a structuralist phoneme, is ideally associated with each sound type that contrasts in some environment and only with these. Let us consider further the status of the phoneme: is it, after all, relevant to phonology, particularly in the reconstruction of recorded historical and pre-historical phonology, despite the fact that it is not a phonologically-based unit?

Relevant here is Lass' citing as a requirement of phonemic theory the following (1984: 24):

Primacy of Phonemes. The “atomic” units of phonology are segmentsized contrastive elements, i.e., phonemes

However, it has long been recognized that segments have a subsegmental structure of contrastive elements, or features of some sort. The classes of segments cited by the nineteenth century historians of languages, for instance, are differentiated by their internal structure: plosives are perceived as sharing a property that results from their shared ‘manner of articulation’ and correlates with a particular acoustic manifestation. And recognition of sub-segmental elements – ‘features’ of some sort – enables us to represent neutralizations more economically than even in the absence of recognition of neutralization, rather than in need of extra segment symbols. The plosives following the onset initial voiceless sibilant in both Old and later Englishes simply lack the properties that distinguish syllable-initial plosives spelled *p* and *b* etc.: the neutralization is neither voiced nor aspirated, expressed by the absence of these as unary features (or of feature values, if you prefer). And the pre-plosive segment spelled *s* in onsets lacks all features but consonantality; it contrasts with the absence of a pre-plosive onset consonant. In preparation for the interface with articulatory implementation of the phonological representation, the empty consonant can be filled in by redundancy rule with the features of the voiceless sibilant and contrast elsewhere, complemented by one concerning the absence of contrast of voiceless/voiced plosives following the sibilant.

Colman (2004) provides a detailed examination of Kentish medial orthographic variation between *b* and *f* and problematic attempts to motivate this. Concerning the variation she concludes that:

in ninth-century Kentish, replacement of / by <f>/<F> is evidence of neutralization of the /f/-/b/ contrast. Where the original graph remains, neutralization has not taken place. The neutralization occurs with the voicing of medial /f/, and the devoicing of final [β]. This voicing and devoicing occurs as a variation which is “free”, other than in its conditioning by regional, social, and political features.

For another discussion of fricative neutralization, see Anderson (1985). And a mysterious massive ‘short’ vowel neutralization in Kentish is observed in Anderson (1988), a development which contravenes any suggested principle of contrast preservation within a system and seems to resist any interpretation based on implementation.

Features can also help us express modesty as historians in the face of lack of evidence for being precise about a segment type. Thus, if it is just of Anderson (1987) to conclude that there are graphs in Wulfila’s Gothic – β , δ , and γ – whose status as voiced plosive or voiced fricative is difficult to decide on the basis of the internal or orthographic evidence, then we can represent the segments, in a suitable notation, as merely voiced obstruents, and lacking contrastive plosiveness vs. fricativity (see below). For illustration of this orthography see e.g. Streitberg (1906: §§19-36) or Kluge (1911: chs. I-V).

We can speculate further in this instance on the basis of comparative evidence from other Germanic languages and by comparing the reconstructed use of these graphs in the source alphabets: Wulfila’s alphabet draws on the sets of Greek, Latin, and Runic letters, though knowledge of this does not help resolve the plosive vs. fricative question, as I conclude below. But at this point I should indicate that we can also appeal to well-supported ‘naturalness’ considerations, again involving assimilation.

Voiced obstruents in present-day languages are often fricative intervocalically (a ‘voice-sharing’ context), even if plosive elsewhere. But this is not necessarily the case; moreover, this does not involve contrast. In Gothic there is, indeed, some suggestion, involving alternation with voicelessness, that voiced fricatives occurred stem-finally (Anderson 1987: 8). But this does not involve contrast; and other speculations involve reconstructing on the basis of reconstructions. Contrast between voiced fricatives and plosives cannot be supported ‘internally’ in view of the paucity of relevant evidence. Despite Busbecq’s (problematic) report of Crimean Gothic, we lack evidence for what Gothic might have become. More specific claims concerning the interpretation of these graphs made on such bases as we have are lacking in humility.

Curiously, Allen (1987: 32) argues that the Gothic orthography of ‘Wulfila... adopted the Greek β , δ , γ to represent Gothic phonemes which in some cases were pronounced as voiced fricatives’ and that ‘in the absence of a phonemic contrast between voiced fricatives and plosives in the Gothic, this need not indicate a fricative pronunciation for the Greek’. But we can at least admit the value of these graphs in Greek to perhaps involve neutralization (or, in phonemic terms, to involve fricative as well as plosive allophones, as Allen implies, without providing evidence); but, though such a distribution is not uncommon, humility suggests that we concede that the evidence is inconclusive, as Anderson (1987) suggests for the Gothic. We can also establish merely a voiced obstruent value for these graphs in classical Greek: perhaps,

in the notation of Anderson (2011c), (2022), we can represent their value as $\{C; \{v\}\}$, where the ‘C; ’, with predominant primary feature simple C, is neutral between plosive and fricative, $\{C\}$ (where $\{C; \}$ has a null filler) vs. $\{C; V\}$, respectively, and secondary feature $\{v\}$ is voicing.

Allen (1987) concludes that a range of phenomena that might be taken to suggest that these Greek graphs were fricative in the classical period of the language (as they are in the present-day) are similarly inconclusive. But he provides no evidence whatever for his (even partially underlined) assertion (p.30):

There seems no reason to doubt that in classical times the value of β, δ, γ was that of voiced plosives, much as the English *b, d* and ‘hard’ *g*, with places of articulation as for the corresponding voiceless sounds ...

The introduction of (Modern) English here is quite gratuitous, given that in Modern Greek the voiced fricatives β, δ, γ are paralleled by the voiceless ϕ, θ, χ , whereas voiceless plosives π, τ, κ have only the marginal voiced congeners with the complex representations $\mu\pi, \nu\tau, \gamma\gamma$. Similarly, but lacking even an attempt at argument, Sommerstein (1973: 2-3) simply assumes the plosive value for β, δ, γ : ‘(b, d, g)’, or ‘–continuant, +voice, +consonantal’).

Another well-known development whose implementation is contentious involves the Old English ‘sound-change’ that is usually rather prematurely labelled as ‘breaking’, introduced in chapter 4. We are concerned, as well as with *eald* from Ch. 2, with forms such as *bearn* ‘child’, *healf* ‘half’, *feoh* ‘cattle’, *miox* ‘dung’, with coda sequences that are spelled *-rC*, *-lC*, *-h(C)*, where *h* is usually reconstructed as a voiceless velar fricative. As well as the corresponding historical long vowels, some other sequences, such as ‘back umlaut’, are also associated with similar ‘vowel changes’; and there is some ‘dialectal’ variation in the relative regularity of occurrence of ‘breaking’, and instances where there is ‘retraction’ to a following syllabic of the same height rather than ‘breaking’ – with this problem.

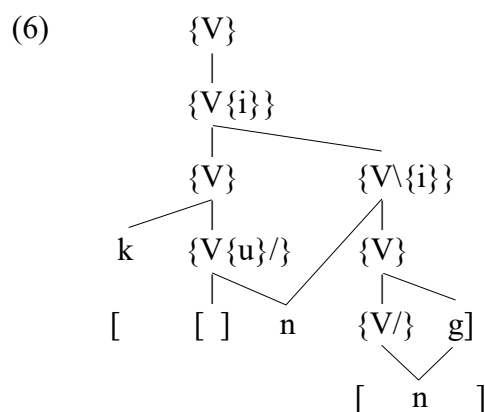
The above codas follow digraphs, two graphs each of which elsewhere can spell a vowel, where the diachronic sources of the vowels concerned appear to be simple transitive/short vowels. Given that the second vowel letter in the ‘broken’ forms otherwise seem to represent back vowels, ‘backness’ is often said to have something to do with what is added by the presence of the following consonants. Say we associate the ‘broken’ vowels with the acquired presence of the feature $\{u\}$, which is also rounded. What then characterizes the ‘triggering’ consonants as a group, given that they share the ‘triggering’ of whatever is the value of such vowel digraphs. Say we characterize the consonants as sharing backness, presence of the feature $\{u\}$, with its precise ‘height’ and thus roundness being determined by the preceding vowel, before the post-vocalic rhotic liquid, spelled *r*, with the non-rhotic liquid, spelled *l*, and with

the voiceless backfricative, usually spelled *h*. The precise implementation of this latter feature is contentious, with competing proposals concerning phonological values appealing to dubious ‘naturalnesses’. But even more contentious is the character of what is the phonological value of the vowel digraphs associated with ‘breaking’, as illustrated by the familiar ‘digraph’ controversy, and perpetuated by appeals to various ‘phonetic phantasies’. Most strikingly, are the sequences of vowel graphs in the examples offered above simple or diphthongs? Modesty based on lack of decisive evidence would again prevent adopting either view – or others, such that the second graph represents a ‘glide’ between articulations of the vowel and the consonant, or a signal of members of what distinguishes this set of ‘triggers’, perhaps to distinguish the ‘triggers’ from what values the graphs involved have elsewhere. Reconstructions of the previous or succeeding history of the digraphs are not decisive here.

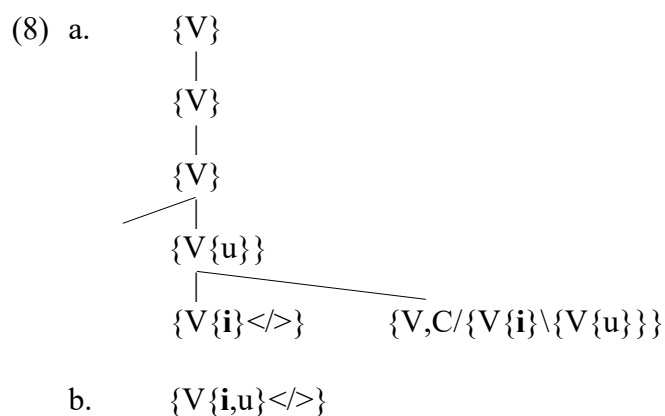
Hogg (1992: §2.24) declares that ‘Opponents of the traditional [diphthongal – JMA] view all begin from the assumption that it is unlikely that OE had a four-way contrast between short and long vowels and short and long diphthongs’. This is certainly not true of Hockett (1959) and is not necessarily important to other non-diphthongal proposals, such as others of those dismissed by Hogg. And the presence of this variety of views and the absence of a phonological characterization of syllable structure in the traditional view might suggest that at the very most the status of ‘broken’ forms is undecidable. This acknowledgement can be expressed by the notation advocated by Anderson (2022a, b) and predecessors, as follows.

Say we interpret ‘breaking’ as the development of a prosody which adds backness, {u}, to the vowels affected, so that {u} is part of the lexical representation of the breaking ‘triggers’, a set otherwise distinguished as {C,V} or (equivalently) {V,C}, i.e. a combination of **C** and **V**, or ‘continuant’. ‘Breaking’ would then be a prosody of the rhyme inserted by the coda as shown in this case of *bearn*: {C{\{V{u}\}}\}} in (8) below.

Recall again, firstly, however, as a model, the *i*-umlaut prosody of the upper ictus of (6) in Chap. 7:



In the lexical redundancy (8a) the rhyme vowel combines a prosodic {u} introduced by the coda and an inherent *i* (indicating ‘front’), i.e. one of {i}, {i;a}, {a;i}.



The specification of the ‘triggers’ lacks contrastive voicing, and {V,C}, a combination of {V;C} and {C;V} allows for both voiceless fricatives and liquids (including semivowels); and the {i} specifies that the rhyme vowel is front, so lacks {u}; the ‘target’ is either transitive or intransitive. The inherent vowel determines vowel height.

But this would not resolve the issues of implementation encountered above: how does the prosodic {u} feature in (8b) share the vowel space with the non-prosodic vowel feature(s), in preparation for implementation? However, the prosody at least establishes a link between ‘trigger’ and ‘target’ and the structural location of the contrast, such that there is no phonological change apart from the introduction of the prosody, which is manifested by both of them. Can we justify a particular implementation internal to the rhyme vowel? That’s what the notorious ‘digraph controversy’ is about. Is the {u} implemented as the second element of a diphthong, as traditionally assumed, or is it combined non-linearly with the ‘target’ vowel, as the simplest implementation of (8b), as argued by Hockett (1959), or is it the second graph in the digraph a diacritic associated with the adjacent consonant, as proposed by Daunt

(1939), or do different digraphs have different powers? (8b) is indecisive concerning the rhyme configuration that is the result of ‘breaking’.

There are also diphthongs apparently derived from (rather different) Germanic diphthongs, such as *seam* and *feond*, spelled as are the alleged diphthongisation of long vowels by ‘breaking’ etc. They raise the further complication that there are also such vowel-digraph spellings, apparently of ‘long’ vowels, that do not occur in the ‘breaking’ context. ‘Broken’ long vowels would then merely add to the set of such vowels, and do not introduce a new contrast. As noted, other contexts also can follow ‘breaking’ spellings; and other spellings of the vowels in such contexts, are such as would suggest ‘retraction’ of the preceding vowel, depending on the ‘dialect’, or provenance, among which the appropriately traditionally labelled ‘back-umlaut’ is prominent.

There is yet another alleged diphthong-creating change invoked in traditional accounts of Old English, illustrated by *scieran* ‘shear’ and *geat* ‘gate’ (*sc* is taken to spell [ʃ]) and *g* [j]). These sequences of ‘vocalic’ graphs are again attributed to a diphthongisation. This involves the even more implausible sound-change ‘diphthongization by initial palatals’ (Lass 1994: §3.94), abbreviated there, and elsewhere, as DIP: on its implausibility see e.g. Daunt (1939), Stockwell and Barritt 1951, 1955), Lass and Anderson (1975: app. III), Colman (1985), and, indeed, Lass (1994:§3.9.4), where other diphthongal interpretation of digraphs are also questioned. Kuhn & Quirk (1953) defend the traditional diphthongal interpretation of these and other digraphs: these sequences of ‘vocalic’ graphs are again attributed to a diphthongisation. I return to this in the Conclusion.

Perhaps more positively, Colman (1988: especially §III) highlights the consequence of a template assumption as a formulation of the constraints on the formation of diphthongs by ‘contraction’ suggested by Luick (1921: §§242, 245.1). In particular, while other digraphs, such as *oa* (as in *onfo(a)n* ‘seize’) in Old English represent only disyllabic sequences, digraphs *ea* and *ie*, by virtue of an existing template (or at least the absence of a ‘counter template’) have three (short or long) phonological significations: diphthongal vs. monophthongal after palatal consonants vs. disyllabic with an initial long vowel (Colman [1988: §V]).

However, as illustrated by the preceding situations, interpretation of the use in the past of even widely used orthographies has its difficulties, given the variety and individual fluctuations, diachronic and synchronic, of phonological systems and spelling conventions. And trying to achieve even the limited goal of identifying the sound-type, if any, to be associated with particular (di)graphs can present many problems – perhaps irresolvable in the present case, given the absence of evidence for internal reconstruction. In some instances the other sources of evidence for reconstruction can be argued to play a part, and comparative evidence for the use of the script, including contemporary commentaries on the language concerned – but not

necessarily uncontentiously. We may have to rest on our modesty in many such situations.

And what if the orthography of a language of the past is unique or similar only to orthographies that apparently represent defunct languages? As we have seen, such was the task facing the would-be decipherers of the Linear B script preserved on tablets in Crete and southern mainland Greece. Chadwick (1987) documents the struggle to finally identify this script as a syllabary representing a form of Greek, and thus to identify many of the forms on the tablets. But what if there is no obvious, or even tenuous independently attested language for such a script? What can be done with scripts apparently associated with otherwise non-attested languages?

Consider the problem of reconstructing the system of the language(s) written in the Linear A script of Minoan Crete, whose descendants (if any) are unknown. Despite some similarities with other scripts – notably Linear B, generally thought, as indicated above, to be a syllabary representing a kind of Greek, and an early Cypriot script, itself problematical – detailed reconstruction is uncertain, perhaps impossible, in the absence of the discovery of a(nother) Rosetta Stone.

Other scripts are less impenetrable, but there is never a simple match between phonology and writing, given the goals and limitations of writing, not to mention the failures of phonological theories and ideas of reconstruction. Many ‘sound-changes’ involve innovation of implementation, not a phonological change: this may be true of the interpretation of the ‘breaking’ digraphs discussed above as representing diphthongs. More modestly, we can restrain ourselves to some sort of a combination of a prosodic {u} with the non-prosodic vowel; the innovation of {u} prosody is the phonological change. The future of its implementation, however, is uncertain.

Chapter 9

‘Naturalness’ and ‘Uniformitarianism’

I have already alluded to the problem of deploying ‘naturalness’ criteria, what I call ‘wishful-thinking criteria’. I noted particularly the conflicts among ‘natural’, implemental tendencies. Moreover, not all linguistic phenomena are ‘natural’, if only because the wear and tear of usage, including the consequences of economy of effort or innovation (wittingly or not), can result in routinizations or contradictions: discrepancies between behaviour and content of segments and/or their colligational representations. Thus not all differences between subsystems are obviously ‘naturally’ based, as Fudge (1969) demonstrates for English and a range of other languages. And there are also, of course, controversies as to what counts as ‘natural’. Unfortunately, as I have noted, some of those cited, for obvious reasons, as ‘natural’ are nevertheless mutually contradictory.

I have indicated that this is particularly problematic with ‘natural’ changes associated with the means of transmission and perception, implementation. Notable here are assimilation vs. dissimilation. The first eases production, the latter differentiation, but it can happen that neither of them is implemented in the appropriate conditions, which of course are not necessarily opposed. In Ancient Greek the equivalent of ‘Yesterday’ was spelled with two medial letters that individually still transcribe two voiceless fricatives, tongue-body and dental: (ε)χθές; around me in southwestern Messinia/Messenia I’ve heard a pronunciation that would be spelled in this way, but also (ε)χτές and (ε)κτές and (ε)κθές. What sequence of changes underlies these variants is undecidable without further evidence or perhaps not at all – but I am not qualified to say.

A further area where ‘naturalness’ has been appealed to concerns linguistic structures and their alleged preference for symmetries, and maximizing the space available to particular subsystems. But these too are implementational, as they are concerned with maximization of phonetic differentiation, providing auricular distinctiveness. Aitchison (1981), following Martinet (1955), Campbell (1959: §30), and others, discusses ‘therapeutic’ changes that remedy asymmetries. Despite the asymmetry of the ‘vowel space’, the long (intransitive) and ‘short’ (transitive) contrastive monophthongs are often reconstructed for particular languages as symmetrical (despite some ‘fuzziness’ in the low vowel area). However, (speakers of) the most familiar varieties of English, for instance, have for centuries resisted (re-)creating symmetry between ‘front (unrounded)’ and ‘back’, and contrastive ‘long’ and ‘short’ vowel systems. Such symmetries are often associated, however, with reconstructions of Old English (see e.g. Campbell [1959: §30]). See too Colman’s comments (2005: 46-7) on other vowel-symmetrists.

But in present-day English the contrastive ‘long’ segment type (where ‘long/short’ pairs also have quality distinctions) we might transcribe as [ɔ:] (as in *law*) has no front equivalent, such as [ɛ:], though there are the pairs [i:/u:] (*tea/two*) and [e:/o:] (*lay/low* – though this latter pair, at least is implementationally diphthongized in many varieties). Moreover, in North American varieties, for instance, the same segregated vowel lacks a ‘short’ congener, despite [i:/ɪ] (*tea/tit*), [u:/ʊ] (*two/toot*), [e:/ɛ] (*lay/let*), [o:/ʌ] (*low/lug*), [ɔ:/ɒ] (*law/lot*), and [ɑ:/æ] in some other varieties, though [ʌ] stands out as back unrounded, and is lacking in some of these varieties. In Scottish English the ‘short’ system is even more depleted, in lacking both [ʊ] and [ɒ]. It also lacks the [ɑ:/æ] contrast in favour of a single ‘low’ vowel that we might transcribe as [a], which typically distributes as a ‘short’ vowel, so that *aunt* and *ant*, with complex coda, are homophonous (but there can be some attempt at ‘length’ in *bra* and *Shah*, given the lack of a coda); and typically in Scottish varieties the partner of (the missing) [ʊ] is fronted. And consider the chaos introduced by the loss of post-vocalic [r] (alluded to above) in common phonological circumstances in some varieties of English. It is difficult to discern any attempt at symmetry in all this, or persistence of an equivalent of ‘social distancing’ among contrasting segments sharing transitivity. Here also belongs what Anderson (1988) terms ‘the Great Kentish Collapse’, whereby short vowels in that dialect of Old English containing the feature *i* (so ‘front’ traditionally) are all spelled as *e*.

Assumptions of symmetry are major participants in discussions of language change; but they constitute what Anderson (2005a: esp. 217-222 & app.) labels as unsupported ‘phonetic phantasies’, principally in reference to Hogg (1992), but also manifested in Welna (1978: §2.4) and many other students of comparative historical linguistics. There is a tradition of phantasies appealed to in discussions of the relationships among Old English, Old Frisian, and Old Saxon (and Old Dutch and Frisian). Indeed, perceived linguistic similarities between English and Frisian, have been commented on since the late thirteenth and fourteenth centuries (Bremmer 1990: 353 n.1, 2009: §220). Thus Schmidt-Petersen and Craigie (1928) declare that: ‘the close relationship between English and Frisian has long been a commonplace of the text-books’. Stenton (1971: 6) bases the following claim on H.M. Chadwick’s *The Origins of the English Nation* (1907: 61-4): ‘linguistic analysis has established a fundamental connection between English and Frisian which is the first certain fact in English history. The Frisian language shared in all the more important sound-changes which distinguish English from German on the one hand and from the Scandinavian languages on the other’. But the relationships involved are not supported by coeval written evidence, and could thus be the consequence of ‘borrowing’.

As an alternative reconstruction, recourse is had elsewhere to the idea of the three (or more) putative languages concerned participating in a ‘dialect continuum’ through which properties can be diffused. But this again cannot be shown not to be a phantasy.

We cannot delimit the continuum or establish persistent places in the continuum for the putative pre-languages, despite Bremmer (2009: 22, n.2): ‘Ingvaemonic, also known as North Sea Germanic, is a grouping within West Germanic from which Old English, Old Frisian, coastal Old Saxon and coastal Dutch is assumed to have developed. Ingvaemonic is generally seen as a dialect from Flanders along the coast to the German Bight, sharing a number of conspicuous linguistic features.’ Rather than tree structures, however, there is adopted for this area the status of a ‘language continuum’. But the ‘assumption’ encapsulated here is another phantasy whose character is not well characterized, not even in assumed present-day instances with a stable location. The same can be said of the similar proposals made in Bloomfield (1933: 58) and Campbell (1939: 78). Some such phantasies adopt the *Wellentheorie* of the structure of historical development (discussed here below), avoiding the lack of evidence for a conventional tree structure. And Stiles (1995: 207) declares, more radically, that ‘as one pursues the entities backwards in time, they dissolve into a haze of uncertainty. The later West-Germanic daughter languages have emerged from this steaming broth.’ Quite so.

On the alleged avoidance of asymmetries, Anderson and Durand (1988a, b) suggest, moreover, that these can be functional. For instance, a vowel left isolated by lacking a mate that other vowels of the same kind possess, can have a special status – for instance, as a default vowel. We return to this area below. Such discussions as Aitchison’s (1981) also, despite disclaimers, give the impression that languages somehow prefer symmetry, whereas, on the other hand, language changes implement the whims or competence of the users of the language variety concerned, including what originate as errors in terms of the tradition of that variety. Here we might include again the innovation in most varieties of English of contrastive [ŋ] word-finally and before most suffixes, as in *sing(er/ing)*, though not medially in simple *linger/finger* or in comparatives like *longer*. The distribution of [ŋ] is bizarre compared with that of the other nasals, rather than being simply non-contrastive.

This is another well-known established asymmetry involving the English nasal consonants briefly described above. The velar does not occur syllable-initially, unlike the labial and alveolar. The nasal contrasts with the other nasals formative-finally (*sing, singing, singer*, but does not occur immediately before comparative and superlative suffixes, where the corresponding voiced stop must intervene. There is similar variation finally in word-initial *con-*, where occurrence of the variants depends on accent-placement and speaker. The nasals agree with the place of articulation of a following voiceless plosive (*limp, lint, link*); here the nasals are associated with coda prosodies, since place is a property of the cluster. The non-velar nasals also do not appear before a coda-final voiced plosive; but all the nasals can be foot medial onset: *mumble, bundle, bangle*.

Speaking anthropomorphically, the English consonant system is thus also in no hurry to remedy the state of ‘marginal’ contrastive units that contribute to

asymmetries. Again, from being apparently ‘allophonic’ in Old English, i.e. occurring only in mutually exclusive positions, the voiced and voiceless dentals in English are almost non-contrastive only finally in words, where only the voiceless segment occurs except for in a few etymologically verb-marking morphophonological formations (*bathe, seethe, lathe, breathe, mouth*) or in the coda of some pluralized nouns (*mouths, sheaths*), often recessive. And initially the voiced segment signals the presence of a subset of ‘function’ words (*that, they, there, then, thus*). ‘Minimal pairs’ are scarce (and *ether/either* ‘works’ only for some speakers); and voiceless medials are mainly loanwords and a few names (*Matthew, Ethel*), and voiced medials are a curious lot (including *mother, father, smother, bother, dither, leather, feather, lather, gather*, and some morphologically complex forms like *smoother, further*), plus the ‘function’ words *hither, thither*.

Innovation by error is, of course, much more obvious in the history of the sense of lexical items: consider again, for example, the recent history of *icon(ic)*, whose sense innovation by English speakers has penetrated the English and even, in my experience, the Greek of some native Greek speakers, while retaining the earlier Greek sense(s). Again in English, the earlier confusions that are now widely adopted of *infer* and *refute*, for instance. Dickens’ dialogues (as in, say, *Martin Chuzzlewit, Our Mutual Friend*, or *The Old Curiosity Shop* – which I happen to have been re-reading recently) document many aberrations of sound and/or sense on the part of individuals (as well as familiar dialectal forms, of course). Widespread now too, for example, is the morphological misconstruing of the long-established classical *-a/-i* plurals (*data, phenomena, magi, alumni* – so that, for instance, a university administrator can address me as an ‘alumni’ and there is vigorous defending of the result of the historical misconstruing). ‘Subscripts for the hard-of-hearing’ offer a cornucopia of misspellings and misinterpretations. And, of course, wide adoption of both mistakes and innovated forms has been accelerated, and indeed have often become even ‘viral’ (in the now common figurative sense), with the development of digital so-called ‘social media’ (the latter form often being another of the renumbered).

The most generalized ‘naturalness’ requirement is what has come to be called the ‘uniformitarian’ assumption, alluded to above. This assumption has been formulated in various ways over a considerable period, with accounts of some of the formations being more cautious than others. Widely cited recently is Labov’s (1977: 231) offering: ‘the forces which operated to produce the historical record are the same as those which can be seen operating today’. This formulation is among the more cautious, provided ‘the historical record’ refers to written languages of the past, and one doesn’t quibble about ‘seeing’ forces, which indeed is now established. Some of this caution is retained by the footnote attached to Denison et al.’s rather vague ‘the uniformitarian principle asserts that the general properties of language do not vary over historical time’ (2012: 1), which makes explicit the qualification that ‘they may of

course have varied over evolutionary time' (fn.1). But the principle is formulated in terms of 'properties' rather than 'forces'. Beal, in the same volume (2012: 126), lacks the caution of limitation to 'historical time', and she talks of 'mechanisms': 'that the mechanisms governing linguistic variation and change operated in past times and societies as they do today'. Mysteriously, she suggests that this principle is relied on by 'socio-historical linguistics and historical sociolinguistics'. There is very little evidence to support the application of these (apparently different) disciplines to languages of the past, particularly if the languages are not recorded. But more serious here is the loss of caution. This also characterizes Kitson's (2012: 35) formulation, which talks about 'causes': 'though the frequency of linguistic phenomena was not always the same in the past as it is now, the causes which operated in human language were' (again from Denison et al. [eds.]). A somewhat more cautious formulation, apart from the allusion to 'historical sciences', is offered by Anthony & Ringe (2015: 200):

The other reason for our confidence is the Uniformitarian Principle (UP), familiar from other historical sciences such as paleontology. As usually applied in linguistics, the UP holds that unless external conditions can be shown to have changed in some way that can be proved to have an impact on human language, we must assume that the structures of past languages, the way they were acquired by children, the changes they underwent, the distribution of linguistic variation in their speech communities, and so on fell within the same ranges as those of languages that can still be observed and studied.

This is closer to what I say below, minus the 'reason for ... confidence', where this last is a variable. So too with Daunt's formulation as relayed in Strang's (1970: xvii) 'Acknowledgements': '... that for the reconstructed language of the past we must never postulate stages or changes that we cannot exemplify, or at least understand, from the evidence of the present', given that 'understanding' is again variable. Closer still is Morpurgo Davies's (1998: 231) formulation of the classical neogrammarian position: 'those mental and physical activities of human beings which impinge on language must have been always the same (the uniformitarian position)', providing that the 'physical activities' participate directly in implementation, either production or recognition. Her work is a constant reminder of how little recent workers in the field (including myself) have learnt.

Lass (1980: ch. 2, particularly App.) offers a detailed critique of uniformitarianism and of appeals to naturalness in general that again does not seem to have inspired caution in his colleagues (or himself) about what can be claimed about the reconstruction of language history. I shall note in addition a rather obvious instance of the emergence of a new factor in language change, as well as introducing the practical

problem posed by the fact that we lack agreed consistent representations of present-day languages, which are anyway increasingly numerically impoverished compared with the no longer spoken.

Certainly, implementational tendencies recur, as their physical character remains constant; but since they typically work in opposition, as observed above with assimilation and dissimilation, it may be difficult to unravel their applications in complex pre-historic processes of change. But language itself and secondary means of implementation such as writing are cultural entities and thus subject to the introduction of new kinds of change and, indeed, mere variations in fashion.

Languages which have an alphabetic record, for instance, are enabled to add to the potential motivations for change the development of discrepancies between spelling and pronunciation. Spelling can retard or reverse sound change; these are different manifestations of ‘spelling pronunciation’. This is particularly evident where a standard orthography has failed to evolve in response to phonological change among some speakers. In the suburb of Leith where I grew up the second syllable in the word spelled *porpoise* was pronounced with a diphthong that gives the Latin letters a value one might expect, as in *humanoid*. Only later did I become aware that the vowel concerned was reduced in many other varieties of English.

Quirk (1962: 309) comments on spelling/pronunciation discrepancies and the influence of spelling:

... a long-established spelling such as that of English gains such prestige among literate people that many speakers believe it to be more correct ‘to pronounce words as they are spelt’, despite the loss of the original relationship between sound and spelling. Thus, spelling itself may influence pronunciation. Compare, for example, the traditional pronunciation of such words as *forehead* and *nephew* /,fɒrɪd, ,nevju:/ with the increasingly common /,fəhed, ,nefju:/; earlier /,lu:ɪsəm/ for *Lewisham* has already given way to /,lu:ɪʃəm/.

Such spelling/pronunciation discrepancies are very common with names, but the treatment of these by language users is often different from that of other words.

In the course of a discussion of the special status of names, Quirk (1974: ch. 8) points to books that seek to list the pronunciations of names in Great Britain, such as G.M. Miller’s *BBC Pronouncing Dictionary of British Names* (constantly updated, and now replaced by Olausson & Sangster [2006]) and Ross [1974]). And Quirk reminds us of such famous instances of name pronunciation as the variants of *Featherstonehaugh* (1974: 125-6):

Some people called *Featherstonehaugh* use a pronunciation more or less corresponding to the spelling; others call themselves ‘festen-haw’, others

again ‘fearsten-haw’, yet others ‘fee-sen-hay’. And then there is what Ross would claim as *the* pronunciation ‘fan-shaw’. But ... spellings representing this least predictable pronunciation – *Fanshaw(e)* – are considerably more numerous (in the London phone directory, for example) than entries spelt *Featherstonehaugh*. And we have little justification for regarding them as synchronic variants of the ‘same’ name.

With British names in general, however, name-bearers tend to be reluctant to ‘reform’ the spelling on the basis of pronunciation, as suggested by the rarity of the *Chumley* spelling as a more transparent rendering of the pronunciation of the surname than what is suggested by the traditional *Cholm(ond)eley*. We have a spelling-influenced type of sound change that is resisted by a particular part of speech and its social role.

Spelling re-pronunciation is obviously a factor of change absent from unrecorded languages. Moreover, even the more cautious uniformitarians, who recognize that different factors may affect historical languages compared with pre-historical, are not modest enough. Spelling pronunciations can occur when at least some items in a traditional spelling system have lost touch with the contrastive phonological system. This happened on a massive scale in English, for instance, with the invention of printing and the standardization of the spelling systems of the UK and USA, and the continuing ineffectiveness of spelling reformers in the face of entrenched opposition based on, for instance, vested financial interests, as well as inertia and nostalgia, and, in terms of practicality, the absence of a standard phonology – or even, as I’ve just alluded to, a current unified spelling system. The identification of local pre-modern standardizations in recorded English is fraught with problems, and spelling pronunciations even more so, as well as the results of periods and dialects where writers permit(t)ed themselves variation in their spelling.

There are other cultural changes that can lead to a novel spelling. This is again unsurprising, given the cultural status of language itself. For instance, the development of ‘smart-phones’ and the so-called ‘social media’, has led to an increasing dependance in the written language on certain modes of word formation, including blending and pervasive abbreviation (especially as initials), and including the use of homonyms, even from outside the core language system, such as the familiar 2 for *to* and the ‘ideographic’ metonym ♥ for *love*. A blend, in particular, can lead to phonological innovation. One example is the development, among some users of English, as expanded by the globalization of ‘English’ (or ‘Panglish’), on the basis of *blog* and *blogger*, of *vlog* and *vlogger*, i.e. a video-using blog and blogger. These introduce an initial cluster that is otherwise absent from most ‘standard’ varieties of present-day English, as far as I am aware (except in obviously foreign names such as *Vlach* and *Vlad*).

This innovation also contradicts Fulk's claim – already dismissed in Ch. 2 – that 'it is a fundamental principle of historical phonology that initially sound changes are governed solely by phonological conditioning, with morphological factors taking effect only subsequently' (1992: 177): in this context a prominent instance of an obfuscating lack of modesty. In the area of language development, proposed 'fundamental principles' should come with a health warning (though this would possibly be too late).

Another recent fashion is the creation of what one might call 'semi-blends' or 'deformed compounds'. Some examples of these are *staycation*, *webinar*, and *covidsplotation*, the second components of which involve a large part of the apparent independent source, but interpretable only by reference to the full form. Happily, a more generous and more successfully blending is *covidiot* – though out-of-context its denotation may be obscure: who qualifies? Change of word class is illustrated by use of a verb *reference* (rather than *refer to*), with (unusually) accentuation as in the related homophonic noun.

Cultural changes have also diversified the range of personal names given to children in various languages and increased the rate of innovation at which the most favoured names change. I have what was a tediously common personal name among my generation in Scotland and often a family tradition. Now it is seldom encountered among recent generations. But we are – or, rather, I am – drifting away from the deconstruction of phonology.

There are constants of perception and articulation, but the role of these in language change is usually partial, even if not contradictory. What are generalizable are perceptual and articulatory phenomena that may favour particular implementational sound-changes: changes, often observed, that favour ease of articulation vs. changes that facilitate the audibility and ease of discrimination of (particularly adjacent) contrastive segments. But such phenomena may be only partially implemented in phonetic or phonological change, and the consequences of their implementation in particular subsystems may be spread over time among groups of speakers. A phonetic change and a possibly resultant phonological change are variously manifested.

A striking instance of change is the 'chain shift', whereby in the instance that follows the ('long'/'tense') vowel contrasts are realigned in an 'orderly' fashion. A much-discussed alleged instance of this phenomenon is the 'Great Vowel Shift' ('GVS') of the history of English. This hypothesized phenomenon is diagrammed by Jespersen (1909) and others as the historical changes in (9), where '↑' is to be interpreted as 'raising' and '← →' as 'diphthongization'; the change is in the distribution of the vowels in the lexicon, so that the relevant content and expression roles are realigned.

(9) ai ← i u → au
 ↑ ↑
 e o
 ↑ ↑
 ε ɔ
 ↑
 a

The ‘inner coherence’ (in Lass’ [1976: ch. 2 (11)] terms) of the shift is intended to be expressed by Carter’s (1975) single ‘rule’: ‘V > [+raise]’, where diphthongizing is apparently a sort of default raise, and which Lass goes on to elaborate upon in terms of ‘metarules’.

Stockwell & Minkova (1988), however, point to problems in the suggested ‘unity’ of this alleged shift, which also assumes a requirement that merger is avoided. The latter phenomenon, again, though natural, is not necessarily implemented, and mergers are not uncommon in language histories. Stockwell & Minkova (1988) also point to histories of varieties of English where diphthongization is not limited to the high vowels. They also invoke Boisson’s (1982) demonstration that varieties of English similarly vary in the shifts that are implemented therein, and the extended period of time over which different components of the ‘shift’ are implemented; these illustrations and Wells’ (1982: vol. 2) coverage of varieties in which history has resulted in a very different picture of the vowels concerned as would result from (9). See too Stockwell (2002).

However, while there is no unitary episode ‘Great Vowel Shift’, I suggest, tentatively, that the various raisings and diphthongings may have in common a perceptual motivation. ‘Long’ vowels, the implementation of whose energy is spread over time, can for that reason be perceived as less low, or compact, or sonorous, than ‘short’ vowels made at the same height. This is an attempt to elucidate somewhat the persistent ‘impulse’ for these changes invoked in Anderson (1992a). One implementational reaction to this perception is to pronounce the ‘long’ vowel concerned, as higher, to ‘raise’ it, eventually phonologized; and another ‘strengthening’ and differentiating is provided by diphthongization, particularly with high vowels, which cannot ‘raise’, but not limited to them, as illustrated by Stockwell & Minkova (1988: §IV) and their references. Varieties of English vary in the strategy adopted and the extent to which merger is avoided. Stockwell & Minkova (1988: 379) object to the ‘abstractionism or idealism’ of typical traditional accounts of the ‘Great Vowel Shift’: what they object to is another manifestation of lack of modesty – though see Lass (1992) for an attempt at a compromise. And such systemic adjustments, though locally variable, may be at least unified, though not identified, by a shared perceptual basis.

Lack of modesty can also be illustrated by the maintenance of assumptions concerning ‘sound-changes’ or ‘sound-laws’ that prevent, indeed, the formulating of

generalizations. An example of this lies in traditional accounts of the pervasiveness of ‘shortening’ and ‘lengthening’ of vowels in Early Middle English. As well as failing to account for exceptions to the ‘sound-changes’ involved (studied in some detail in Ritt (1994; also 1992), such accounts are prevented by the traditional idea of change by well-defined unitary ‘sound-changes’ from offering any generalization concerning these roughly contemporary changes in the occurrence of the same vowel-length contrast.

Ritt’s calculations of probability of implementation of these changes attracted a relevant comment from Benskin (2001: 214, note 66):

...Ritt 1997 <a later edition of (1994) – *JMA*> ...isolates (pp. 33-8) features which correlate significantly with lengthening and non-lengthening <and shortening and non-shortening – *JMA*> in the apparently eligible lexis; the material from the Corpus compendia fits unremarkably into these patterns. (It should be noted, however, that what Ritt regards as probabilities are mostly outcomes: the probabilities coeval with the engendering of these outcomes are not themselves accessible. Astonishing in the table on pp. 129-35 <concerned with ‘open-syllable lengthening’ – *JMA*> is the near absence of *u*-words, and the absence of *i*-words altogether.)

We should certainly remind ourselves that our main source of evidence for the implementation of these quantitative changes, given particularly that vowel length is seldom indicated in texts, is the present-day language, but too little use is often made of the range of present-day varieties and the complexity of their evolution, along with what ‘principle’ or ‘law’ might be involved. Language change involves lexical diffusion of the change and failure of diffusion in some lexical and user potential targets, as well as implementation preferences and, in the present case, gradual loss of stem-markers.

What emerges from the vowel lengthening and shortenings of Early Middle English is the development in this period of a preference for consistent initial accented-syllable types, with the loss of some word form types as a consequence. The maximal coda is one adjoined consonant segment modifier, but there must be a complement to a vowel that is transitive ‘short’ (and not reduced), though it may also have a modifier. What happens in early Middle English is the imposition of restrictions on the structure of tonic syllables in particular, and involvement of the factors of rhyme transitivity and ‘weight’ alluded to in the preceding.

Thus, as a result of ‘open-syllable lengthening’ (an unfortunate term), a ‘short’/transitive tonic-bearing vowel preceding a foot-medial ambisyllabic consonant (Anderson [2022a: ch. 11]), such as typically disyllabic OE *ofer*, is bolstered by length. In neither stage can the syllable be described as ‘open’: in the transitive *ofer* the medial

consonant is ambisyllabic, satisfying both the initial transitive vowel and the onset of the second syllable. The transitive (complement-requiring) vowel is made intransitive and in general can stand on its own with or without an adjoined coda (cf. Modern English monosyllabic *by/byte* with **bi/bit*): it does not need a coda. The tonic syllable acquires independent weight as the head of the trochaic structure (Anderson [2022b: ch. 28]).

By ‘lengthening’ before apparently complex homorganic codas such as that in OE *bindan*, transitive vowels are also ‘lengthened’. But only the first segment in the coda of the first syllable depends on the tonic vowel and satisfies its transitivity; the second consonant in this sequence merely satisfies the onset of the second syllable, in accordance with ‘onset/initial maximization’. The tonic syllable is thus still rather light: codas of ‘short’ vowels are eligible for an adjoined consonant as well as a complement, but the consonant spelled *d* here is apparently syllable-initial, given onset maximization. Moreover, Anderson (2022a: ch. 13) suggests that the homorganicity prosody linking the components of this coda is a marker of phonological-compound status; the coda is equivalent to one segment, and only its second component participates in ambisyllabicity. That is, only the head of the coda, the oral stop, participates in it. But the initial syllable is likely to figure as a sentential tonic. However, the vowel in *hand* and the like can fail to be susceptible to ‘homorganic lengthening’ on implementational grounds disfavouring low vowels.

The ‘lengthening’ changes we have looked at are associated with the immediate environment of the tonic vowel affected: they make the first syllable ‘heavier’; we have a more salient trochee. However, ‘shortening’ of the tonic occurs before two further syllables in the same word form – though this may coincide with at least two non-homorganic or identical subordinate consonants following the accent, as in *wisdome*, or *seolliche*, Modern English *wisdom* and *silly*. And the former environment may occur in simple forms, as in *dust* ‘dust’ or *fyst* ‘fist’. There are moreover many exceptions again. Ritt (1994) identifies various implementational factors, as well as morphological and social evolutions.

The ‘quantity-adjustment’ changes as a whole might then be regarded as leading to restricted shapes to accented word forms: ‘long’, or intransitive, accented vowels prefer to have minimal codas, with one adjoined (but possibly compound) consonant; and short accented vowels have a complement and a modifier, or such a tonic has two adjoined syllables; in other situations, e.g. in accented monosyllables, either length may occur. The picture is complicated by word forms differentiated by presence or absence of a syllabic inflection and the variable influence of stem-forming [i], as well as other variations discussed by Ritt (1992). But, unlike the situation with some implementationally favoured changes, which are often in competition, as with assimilation and dissimilation, these quantitative changes might be said to co-operate in establishing the possible environments for initial accent-bearing forms, both

primary (tonic) and secondary (typically the ictus of the second component of compounds). Accentuation prefers certain weights in the (sequence of) rhyme(s). Heaviness is associated with ‘long’ vowels and/or with the rhyme structure, where typically presence of weight is associated with a certain complexity of the rhyme or presence of two following syllables. As noted, sonority has a role in whether or not these length changes are implemented (Ritt [1994: ch. 8]). And the reinterpretation of roots as word-forms rather than stems, is encouraged by Old English changes (Ritt [1994: §8.2.1]). But the reconstructible details and the scope of historic length adjustments is, by contrast, an indication of how little, in comparison, we can reconstruct about the phonology and its development in pre-historical reconstructions.

Consider now the word-form accent that emerges in the Modern English system. With continuing reduction or disappearance of inflections and the introduction of neo-classical and Romance ‘borrowings’, the pattern developed in early Middle English disintegrates in favour of variants of the complex accentual regularity illustrated in (10) described in Anderson (2022b: ch. 28). (10a) formulates the accent placement that different contentive word classes (adjectives, verbs, and nouns) contract. (10b) indicate eccentricities of intransitive (‘long’) vowels. SH marks a ‘superheavy’ vowel, with (non-ambisyllabic) modifier(s); H is ‘heavy’, as in (10.b.iii), different from Old English, but also basically trochaic.

(10)a. THE PHONOLOGY OF WORD ACCENT

$$\begin{array}{ccccccc}
 & & & & & & <\{V_4\}>_{*1} \\
 & & & & & & | \\
 & & & & <\{V_4\}>_1 & <\{V_3\}>_1 \\
 & & & & | & & | \\
 \{V_1\} <\{V_1\}>_1 <\{V_{SH}\}>_2 <\{V_1\}_E > \# & \Rightarrow & \{V_1\} <\{V_1\}>_1 <\{V_1\}>_2 <\{V_1\}_E >
 \end{array}$$

b. FINAL EXTRAMETRICALITY IN ENTITATIVES

i. $\{V_{INTR, SH}\}, \{V_{\{1\}}\} = *(10.bii)$ ii. $\{<N;P>\}$
 $\vdots$
 $\{V_1\}_E \# = \underline{\quad} \underline{\quad}$ iii. $\{V_3\}$
 $\mid$
 $\{V_{INTR, H}\}_E = \{V\}$

Anderson (2022b: ch. 28) discusses other regularities, prominently those associated with the accentuation of nouns. In their case including most a heavy (modifier-less) final rhyme of a trisyllabic noun is marked as extrametrical, / /_E, by the redundancy in (10b.ii), as in *attórn/ey/*_E, whereas a superheavy bears a tonic or ictus in conformity with (10a). (10b.i) thus rejects extrasyllabicity for a final superheavy intransitive vowel in nouns: thus *fárthingàle*, *domaín*. See further Anderson (2022b: 42-45) on the examples given here as (11):

- (11)a. ántelòpe, fárthingàle, domaín
 b. filigrèè, pédigrèè, canoé, tattoó, degréè, decreé
 c. casino, attórney, alúmni, búffalo, bróccoli, rádio

Whatever the (de)merits of these formulations, they do incorporate the basic factors involved in accent placement: word class (e.g. $\{N;P\}$, noun), transitivity ($\{INTR_{(ansitive)}\}$, ‘long’), weight ($\{SH\}$, superheavy; $\{H\}$, heavy), elements that require either their co-presence or co-absence (marked by numerical co-indexing ($V_{>1}$, $>V_2$), and extrametricality ($\}_E >$) in the case of non-heavy final rhymes in a noun. Accent placement is not now typically root-initial (as it was in Old English), but its location is more obviously dependent on the structure of the end of the word form rather than the beginning.

There is, of course, rather more to the accentuation story, including the role (or non-role) of certain affixes in accentuation, morphological triggers, as well as word class and phonological weight. The most striking example of this is a restriction that Old and Modern English share, despite all the other changes. Just as there are predominant extrametrical verb-prefixes in Modern English, such as those in the non-trochaic *perféct* and *présént* – compare the non-verbs *pérfect* and *présent* –, so we can reconstruct Old English *onsácan* ‘deny’, *oþgán* ‘escape’ – cf. non-verbal *ándsàva* ‘apostate’, *ūþ-gènge* ‘evanescent’. Lass (1994: 92-3), however, reminds of exceptions where the accentuation of the base is retained in the derived form, be it verb or noun, as well as compound forms based on adverb + verb. Also, though in both stages of the

language, accentuation of word forms is basically trochaic, in Old English the trochee is initial, provided we allow for verb-prefix extrametricality, while in the modern language accentuation works from the end of a form (cf. Anderson 2022b: ch. 28), as is illustrated in (11c) above.

Chapter 10

Temporal ascendance and descentance in time of related languages

Where there are no texts in an orthographic notation available for reconstruction of the language, or rather phonology, given our focus here, as with the ancestor of Old English, this ‘language’ can be said to be a pre-language, a non-recorded language. As acknowledged above (Chap. 4), internal features of a synchronically reconstructible language, such as morphologically conditioned alternations, may enable us to reconstruct aspects of an earlier stage of this language. If a number of synchronically reconstructible languages show sufficient similarities, we can supplement internal reconstruction with comparative evidence (Chap. 9). And such a set of languages may enable us to reconstruct a common ancestor or prior-language. And we may be able to partially reconstruct higher-order prior-languages on the basis of similarities between first-order prior-languages. This is illustrated by the development of ideas concerning the relationships within most of the language families of the Americas, for instance, as well as most of those in Indo-European, which has long been focused on.

We have ‘cross-linguistic internal reconstruction’. Higher order prior-languages, however, permit increasingly less secure and less extensive reconstructions of the language and especially of the spatio-temporal coordinates of its users. And no pre-/prior-/proto-language is a ‘real’ language, or has a ‘real’ phonology. Campbell (1959), for instance, beside his representational failure, doesn’t reconstruct a ‘real’ language, or even phonology, but suggestions of some aspects of its development that may have been associated with manifestations of what became historical English. Prior-languages are aspects of stages in the phonological development of several pre-languages, perhaps in the form of what has been described as a ‘steaming broth’.

The ‘law’ that is claimed to differentiate the consonantal phonology of Germanic languages from that of other Indo-European groups of languages, under the rubric of ‘Rask-Grimms law’, is perhaps among the least worthy among ‘sound-changes’ of the ‘law’ appellation. It offers no motivation for this ‘event’. The ‘law’ can be presented in traditional terms as in (12a) (from Lass [1994: 20, §2.5]). Ringe (2006: §3.2.4 (i)), for instance, provides fuller exemplification, though in another variant of the same curious traditional segmental notation for pre-historic forms.

- (12)a. ‘A. p, t, k > f, θ, x
 B. b, d, g > p, t, k
 C. bh, dh, gh > β, ð, γ (later b, d, g)’

- b. A. $\{C\} \rightarrow \{C;V\}$
 B. $\{C\{v\}\} \rightarrow \{C\}$
 C. $\{C\{v\{c,v\}\}\} \rightarrow \{C;V\{v\}\} \quad (\rightarrow \{C\{v\}\})$

His representation is one which includes a mixture of Greek and Latin letters, and macrons, subjoined hooks, plus occasional ‘phonemic’ brackets’, but no indication of contrastivity.

Many relevant issues concerning (12a) have been raised by Theo Vennemann, whose website lists the relevant publications. Vennemann (1985), for instance, argues for an alternative, more comprehensive formulation of the German as well as the Germanic shifts, but again presented with a combination of traditional and innovated ‘letters’, including in this case capitals and various further diacritics.

There is no obvious motivation for the occurrence of these alleged pre-historical ‘changes’ in Germanic; they are sometimes analysed as constituting a ‘chain-shift’ of ‘obstruents’ (i.e. segments with dominant **C** in the notation in (12.b) from Anderson [2022a: ch. 12]), where a movement towards $\{C\}$ displaces it; but the motivation of the ‘chain’ is not determinate in (12a). And the pre-historic ‘shifts’ are in the usual quasi-litteral tradition.

If we compare Lass’s description (or any other quasi-litteral account) with a featural notation such as (12b), the latter is at least more transparent, as well as explicit in expressing what the members of the lateral and vertical sequences have in common. But it may be that an explanatory account is unattainable, given what evidence we have. One characteristic that emerges from (12b) is that that the ‘changes’, indicated by arrows, are featural simplifications of contrastive segments, unless this is not possible: the ‘input’ consonants of A in (7b) are already the simplest in the system of ‘obstruents’. But the overall result is to eliminate the marked (most complex) value ‘aspirated voiced plosive’, or, more explicitly, $\{C\{v\{c,v\}\}\}$ – for what it’s worth.

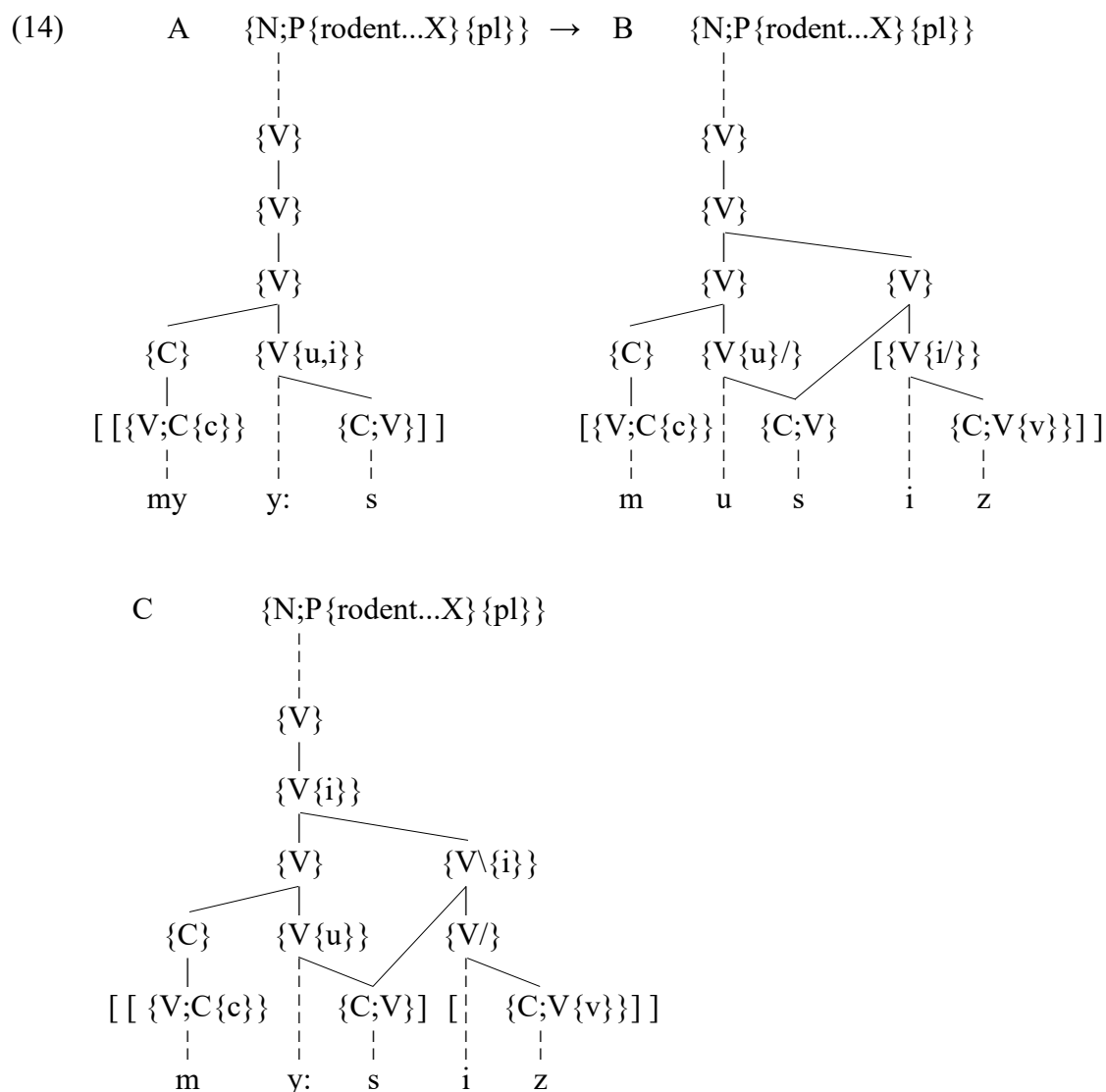
However, our main concern in this section is the reconstruction and representation of descentance/ascendance through time and the interaction of phonological systems, again susceptible to a characterisation in terms of a dependency relation. We can represent linguistic descentance, at its simplest, indeed oversimplified, form, as in the simple directed connected graph in (13a), where the arrow points to the past, and where the node labels ‘A’ and ‘B’ abbreviate bundles of reconstructed linguistic properties, some of them in common, some different; and the arc (or ‘directed line’) represents a combination of change and continuity over time, where the arc points to the past as its direction and the line shows continuity of the referents at the nodes:

(13)a. A
 ↓
 B

‘A’ and ‘B’ here and such symbols in what follows are not associated with a particular-language history; they stand for exemplary bundles of systems that would have contrastive content, bundles including subsegmentals, and some of these including neutralizations; and some involve prosodies. B depends on A; A is the head of the arc, and as such is preceded by B in time. That is, I adopt at this point the Strangian strategy based on the observation that in the process of reconstruction the more temporally accessible determines much of the less (cf. Strang 1970: §16). The contents of A and B are reconstructed on the basis of some of the kinds of evidences we have outlined, excluding mere phantasies. If A is recorded or even current and B not, then our reconstruction of the latter will be more markedly incomplete, though aspects of A may contribute to the reconstruction of B, as manifested in ‘internal reconstruction’.

Even if we neglect syntax, apart from valencies (‘argument structure’) perhaps, characterising the contents of A and B in (13) would involve contrasting the representations of one lexicon with another, as well as formulating individual connections that satisfy etymologies. It’s not just a question of list of contrastive segments and perhaps accentual generalizations, with listing of a few lexical items that are claimed to share approximate ‘sense and sound’ apart from arguably plausible differences in either. There needs to be shown historical lexical links between items at different stages such as A and B in (13a) that are representative of a large enough number of others to enable the prediction of the form of still others, even if unattested.

A simplified but more transparent representation of such a simple lexical link as is abbreviated in (13a) might be expressed as in (14), with traditional values attributed to the nominative plural of the ancestor of word form *mice*.



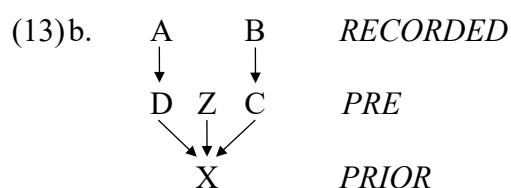
Here again the arrow points to the past state of affairs, where the sequences at the top is a much abbreviated representation of the content planes, and at the bottom is a phonetic implementation, below the phonological representations, with the morphological structure is indicated by the square brackets, where the outer bracket pair encloses the whole form, and inner encloses the stem; and the inflection thus falls between the closing ‘]’]. C represents the intermediate prosodic stage, where the suffix demands a {i} on its governor as a prosody; the intervening consonant may have had a paralysed implementation. It’s only with stage A that the vowel that is implemented as [y] is contrastive.

The labelling of such stages as involving presence of pre- and not is helpful, given the difference in available evidence; but in relation to recorded languages, such a relabeling of ‘stages’ (‘Old English’, ‘Middle English’, ‘Late Middle English’, ‘Early

Modern English’, etc.) is linguistically relatively arbitrary and it is usually motivated by extralinguistic factors as much as by, say, reconstruction of the occurrence of drastic (including social) changes in the language concerned and its culture. Moreover, this labelling ignores the variability over time (and space) we can reconstruct for such periods and any associated shift in what is regarded as ‘standard’, and it thus has only a fuzzy status as a representation of descentance.

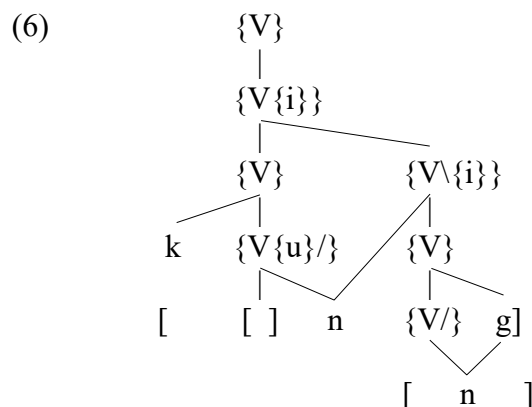
If B and C in (14) and the like belong to pre-language, we are dealing merely with bundles of reconstructible properties, properties whose individual introduction may be indeed disparate in time, as is recognized, indeed highlighted, in traditional accounts. As I have insisted, pre- and subordinate prior-languages are not ‘real’ languages, and even transcribed languages of the past and their cultures are necessarily accessible only to a limited extent. And indeed reconstructions of present-day languages, or just phonologies, can be overwhelmed by a mass of variability in the evidential data unless they adopt some principled limitation in scope, including the maligned study of ‘my language’, or if the goal of the research is indeed to investigate the variation.

If we are aiming at a prior-language then we typically need more than one arc at the very beginning, and eventually a shared ancestor, as represented schematically in (13b), where again the pre-historical node-labels are not ‘languages’ – and their unitary labelling in (13a) – is misleading, given that much of pre-historic structures cannot be reconstructed, and parts of any language may be ‘borrowed’ from an unrelated or distantly related source.



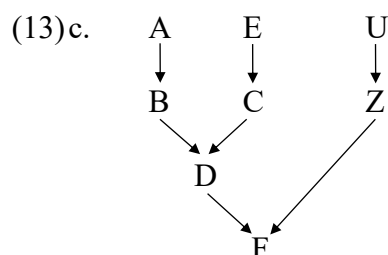
The records of A and Z may not be coeval, even if their pre-languages are thought to share exclusively much of their phonologies; indeed, typically they are not coeval, as with Old English and Old Frisian – and there are apparently shared properties with Old Saxon, which also shares properties with High German. Of course, A and Z follow temporally their respective dependents. And, as observed, the reconstructed properties of B and C may be reconstructible as relatively timed. B and C differ systematically from each other in some properties and share others, and their patterning enables us to reconstruct properties of D. The left-to-right ordering is not relevant, but arbitrary, in such a simple representation. And it is not clear how to express systematically the volatile sharing of properties that is attributed to the pre-historic reconstructions alluded to here.

In synchronic syntactic and phonological structure the head-dependent relation is expressed by directed lines that typically diverge, and only occasionally in phonology converge, as, for instance, in foot-medial forms such as that in the synchronic stage in (6) above in Chap. 4, where the first nasal is reconstructed as shared between the first and second syllable (ambisyllabic), and the second nasal is co-dependent on the rhyme-head and the coda-head of the second syllable.



In diachronic reconstruction, however, such as in the representations in (13), the dependency lines are convergent in accord with distancing in time from the present: they have a unique dependent and non-governing node.

Such relations among pre-histories, however, have traditionally, once established, been represented graphically as a single-rooted inverted tree of variable extension, the *Stammbaum*, with the highest-order prior-language as ‘root’ or proto-language, with descendants diverging or branching from each other in time – which we will return to. This reflects the result of reconstruction invoking temporal distancing from the present à la Strang, where the recognized root is maximally dependent. What is traditionally represented is the proposed descentance from pastness of reconstructed ‘stages’ rather than ascendance in pastness as in the process of reconstruction. However, we began this discussion in constructing a converging graph that, in accordance with our Strangian perspective, reflects the temporal order of active reconstruction, and is more appropriate in this respect. Given appropriate evidence, we can add further mergers to (13a, b).



The reconstruction in F (13c) depends on evidence from D and Z. But reconciling aspects of the latter pair with their common descendant may require refinements of how they are reconstructed, particularly in the light of the construction of D and vice versa.

The belated reconstruction of Hittite and sister languages and especially of its so-called ‘laryngeals’, for instance, led to refinements in the analysis of its traditional ‘Indo-European’ co-descendant and its descendants. In the absence of decisive counter-evidence, the notation of (13c) allows for the possibility of non-binary mergers; but typically there is evidence for staging mergers so that there is a set of binaries. In the case of a prior-language (reconstructee), the comparative evidence from its immediate predecessor(s) is obviously all-important for its reconstruction, and the evidence from different predecessors may suggest less or more ‘closeness’ to the prior-language, not represented in the traditional tree.

It is misleading to think of these labels as abbreviating only phonological, particularly segmental, properties. Much change in phonology, as well as reflecting implementational phenomena (such as assimilation/dissimilation), is associated with morphological structure, as well as prosodic phenomena borne by a node in the phonological hierarchy of lexical items crowned by a tonic type or by a morphological hierarchization. Thus, phonology cannot be reconstructed without acknowledging these factors. Indeed, the phonology of sentences – pre-utterance phonology – is bound up with syntactic functions. This, and the reconstruction of syntax itself, must be speculative in pre-historical languages, in their lack of alphabetical-, or even syllabary- or character-based, written texts.

All of the limitations that come from neglect of such potential non-phonological evidence are associated with the reconstructions associated with the orientationally counter-Strangian Stammbaumtheorie, alluded to above as the most familiar tree-graph-based representation of descentance to present-time, the highest order prior-language is an *Ursprache*, which splits into a number of *Grundsprachen*, which in their turn split into language families and eventually into actual languages and dialects through a process that can be repeated again and again’ (Morpurgo Davies 1998: 170, commenting on Schleicher’s *Compendium*, first published 1861-2). In this case we have a diverging directed graph, an inverted tree to a point at its foot. As observed, the Strangian orientation reflects the progressive discovery of earlier stages in development, involving ascendance in time; the ‘Stammbaum’ displays the results of such discoveries from the perspective of the proposed ‘root’ of the developments, involving an ascendance in time, towards the present.

The difficulties in reconstructing higher-order prior-language relationships have led to continuing disputes, already alluded to (in Chap. 2, for instance), over the groupings of ‘families’ within Indo-European, for instance, despite the long history of literacy among many of its descendants. There is, for example, the possibility that

shared properties might have been independently developed, or might involve ‘borrowing’ between related prior-languages, or indeed unrelated languages. And the Romance languages illustrate overtly a situation where Latin continued to be spoken and written while these languages were diversifying and inter-borrowing, and Latin itself was developing.

Such and other perceived difficulties, along with a prematurely developing interest in social variation, prompted the development of the ‘Wellentheorie’, promoted by Schuchardt (1866-8), Schmidt (1872) and others. According to this conception (again in the words of Morpurgo Davies [1998: 286]) ‘the relationship of the Indo-European languages’ would be represented as

...situated on concentric rings with the parent language at the centre and the individual languages more or less distant from it but still linked to each other; the model was that of waves in water which spreads from a central point in concentric rings which progressively become weaker and weaker.

Computation of relative temporal ‘closeness’ of languages is not straightforward, however. ‘Wave-theory’ introduces new problems. Such a picture might be compatible with the development of the members of the Romance family. But it is not clear how the links between the predecessors of the descendant languages integrate with the wave metaphor, or how indeed it resolves the problems mentioned above, except in terms of combination with a tree graph: given possible collateral influence, this would not be a proper tree (on such see again Marcus [1967: ch. VI, §3]), and one involving subjunctions, as exemplified by the examples in (14).

The use of the ‘tree’ metaphor, however, as adopted in graph-theory and elsewhere is also unfortunate, for yet other reasons, in relation to the representation of linguistic temporal descentance. So too is sex-specific talk of ‘mother-daughter’ and ‘sister’ relations among the nodes in a ‘tree’ (as I did above), and also talk of ‘generation specific’ ‘parent’ languages, even apart from the combined use of these also manifesting a mixing of human and vegetable metaphors. Here, and elsewhere in the analysis of language (and some other phenomena), strict retention of the properties of the metaphorical vehicle ‘tree’ inhibits the recognition of borrowing from un- or distantly-related sources alluded to above; and introduction of a notion of ‘grafting’ is not very helpful.

What is most unfortunate is that such metaphors, and others concerning ‘language filiation’ and even one sense of ‘descentance’, may evoke their deployment in describing biological phenomena (as discussed in Hoenigswald & Wiener [1987]); but reconstructible languages and their descentance in time are cultural not biological phenomena. The development of language, on the other hand, no doubt contributed to biological developments to cope with it, and was made possible by previous biological

developments in humans. Description of lines, their junctions, and their orientation in terms of a subset of family relations may be mildly amusing, or, intentionally or not, offensive, but it may be misleading, as is the use of even ‘tree’. Both types of phenomena, biological and cultural, exist and change in time, but we should be wary of the pervasiveness of interpretations of language histories based on biological evolution, both at earlier periods and currently. The biological changes that made possible the development of language predate that cultural development (Hurford [2007, 2011, 2014]), though the discovery of language may have promoted some mental developments. The reconstruction of grammars, however, need not appeal to language-particular biological phenomena, such as the development of the conscious mind, as explored in Wilson (1937). We can indeed derive some information about migratory patterns of ‘tribes’ from present-day DNA distribution. But ‘tribal’ and language connections can change over time, by conquest or less brutal interaction, for instance, as well as by internal changes in fashion; and culturally specific artefacts can ‘migrate’, by virtue of trading or plagiarism/counterfeiting.

This cultural status of language and its history is as important a limitation on what we can attribute to reconstructions as the recognition that the labels on the above ‘trees’ are not associated with ‘real’ languages, but with (partial) reconstructions. And, as we have noted, when the reconstruction is that of a prior-language there is a severe limit on the linguistic evidence that can be brought to bear on it, and particularly on the temporal, geographical, and social circumstances of its users, though the constructible senses of early vocabulary may be suggestive of their culture and a location. Even in non-current but recorded languages such as Old English, and despite careful studies such as Whitelock (1951) and contributions to Rumble (2006), access to the culture of language users is limited.

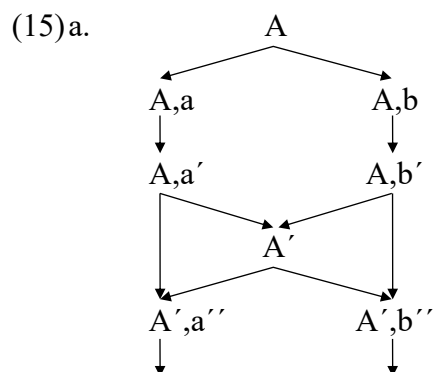
Representations (13a)-(c) are all minimal ‘proper trees’, with a single ‘root A’, where all vertices, or ‘nodes’ are connected; the trees are directed and do not contain cycles – where there would be a path, or sequence, of arcs such that a single node initiates and terminates the path. The arcs, or directed lines, iconize time, and are not associated with genetic or other biological change but simply with the relative time of the reconstructions associated with connected nodes. Despite the dangers of shared metaphors – such as in the representation of genetic relations and language change – we should recognize that the same metaphor may be appropriate in the representation of phenomena belonging to different conceptual domains. That we can use a ‘tree’ metaphor both with respect to biological developments and language and other social histories merely reflects that we can perceive and characterize similarities in their manifestation, in this case involving change, particularly divergence, through time. Such considerations are placed in a well-contextualized historical context in contributions to Hoenigswald & Wiener (1987), particularly that of Rulon S. Wells. Problems arise when particular developments are assigned to an inappropriate

metaphorical domain, or vehicle, as with the invention of language: Wilson (1937) entitles his book *The Miraculous Birth of Language* – though use of the ‘birth’ metaphor is indeed widespread.

The representations in (13a)-(c) represent the skeleton along which reconstruction is registered: it is the skeleton of the methodology of reconstruction, but it does not explore the difficult decisions about what might have motivated mergers. Consider the continuing controversies and phantasies concerning the relationship among the ancestors of Old English, Old Frisian, and Old Saxon. The representations to follow, (15a)-(c), represent the results of reconstruction, and to indicate the sources of ‘properties’ characterizing the bundles at successive vertices. These ‘properties’ are sequenced subsystems, including suprasegmental configurations and featural decomposition, not simply sets of segments. In (13a)-(13c) the arrow, directed line, points back in time; in (15a)-(15b) the reverse. The dependency relations are reversed in the traditional approach, regrettably, given the direction of reconstruction. Let me, nevertheless, try to illustrate the resulting relationships among pre-languages when representations are not restricted to minimal trees.

Even in the prior-linguistic domain the introduction of some of the ‘properties’ that characterize a particular reconstruction of a language may be ordered in time: a property may presuppose the existence of another: one change may ‘feed’ or ‘bleed’ the other. Reconstructors can often establish a relative chronology for the ‘sound-changes’ that produced a particular reconstructed phonology. With respect to A in (10b) the times of introductions of components of its reconstructed phonology can often be partially ordered. Similarly with the history of B and C. However, as I’ve noted, it seems to be the case that such descendants as B and C, while having innovated in such a way as to partially distinguish them from each other, at a later stage can both undergo a distinctive change: their ‘properties’ partially converge further (by various mechanisms). A ‘property’ is added over time to those preserved from A that they have in common. The nodes in trees are labelled with the components of a language of the past that can be reconstructed as in common with or different from components of other nodes. These ‘re-directed’ trees are ‘orthodox’, or ‘counter-Strangian’.

We might represent the result of introducing such properties as in the ‘counter-Strangian’ (15a), which analyses the labels on the nodes into complexes that indicate both sharing and non-sharing of reconstructed ‘properties’, however this sharing or splitting may have come about (independent developments, ‘loaning from one to the other or from a different source, even merger).

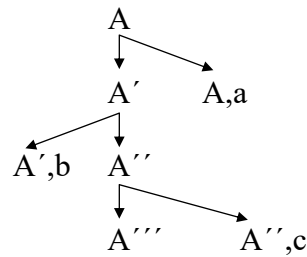


A is the set of properties that a and b initially have in common; a and b are subsequent properties not in common; a' and b' incorporate further changes. At least one of the descendants incorporates further changes; (15a) assumes that both do. A' is a 'property' that is later shared by a' and b'; it thus has two heads. Further changes are made in a'' and b''. If a descendant doesn't acquire more than a few further features, it will remain closer to the root than is a co-descendant that does; and the set of A properties will contract over time.

This merely gives a simple graphic dependency expression (though with potential double heads and mergers) to the diachronic relations we can reconstruct; they do not indicate the circumstances from which these relations arise, which, particularly in prior-linguistic territory, are typically unknowable – though there is lots of room given up to such phantasies as have been indulged in historical linguistics (see again e.g. Anderson [2005a]).

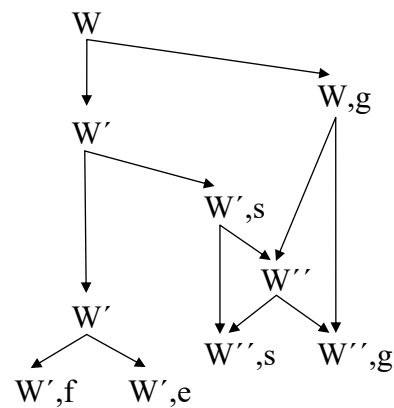
However, by allowing convergence in a basically divergent tree we can allow for shared developments. And along with a notation for node labelling such as in (15a) we can go some way to meet the 'problems' that were invoked in the arguments in favour of *Wellentheorie* and other problems introduced above. This depends on regarding the node labels in (15a) etc. as not denoting 'real' languages but collections of the systems that are hypothesized as distinguishing a 'stage' but also connect it with other node labels. This is all that can be reconstructed, but from which one or more 'real' languages may eventually develop. It allows the descendants of an *Ursprache* to separate from it at different stages in time, i.e. on the vertical axis, as in (15b).

(15)b.



Moreover, we can combine the configurational possibilities illustrated in (15a) and (15b) in such as (15c), which includes a partial merger and diversion within that. We end up with four descendants of the *Ursprache* at this point, where it's not necessarily the case that there is a difference between the upper and the lower W's, but there could well be.

(15)c.



There are also nodes in (15a, c) that two arcs converge on, thus violating the so-called 'single-mother' restriction (another sex-specific metaphor) on directed trees. But this restriction is questionable when applied to graphs of historical developments (as well as elsewhere). Indeed, Theo Vennemann's website again displays a bibliography including a large variety of interlinguistic phenomena **adoptions**, mainly **substratal** and **superstratal**.

Given that language change has been associated with convergence as well as divergence, such a restriction as assumed in traditional cladistic structure is not appropriate. This allows us to include kinds of evidence for convergence that has been offered, in the context of the modesty-based proposal I have been making here concerning the character of reconstructions. But latterly I have adopted, for ease and familiarity of illustration, the graphic counter-Strangian perspective of the traditional *Stammbaum*. From a Strangian perspective, its reconstructionally 'upside-down' trees systematically reverse ontological status.

It is important always to bear in mind that A, W, etc. in the above representations do not label 'real' languages but merely structured bundles of reconstructed contrastive

‘properties’. Thus (15a) is not to be interpreted as saying that a language A, from which developed A_a and A_b somehow later developed, as A’, together with a’ and b’ into more recent versions of the latter. The scenario is simply that the set of ‘properties’ associated with A that are in common between a and b is later amplified. This situation is obscured by the labelling of higher-order prior-languages with what can easily be misconstrued as names of ‘real’ languages, such as ‘West Germanic’. ‘West Germanic’ is simply a bundle of reconstructed ‘properties’ based on evidence from its potential descendants, and reconcilable with evidence from its potential co-descendants of a yet higher-order prior-language.

Such complex labelling should also be substituted for A and B in (15a)-(15c); they were omitted initially only for the expository reason of staging the introduction of the different aspects of the notation. The configurations in (15) represent possibilities: they are not (phantastic or otherwise) representations of actual histories. The work of reconstruction lies in determining what combination of configurations or systems, including suprasegmental (prosodic), is most appropriate to the characterisation of particular diachronic developments, though the evidence typically turns out to be indecisive. I leave the testing, with attested material, of the provisional notation developed in this chapter (particularly as deployed in [15a-c]) to someone more knowledgeable, of which there are many, concerning the history of languages.

Chapter 11

Conclusion and inconclusions

What unites the treatment of the various topics touched on in the preceding is the conviction that progress in our understanding of reconstruction, specifically of phonology, depends on recognition of the non-grammaticality of phonetics and the adoption of due modesty concerning the status of what we can reconstruct in phonology on the basis of the kinds of evidence available. There are, for instance, implementational pressures that can be operative in language use, but even they may be infringed – unsurprisingly, given that there are contradictory pressures. And there is usually little evidence for the variational complexities that help to explain how it comes about that there are apparently no exception-free ‘sound-laws’ or ‘fundamental principles’ of reconstruction. Much of history cannot be reconstructed determinately: we cannot know why or exactly how and why things changed: reconstruction of phonological histories is necessarily incomplete.

It is now my view, for instance, that the ‘wars’ over interpretation of the Old English vowel digraphs of the 1940-50s and thereafter were indecisive. All that we can reconstruct here is, firstly and in all modesty, that there were three common digraphs, <ea>, <eo>, <io>, earlier <æu>, <eu>, <iu>, with ‘long’ or ‘short’ contrastive values different from those of monographs, and mostly plausibly contrasted with each other in initial ‘height’, but some contrasts are lost in different dialects. The same digraphs, with ‘long’ and ‘short’ values occur in particular contexts and are non-contrastive as long as the context is preserved, or they are prosodic. Further, there is a sequence <ie> that occurs in contexts where the occurrence of *i*-umlaut is also reconstructed. Given that the context of *i*-umlaut is frequently lost pre-historically the value of this sequence too seems to be potentially contrastive.

But the graphic picture is further complicated by the occurrence of the sequences <ea> and <ie> after consonants that by their spelling and/or subsequent development may be suggested to have palatal values. Hogg (1992: §5.56-5.73) offers perhaps the most fervent defence of diphthongal interpretations in this case, and specifically diphthongization triggered after palatal consonants, despite the indecisive nature of the evidence for this diphthongisation in this context (i.e. DIP – diphthongisation after initial palatals). He makes much of the difficulty of constructing a non-diphthongal etymology for the form (in quasi-littera-speak) *ċȳse* ‘cheese’. He traces its development from classical Latin *cāseus*, despite the lack of evidence for when it was ‘loaned’ and from what variety of Latin or if via some intermediary. If we take the starting point of the root vowel in pre-Old English to be ‘front’ rather than ‘back’, there is no problem in tracing a conventional etymology for English that does not have recourse to an inexplicable diphthongization; rather, the spelling *ċȳse* signals the frontness of the consonant before a front vowel, as indicated by the ‘dotting’ of the

initial graph by Campbell and others, and consistent with the subsequent developments, such as initially in *cheese*. The status of these digraph spellings, in particular, remains controversial: it may again be undecidable; what is certain is that arguments based on the ordering of ‘sound-changes’, undifferentiated as contrastive or not, and almost entirely framed in terms of segments and their representation, are unequal to the achieving of more decidability concerning their phonological value.

Unfortunately, in this way, Richard Hogg’s phonology (1992) does not, indeed cannot, make any progress in resolving these earlier digraph wars and other controversies, and its main addition to Campbell (1959) is an annotated bibliography of more recent work and formulations of sound-changes, already expressed in prose, that are unfortunately reminiscent of ‘generative’ ‘rules’, but presumably to be regarded as ‘not muddled by theoretical squabbles’. And adding the confusions of phonemics to the tradition doesn’t lead to any clarity concerning where contrastivity lies – quite the contrary. Advancement here depends on the recognition that the object of historical phonology is the establishment of (a succession of) phonological systems of contrast which themselves are a set of subsystems, and on acknowledgement of the role of neutralization and of suprasegmentals and of morphophonology and morphosyntax in phonological representation and in phonological change, as well as the role of phonetic implementation in this.

One encouraging aspect of concerns with the history of phonologies is the extent of their escape from the major damage to the study of language inflicted by the ‘generativeness’ coup of the last century. My esteemed colleague Olga Fischer asks in the first part of the title of a recent review paper (2008): ‘Is there life beyond generative syntax?’ From my perspective, the answer is obviously ‘no’ – or, rather, the question is unanswerable: there never was life in ‘generativeness’, even if the question is applied to the study of language and ‘generativism’ as a whole (not just syntax). This is because aggressive ‘generativists’ disdainfully dismissed or simply ignored approaches to language that weren’t committed to what amounted to attempts to resolve the self-inflicted initial problems of ‘generativism’: principally, successive attempts to avoid the consequences for an autonomous syntax of the introduction of the massively over-powerful notion of ‘transformation’ (see e.g. Anderson [2005b], [2011a: Parts I-II]), and the confusion of synchronic phonology with its history and the morphophonological consequences of the latter, all finally abandoned in favour of biolinguistic phantasies and/or a primitive computer game involving identification of the ‘optimal constraints’ that, out of an allegedly universally available set, characterize different languages. After more than half a century of ‘generative’ self-abuse and its final implosion – RIP, along with DIP – we are now, as a consequence, well into a century of an illiterate withdrawal to the peripheries of language structure that avoids addressing what I see as the major issues in the study of languages. I hope, however, that grammar and phonetics will survive.

Earlier I cited Martinet's (1955: 27) observation concerning the conservative character of philological studies. Current studies of the history of languages – and, indeed, of the study of language in general – are, however, not only conservative but deficient in historicity. There is a lack of awareness of the achievements as well as false paths of the past (cf. e.g. Fortis [2018: 113, 187] on the (non-)history of 'localism' in the development of research on language). Even quite recent contributions have disappeared under the current cliquey floods of informal articles and (commercially desirable) textbooks and handbooks. For just a couple of examples relevant to the areas we have looked at, where are recognitions of the role of Hubmayer (1986) on phonetic motivations in the history of languages or of Yamada (1994) in the tradition concerned with the different behaviour of accented and unaccented syllables with respect to historical 'strengthening' and 'weakening'? On the other hand, the poverty (to my knowledge) of substantial allusions here is perhaps just one instance of the effect of the continuing obsession of historical reconstructions of phonology with the history of segments of questionable status.

Given the above assessment of recent history and the problems introduced by the littera and its conceptual consequences, much of what proceeds in my plea for modesty is negative; and also prominent among these problems are what I have characterized as 'phantasies', empirically unsupported proposals of structural change. In fairness, I should now admit to a phantasy of mine own (though not necessarily the only one). My brief look at studies of Old English metrics suggests that traditional accounts of this are undersupported, as well as confusing both phonology and phonetics and lexical and pre-utterance phonology: something of the variety of suggestions is provided by Amos (1980: ch. 1, especially §1.5). But I haven't offered here any alternative account, but merely the need to make particular corrections. However, any attempt at substantifying my phantasy would require a book-length offering by itself. Thus, I have offered here merely some observations that I take as necessary and, indeed, familiar in treatments of metrics in relation to languages other than Old English. These observations allude to the variety of properties to take into consideration in distinguishing types of versification. Germanic has been fortunate in avoiding the syllable timing of some other European Languages.

At the end of Ch. 2 above I offered the familiar observation that Campbell (1959) and others provide not a phonology of Old English at any particular stage but a set of ordered 'changes'. And even as recently as Ringe (2006) it is claimed concerning Proto-Indo-European that '<a> remarkable amount of the phonological rule system of PIE can be reconstructed' (p. 11)', and he goes on to present a number of changes in letters representing 'phonological' segments that constitute the 'most important rules' (p. 11). Where is the phonology? Is this a segment-based crypto-generative phonology, after all? Certainly, there is a short (pp. 6-11) preliminary section §2.2 labelled 'PIE phonology', but it consists merely of sets of such letters whose members are claimed

to be ‘contrastive’. Phonology needs to be distinguished from both spelling and phonetics; the latter is implementational not grammatical.

A final remark on reconstructing languages, which, I trust, accounts for my focus on phonology. The most convincing reconstructions in this wider area are of morphophonological and the overall structure of the phonological plane and, within that, the segmentation and subsegmental content of the phonology of minimal signs (word forms). Other attempted reconstructions, particularly of the syntactic plane, are much more difficult to compare than reconstructions of phonological phenomena, even those of historical languages.

One reason for that, I suggest, apart from the greater complexity of syntax, is, partly because of this complexity, and the lack of an agreed theory of syntax, and its reconstruction, and of its relation to the lexicon and phonology and morphology and meaningfulness, in particular; these lacks, again in my view, are getting to be more and more apparent. Recent descriptive products in the areas concerned are more and more diverse, tunnel-visioned, and undisciplined, whereas the divergences in phonological theory are more clear-cut – to me at least (on why cf. e.g. Anderson [2014a]), though I may be mistaken. And, predominantly, present-day historical ‘syntax’ is concerned with identifying so-called ‘constructions’, i.e. in the proper sense, idioms. This situation makes it very difficult to evaluate characterizations of reconstructions of the syntactic plane of any language, present-day included. They are variously phantastic – and the reader will by now understand that that’s not meant to be a compliment.

Appendix I

The representational failure of theories of phonology

Part of what I have presented above is an illustration of areas where the domain of phonology is often misrepresented: the (synchronic) spreading of ‘harmony’, for example, is not phonological. Defining the domain of phonology is a perennial problem, both in its role in synchrony and history, though its character and justification for it may be and often are neglected. The classical concept of the ‘littera’, for instance, which persisted in discussions of phonology, or ‘pronunciation’, in the centuries that followed until the twentieth century, and even presently (though rarely acknowledged, but see e.g. Lass & Laing [2012: note 7]), is disastrous to attempts at the gaining of any understanding of the domain of phonology.

The littera is a complex unit consisting of a graphical shape, a phonic value, and a name. In application its value is not necessarily phonologically relevant, given its segmental unit status and the bi-unique relation between shape and value. The littera cannot cope with neutralization and prosodies, for example; and, as utilized, it encourages the tendency for spelling to conventionalize, to lag behind ‘sound-change’. For discussion of these limitations and their consequences for phonological theory see Anderson (2014: §1).

This conformity to written forms also contaminates ‘phoneme theory’. Swadesh (1934: 35) makes clear, approvingly, phonemics can be argued to provide the basis for an optimal alphabetic orthography. But phonemics is thus not a theory of phonology, since, in wishing to achieve the equivalent of littera economy, it fails to recognize neutralization of contrasts in particular places in the poles of expression, requiring distinct symbolisation to indicate this. And it fails to characterize prosodic contrasts, which, as argued above, need to be recognized as non-segmental, not littera size, as well as failing to recognize segmental componentiality, which is associated with the preceding limitations.

Phonemics fails as phonology (Anderson [2014: §2]), as does the classical ‘generative phonology’ of Chomsky and Halle (1968) and descendants, which even extends the domain of the failure with its confusion of lexical phonological contrasts and morphophonological alternations (Anderson [2014: §3]). The introduction of ‘feature geometry’ (e.g. Clements & Hume [1995]) further intrudes, in terms of this ‘geometry’, the representation of phonetic considerations in phonology. ‘Optimality theory’ (e.g. Lombardi ed. [2001], Prince and Smolensky [2004]) also preserves much of the confusion of earlier generative phonology and adds even more confusion among the morphophonological, the phonological, and the phonetic, in terms of focus on the genesis of the selection of a language’s constraints from the (allegedly) universally available set. These constraints may include, among much disharmony, implementational considerations, as in Pater’s (1999) problematical ‘*NC_o’, which in

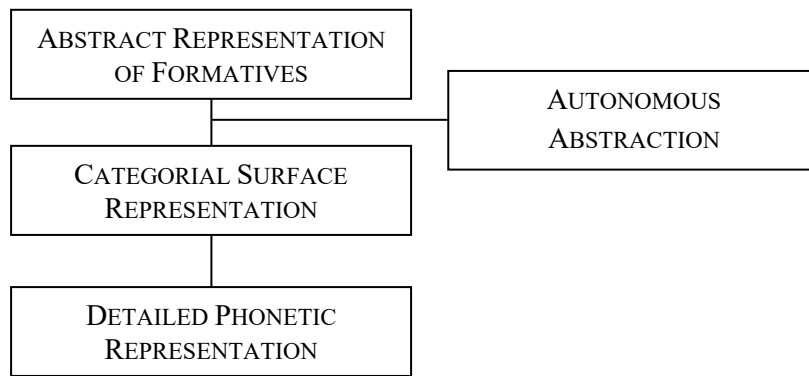
the words of Lombardi (2001: 6) ‘penalizes a cluster of a nasal and a voiceless stop, a frequently preferred cluster’.

And confusion between the phonological and the phonetic prevails in other recent literature, as in discussions of ‘vowel harmony’. As we have seen in §2 above, the ‘spreading’ of a feature (value) through temporally successive vowels is not a phonological phenomenon; rather, a single contrastive feature associated with a word form is implemented phonetically, in articulation, over the sequence subordinate to it, with varying perceptibility. And ‘articulatory phonology’ (e.g. Browman and Goldstein [1992]), indeed, is a contradiction, and paradoxically devoted almost entirely to phonetics. So too ‘laboratory phonology’ (in e.g. Cohn, Fougeron, and Huffman eds. [2011]).

The mixture of synchrony and diachrony of Anderson and Jones (1977), apart from suffering from the influence of (de-)generative phonology, does not distinguish consistently between phonological changes and phonetic; this is not excusable in view of the observation that many traditional ‘sound-changes’ are diachronically complex, involving phonetics, as in the assimilation of umlaut, and then, with possible loss of the conditioning environment, phonologization of a new contrast or at least a new instance of one. And much of the ‘dependency phonology’ of Anderson and Ewen (1987: Part II) is concerned, without acknowledgement as such, with the articulatory implementation of a wide range of only potentially contrastive elements. So too with van der Hulst’s (1994, 1995, 2020) ‘radical CV phonology’.

‘Government phonology’ and its descendants (as described in Harris [1994], Brockhaus [1995], Kaye [1995], and Harris & Lindsey [2000], for example) have misinterpreted phonology in opposed directions, with, on the one hand, emphasis on the mysterious abstract properties, ‘charm’ and ‘licensing’, and ‘empty’ categories or positions, as well as a sense of ‘government’ that does not seem to relate to traditional (non-Chomskyan) usages, and, on the other hand, incorporating as phonological what are diachronic implementational phenomena such as fortition/lenition. As a slightly more recent variant (‘CVCV’) manifestation of this tradition, see e.g. Scheer (2004).

Davidson-Nielsen (1978) provides a survey of approaches to neutralization and (less so) archiphoneme that concludes with ‘A Theory of Neutralization’ (§10) based on the ‘generative’ framework that is advocated by Rischel (1974). This is based on the set of representation types assembled on Davidson-Nielsen’s (1978: 218), which is drawn from Rischel (1974: 364).



In accord with their generative orientation, both Rischel (1974) and Davidsen-Nielsen (1978) fail to distinguish adequately between phonology and phonetics and lack an account of the role of suprasegmental phonology in the phenomena they discuss.

More positively, Anderson (1992b: particularly §5.3) and (2011c: §4.2.3) discuss syntactic phenomena he suggests to be analogous to phonological ‘prosodies’, as anticipated in Anderson (1965). But these are crucially different from phonological harmonies and prosodies, in that while the ‘implementation’ of a putative syntactic prosody such as ‘agreement’ or ‘tense’ is grammatical, expounded within the grammar, phonological harmonies and prosodies are implemented by articulation, and conveyed via the auricular mechanism, i.e. extra-grammatically.

Appendix II

From weak form to inflection in English

There are words that have, as well as an unweakened form, a ‘weak’ form that typically occurs in informal situations. In English the weak forms of the copula and some auxiliaries have undergone a re-analysis as components in a set of novel lexico-syntactic structures in which they serve as inflections. The consequence is thus an unusual kind of reinterpretation, ‘inflectionalisation’, of the morphosyntax of these forms. Invocation of ‘cliticization’ does not throw any light on these developments. The development of weak forms is a very common phenomenon in the phonetic implementation of the expression of signs. In informal speech, commonly used and semantically light word forms may lose phonic ‘bulk’, contracting what they share phonetically with the inherited non-weak form. Weakening is manifested as: loss of accent, or of ‘length’ of vowels or reduction to ‘central’ vowels or loss of a vowel, or de-clustering of consonants or loss of weakly articulated consonants. Obendorfer (1998: ch. 6), for instance, introduces part of a ‘weak-form scale’ (p. 185), beyond the left of which would presumably be a monosyllabic that is at least ‘unweakened’.

WEAK-FORM SCALE

o	1	2	3	4	5	o
	full vocalic syllable	semi-reduced weak vocalic syllable	syllabic schwa	non-vocalic syllable	non-syllabic segment(s)	

I take it that there is for a weak form a corresponding non-weak form with a shared sense and with a similar realization (though these sharings may eventually diverge). On weak forms in English see too Jones (1962: ch. XVI) – though the phenomenon is particularly subject to social variation, as well as (like that book itself) to change.

We can distinguish weak forms from semantically light forms, such as the verb in English *take a walk/break*: illustrating one of the **lexical periphrasts** of Anderson (2022a: ch. 22 & Commentary thereon). Weak forms are the product of the diachronic implementation of phonetic weakening of a form, as in varieties of English with (in rough transcription) [mɪ faðə] rather than [maɪ faðə], or, with another kind of informal second form, [mɪ dad/daddy]. These all involve the distribution of contrastive segments in the lexicon, and the last often functions as a name for particular groups. But the former, functional type of alternative may have consequences beyond merely the provision of a phonetic or minimal locational phonological change associated with lesser formality.

Despite, or because of, a widespread tradition that the above may bring to the mind of the reader, I avoid the term ‘clitic’ in what follows, despite the brief passing mentions in the discussion of similar phenomena in Anderson (2022b: chs. 35, 42), for instance. I have become ever more aware of the variety of usage of this term – which has come to rival in this respect the term ‘grammaticalization’. For my part, I understand the latter term as denoting becoming part of ‘grammar’ (rather than being, say, a grammar-internal change). And ‘weakening’ is interpreted here as purely an aspect of implementation, or phonetics; it is associated with (diachronic) loss of bulk in phonetic implementation, particularly in informal situations. That is my concern here, and the possible morphosyntactic consequence, a grammaticalization of the result of the phonetic weakening. But I address my problems with the use of ‘clitic’.

Obendorfer (1998: particularly §2.7) illustrates the confusion in the use of the term ‘clitic’ in work on both phonetics and morphology, and particularly in the ‘generative-phonology’ tradition following from the work of Zwicky, particularly (1994) – as well as other contributions to the volume that this work appears in, a tradition culminating more recently in S.R. Anderson’s (2005) detailed account appealing to optimality theory, which is of little interest outside this tradition.

The entry for ‘clitic’ in Carr’s (2008: 28) *Glossary of Phonology* is particularly unhelpful in exemplifying the alleged concept of ‘clitic’ with personal pronouns in French, as in *Je te le rends* ‘I you it give-back’ (‘I’m giving it back to you’), with a nominative personal pronoun followed by two obliques (‘dative’ and ‘accusative’). Such oblique singular pronouns are certainly weak on Obendorfer’s scale, each being expressed by a minimal vowel, except *la*, with feminine grammatical gender, as well as nominatives other than *je*. These phonetically light forms are without coda and with a single-segment onset; but these forms are the normal (not necessarily informal) forms; they are not synchronic weak forms, alternative to unweakened congeners. When addressed to a single person, the distinction between *vous* and *tu* is formal vs. informal, but *tu* is not a weak form, though it may not be as heavy as *vous*.

Some of these light forms can, certainly, be even weaker before vowels: thus, *J’aime tout cela* ‘I love all that’, *Je l’aime* ‘I love it/her/him’. And even more weakening occurs in *J’t’aime*, where both the oblique pronoun and the nominative are vowel-less. On the other hand, as alluded to above, the second- and third-person nominatives singular certainly have a phonically heavier form as the norm, with no matching weak form: *Tu l’aimes* ‘You love it/him’; so too *il/elle*. The plural pronouns *nous/vous/ils/elles/les* are also scarcely phonically light, though the final sibilant is present only before a vowel. The non-plural pronouns, however, are light; and they are phonologically weightier only if used disjunctively (emphatically) or ‘absolutely’ (*Moi?!/Toi?!/Lui?!/Elle?!/ ‘me/you/him/her’*) or when overtly governed by a preposition (*pour moi*, etc.) or used as a form of address (*tois!*).

I have omitted mention here of the French reflexives (*se* etc.), or the neuter dative (*y*) or genitive (*en*) pronouns: for illustration of these and further details see e.g. Ferrar (1967: ch. XV). However, the most obvious distinguishing feature of the pre-verbal oblique pronouns, including those just-mentioned, is, indeed, their distinctiveness in placement compared with that of equivalent determiner phrases (and even names – said to manifest the status of the former as ‘special clitics’).

On the other hand, the alternations in the composition of the weak pronouns and the masculine article *l(ə)* are akin to the more general phenomenon in French known as ‘liaison’, again mentioned in Carr (2008: 92), for instance. This phenomenon, which has been widely and controversially discussed, is described in brief by Carr as follows:

Many **word-final coda** consonants were **elided** in the history of French, as in words such as *capot*, pronounced [kapo] <‘bonnet (of car, in non-American English)’ – cf. *capote* ‘soft-top’, with pronounced final stop – *JMA*>. But some word-final consonants in certain words have remained as **floating consonants**, and will be realised under certain conditions when there follows a word with an **empty onset**, as in *les amis* (‘the friends’), pronounced [lezami]. Where the following word does not have an empty onset, the floating consonant is not pronounced, as in *les gants* ‘the gloves’, pronounced [legã].

(The emboldenings here are Carr’s, and they mark a term with an entry in his *Glossary*.) The light nominative forms of the pronouns, as well as the heavy forms, ‘invert’ in simple interrogatives and there is ‘liaison’ when appropriate, as illustrated by *Peut-il-venir?* ‘Can he come?’, but not in *Il peut venir* ‘He can come’.

The nearest English equivalent to this phenomenon perhaps occurs in varieties that show non-rhoticity or ‘[r]-loss’ in, for instance, the ‘linking-[r]’ in the second instance of the first form in *near the bridge* vs. *near a bridge*, or the ambisyllabic rhoticization between the first form and *and* in *law and order*, with so-called ‘intrusive *r*’. But the status of both of these phenomena, as involving instances of external **sandhi**, is rather unlike the characterization of English ‘weak forms’ that I look at in what follows here, though they too are sometimes designated ‘clitics’, as are oblique pronouns with a syntactic placement in Old English similar to those in French (e.g. in van Kemenade [1987], Koopman [1992]); but problems with alleged Old English ‘clitics’ are expressed by Colman (1991: 74-7). French liaison and the rhotic behaviour in some English varieties are both **implementational** (phonetic: Anderson [2022b: 622]), however: phonetic not phonological, not grammatically contrastive. The presence or absence of /<r>/ is decided at the phonology/phonetic interface. So too with the isolated liaison of *an* (for *a*, the article) before a vowel, and in the obsolete usage illustrated in *an hotel/history*; I have also encountered loss of [h] in the former.

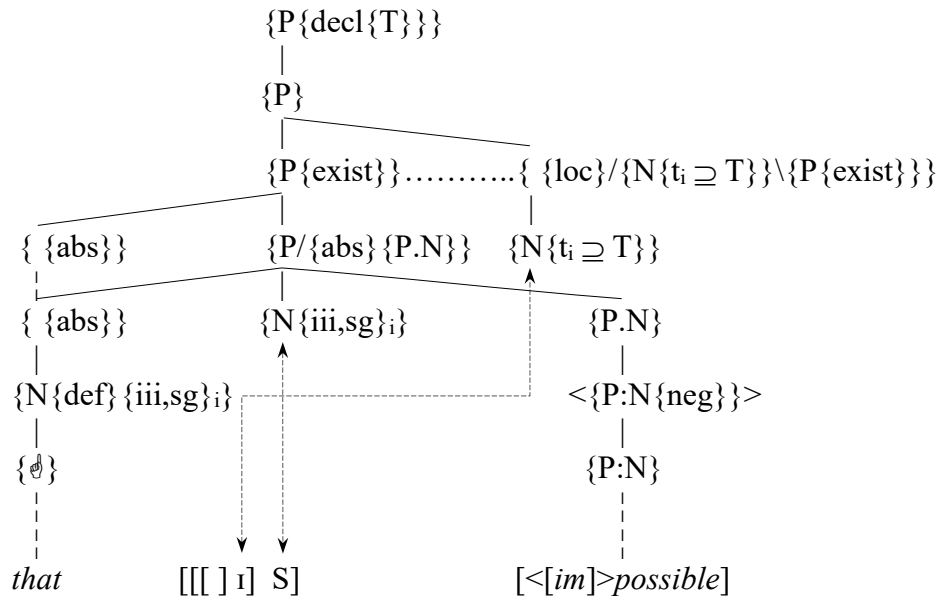
I'm also not concerned here with the absence of word-form expression of subject-pronoun + finiteness-mood illustrated in *Coming?* or *Better?* or *Love you.*, interesting as this is.

What shall be our concern here is again with what precedes a verbal or predicative contentive category (unhelpfully designated 'lexical' in some traditions), but in English it is the traditional auxiliaries and copula that are my focus, and unquestionable weak forms that share their unmarked relative positioning with their corresponding non-weakened congeners. Thus, Anderson (2022b: ch. 42 & Commentary) discusses the status of the weak form of the unweakened copula in (1a) that we find in (1b). In the latter the termination of the form *that's* is apparently a weak form of a copula, attached phonetically and in spelling to the immediately preceding form:

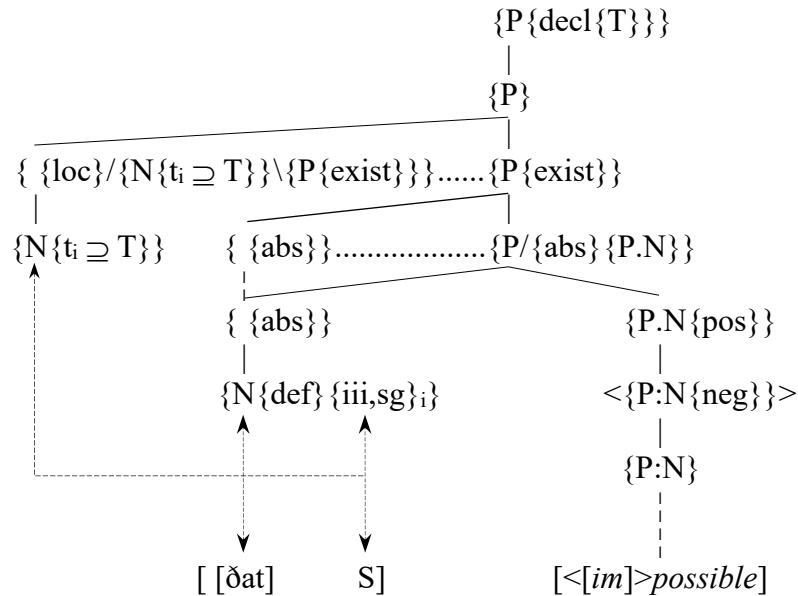
- (1) a. *That is (im)possible*
 b. *That's (im)possible*
 c. *Is that (im)possible?*

That('s) contains the ictus head (secondary accent) of the sequence on which the first syllable of the following negative word, if that prefix is present, is dependent; and in all of (1) the vowel spelled *o* typically bears the tonic (primary) accent. We find an 'inverted' relationship between the first two forms in (1a) in (1c), expounding the interrogative mood, but there is no such alternative to (1b). In (1c) there is 'inverted placing' of the subject and the copula, the finite head of the sentence. (On subjects see Anderson [2006: §6.3, 2022a: 68-73].) But what is the morphosyntax of (1b), in the light of the failure of 'inversion'? Let us look firstly at the relevant structure of the unweakened (1a) suggested in (2a).

(2) a.



b.



The category-bearing nodes in the syntactic representation in (2a) are linked by **dependency** arcs (directed lines) each of which links a head or governor dependent, where the head is higher than the dependent; the nodes in the structure bear information concerning syntactic categories, except the lowest categorizations joined by an **association** (discontinuous) line. The syntactic categories are the core of the content poles of the minimal sign (word); the other pole of the sign linked to them by an association line is expressive, (morpho)phonological. The morphological content of the (adjectival) contentive is largely ignored, apart from that of the optional negative

derivative of the basic adjective, where the optionalities of morpho-syntax and morphology are linked by the co-presence of the angle brackets.

The representation is further complicated by a distinction between dependent and sequenced and dependent and unsequenced. Thus we can have an **adjoined dependent**, differing in syntactic placement from its head, or a dependent that is unsequenced (where graphically the arc is vertical), so a **subjoined dependent**. The latter here is lexical; the former typically represent syntactic dependencies, given in the lexicon or inserted at the **lexicon-syntax interface**, in response to the **valencies** of the categories labelling the nodes; these **participants** of a verb ($\{P;N/\}$) are indicated after the '/'. The category that a category may be adjoined to as a **circumstantial** (modifier) follows '\`' in the valency of the circumstantial.

Both valency types occur in (2a) in the locative construction $\{ \{loc\}/\{N\{t \supseteq T\}\}\{P\{exist\}\} \}$. $\{t\}$ here, the time of what is predicated, includes that of the speech act $\{T\}$: $\{t_i \supseteq T\}$. (2a) also shows both subjoined and adjoined dependencies: the former are illustrated by the categorization of *(im)possible*; the latter by the line between the lowest $\{P\}$ and that categorization. The latter are thus typically **lexical dependencies**, stored or combined in the lexicon as the **content pole** of (in the present cases) the minimal signs (words) whose other, expressive, pole is **(morpho)phonological**. As indicated, the categories enclosed in $< >$ brackets are possible absence of the derivation of the negative adjective from the positive, and so the optionality of the prefix in *(im)possible*.

The last line in (2a) provides a fuller representation of the morphophonological pole only for the copula; and the morphology of the other words is simplified. The brackets indicate the morphophonological structure of the word form that is spelled as *is*, where a distinct **stem formative** is lacking, and which is inflected for tense (cf. *was*); and it has a further person-number suffixal **inflection** enclosed as in '[S]', where the capitalization as 'S' indicates a variation in the phonological realization of the person-number inflection: overall we have $]_1[S]$ attached to the stem as in (2a).

The two-headed discontinuous arrows provide interface connections between morphological and syntactic structure, but, as noted, the morphology of the first and last words is not our concern here. The dotted horizontal lines between two categories indicate non-sequence; they are realized by a single word form, built in the morphology of the lexicon or in the interface. Thus, the tense categorization, beginning with the locative, is not syntactically separate from the existence/truth $\{P\}$; and they both are subjoined to the mood $\{P\}$, here declarative. The bare $\{P\}$ above the existence $\{P\}$ indicates that it satisfies a circumstantial rather than a participant of $\{P\{exist\}\}$, namely the tense of what is being stated to be the case.

In preparation for further explication of the diagrams, the Table below lists the representation of the simple major syntactic categories, as proposed for English in Anderson (2007: ch. 8, 2022a: ch. 9), except for personal pronouns and members of

the onomasticon within the lexicon. The categories in the table are distinguished by the presence of one or two substance-based features; and one or other of the two features may preponderate, indicated by its preceding the ‘;’, or two may be reciprocally preponderant, as with adjectives, with the features linked by ‘:’. Each **functional** category has a **contentive** congener (often, as noted, misleadingly labelled ‘lexical’, despite the presence in the lexicon of the functional categories), except the ultimate relational among them, the **functor**, which also lacks a major category label; but all the functional categories are necessarily complemented, as indicated by their following ‘/’ in their valencies. The (combinations of) capital letters in the Table represent the semantic feature combinations that define the major syntactic categories, which determine the basic distribution of the words in accordance with their meaning. But, for simplicity, the icon ‘{ $\emptyset$ }’ in (2a) abbreviates the pronoun content: on which in detail see Anderson (2022a: 136-140).

Functional	{N}	{P}	{P.N}	{ / }
	determiner	finiteness	comparator	functor
examples	<i>the, that, a</i>	<i>may, be</i>	<i>more, almost</i>	<i>at, to, from</i>
Contentive	{N;P}	{P;N}	{P:N}	
	noun	verb	adjective	
examples	<i>person, hill</i>	<i>fall, walk</i>	<i>(im)possible</i>	

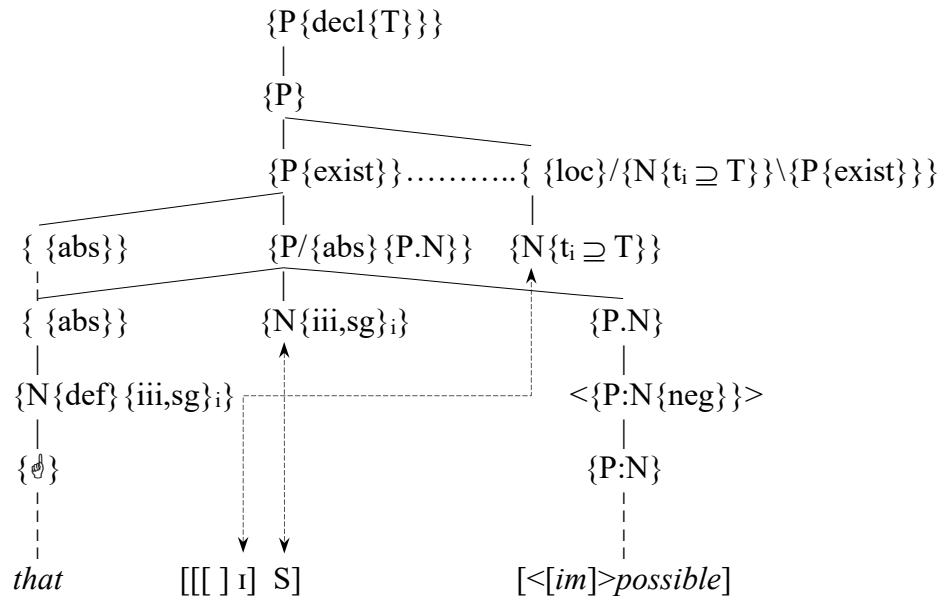
Major Syntactic Categories

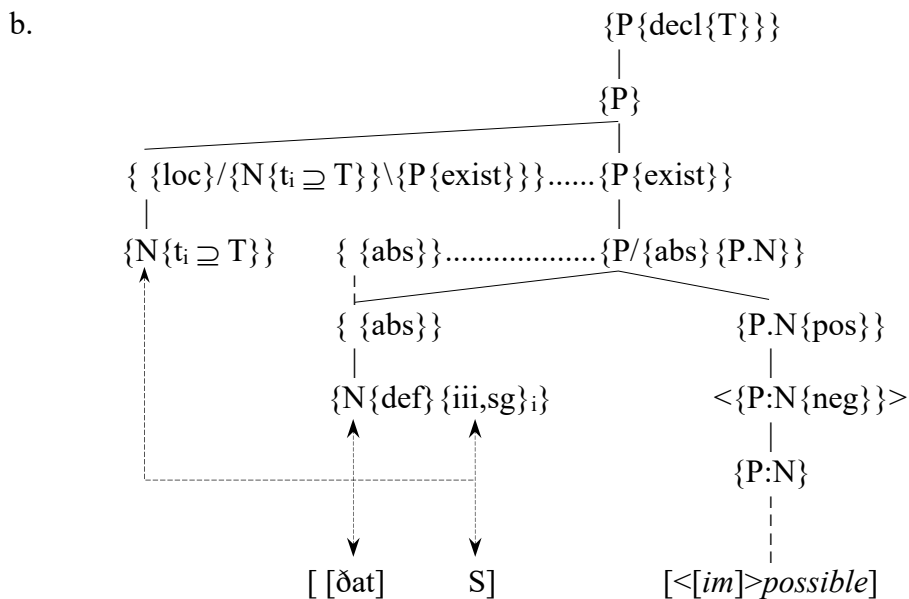
The finiteness feature **P** is relational and the topmost {P} in (2a) bears the mood and can head an independent sentential structure; in combination with **N** and superior to it, as indicated by {P;N}, it is more weakly relational and is the (optimally dynamic) verb, the head of a predication. **N** is referential, as with determiners and pronouns; and {N;P} is a noun, which is merely denotational rather than referential (Lyons (1977a: ch. 7), but non-dynamic and optimally discrete. Adjectives combine both {P;N} and {N;P}, indicated by {P:N} or {N:P}. Thus the two components are weakened: adjectives designate attributes rather than discrete entities (nouns) and are preferably non-dynamic, but stative. {P.N}, combining simply **P** and **N**, marks the comparator, such as *more*. Other parts of speech, such as adverbs, and the pronoun in (2a), are inherently complex, necessarily involving more than one of the categories in the Table. The lower case letters in (2a) label minor categories that discriminate minor distinctions in meaning and distribution. As indicated, participants (complements, including subjects) satisfy the valencies of instances of categories, as in {P/{abs} {P.N}}, where the functor, as the empty major category, is represented, as usual, by the relevant minor feature, here the neutral semantic relation {abs(olutive)};

and circumstantials (modifiers) seek a head, as indicated by the ‘\’ introducing the last item of the temporal locative $\{ \{loc\}/\{N\{t_i \supseteq T\}\} \backslash \{P\{exist\}\} \}$.

Let us now look more closely at the differences between the representations in (2), beginning with a repeat of (2a), for ease of comparison. This configuration is that of a predicative construction and, as such, is dependent on a member of the copula paradigm, here that expounded as *is*. In $\{N\{iii,sg\}\}$, on the other hand, $\{iii,sg\}$, are features of the subject $\{N\}$, with a concording replica incorporated in this lowest (predicational) $\{P\}$; and $\{N\{t_i \supseteq T\}\}$, which is a locative modifier ($\backslash$) of the existential (= truth – Lyons [1977b: §15.7]) $\{P\}$, provides the time of the existence $\{P\}$ above the copular $\{P\}$. Concord is expressed by the coindexing with subscript ‘i’ of the concording minor features in (2), here marking coreference.

(2) a.





As again observed above, the syntactic structures are built in the interface between the lexicon and the syntax on the basis of the categorial and structural information stored in the lexicon, including valencies. The valency sequence in the copular $\{P/\{abs\}\{P.N\}\}$ indicates that this finiteness category is to govern (as expressed by ‘/’) an absolutive functor (i.e. the least marked functor) and a comparator, which here, as pos(itive), is not expressed overtly (syntactically) but does have a lexically dependent adjective. We have an overt comparator in both *lovelier* and *more lovely*, and in *much lovelier* and *much more lovely* we have modification of a comparative comparator (see Anderson [2022a: ch. 23]).

The syntactic structures in (2) are headed by the mood $\{P\}$, here abbreviated as ‘declarative’, so associated with the indicative structure, as is usual. And subjoined to it in both structures is a line of other types of $\{P\}$, the first of which is a reflection of a temporal locative modifier of the existence (truth) $\{P\}$ below, as abbreviated by $\{P\{exist\}\}$; the modification is fulfilled by adding this empty $\{P\}$ above the truth $\{P\}$. Below that is the copular $\{P\}$ that governs the predicative adjectival and what it is predicated of, where the functor (semantic relation) is signalled by position, and takes a subjoined dependent.

The mood $\{P\}$ is a massive abbreviation of the representations such as that in (202) of Part IV of Anderson (2022b), which includes even the phonological exponence as well as the internal structure of mood and tense. The capital ‘T’ here is simply an abbreviation of the tense configuration of the speech act: so, as noted, $\{t_i \geq T\}$ tells us that the time of the scene whose truth is declared includes the time of utterance. The empty temporal $\{P\}$ below it has only the existential $\{P\}$ as a dependent. The latter lacks an adjoined complement in its valency, and so a free

absolute, lacking a dependent of its own, is introduced by a general convention of the interface forbidding predications that lack an absolute. The existential {P} therefore shares its dependent with the functor of the copula that here is the absolute, by the subject-selection convention; in the present case, the valency-lacking absolute of the existential shares its dependent with that of the absolute that is part of the valency of the copula, where sharing is indicated by their being joined by the discontinuous vertical association line rather than a dependency arc. In (2b) and subsequently I omit the internal structure of the pronoun.

The categories joined by a horizontal dotted line belong to the same word; they are part of lexical structure; and the difference between the positioning of the tense phrases in (2a) and (2b) has no syntactic significance, but merely reflects a graphic convenience. The *be*-form in (2a), however, has a bizarre monosyllabic morphological structure, with two inflections, tense-marker + person-number marker, and no distinct expression of a stem. In the case of *be*-forms there can be identified a stem only in non-finites (*being*, *been*, [*to*] *be*) and the (finite) imperative (*be quiet*), i.e. forms that non-elliptically lack or may lack a nominative subject, as well as functioning as (so-called 'present') subjunctive. This situation contributes to the development of the structure that I suggest for (1b), as represented in (2b).

The sequence *that's* in (1b) is given a distinctive grammatical status: specifically, in (2b) the syntactic subject, including as usual the person-number suffix, is dependent on the copular {P}, but not sequentially distinct from it, as indicated by the discontinuous horizontal lines above: they form a single word. There is thus only one occurrence of person-number, so no concord. The tense of the sentence is expressed by the absence of a past element such as that in *was*: cf. *is*/'s. Moreover, the morphophonological behaviour of what is spelled as the final, inflexional -s in (2b) parallels that of other inflections, nominal as well as verbal.

A final syntactically significant sibilant is common in English word forms, and represented as [S] in (2b), and it behaves as do the suffixes marking nominal plurality and the genitive suffix and verbal 3rd person singular, in varying in accordance with the preceding, stem-final, segment, as illustrated in the variants of (3).

- (3) a. *Bret's impossible/left*: voiceless suffix: [s]
 b. *John's impossible/left*: voiced: [z]
 c. *Madge's impossible/left*: syllabic: [əz]

A copula or an auxiliary (here perfect) may be involved. Compare the genitives in (4).

This last sub-generalization is allowed for in (6) by the optionality of the **V** of fricativity. However, the parallels between syntactic genitives/plurals and the immediately above auxiliary and copular inflections lend support for this shared redundancy, where, as elsewhere, the angle brackets enclose optional elements. The shared generalization in (6) raises interesting questions about the development of the syntactic inflections with implementations dependent on interaction with the final segments preceding the inflected sequence.

Given, however, the limited syntax of personal pronouns and their diverse morphology, their genitives are also listed individually in the lexicon. Their idiosyncratic morphology is apparent from comparison in (7) with the non-genitive forms, though there are a couple with final *s*.

- (7) Genitive: *my your her his its our your their*
 Non-genitive: *I/me you she/her he/him it we/us you they/them*

Their listability as a whole word is indeed an indication of lexical status, as members of a ‘closed-class’: they do not involve syntactic inflections.

The weak presents of the copula in (8) are similarly lexically listed, and indeed show some similarities with the genitives, even if involving reversed components in the case of *my* vs. *I’m*.

- (8) Weak present copula: *I’m you’re she’s he’s it’s we’re you’re they’re*

Names, despite their syntactico-semantic similarity to pronouns, obey the genitive generalization in Anderson (2022a) alluded to above and are inflected with the same weak copular variants as any other form – as illustrated in (3-4).

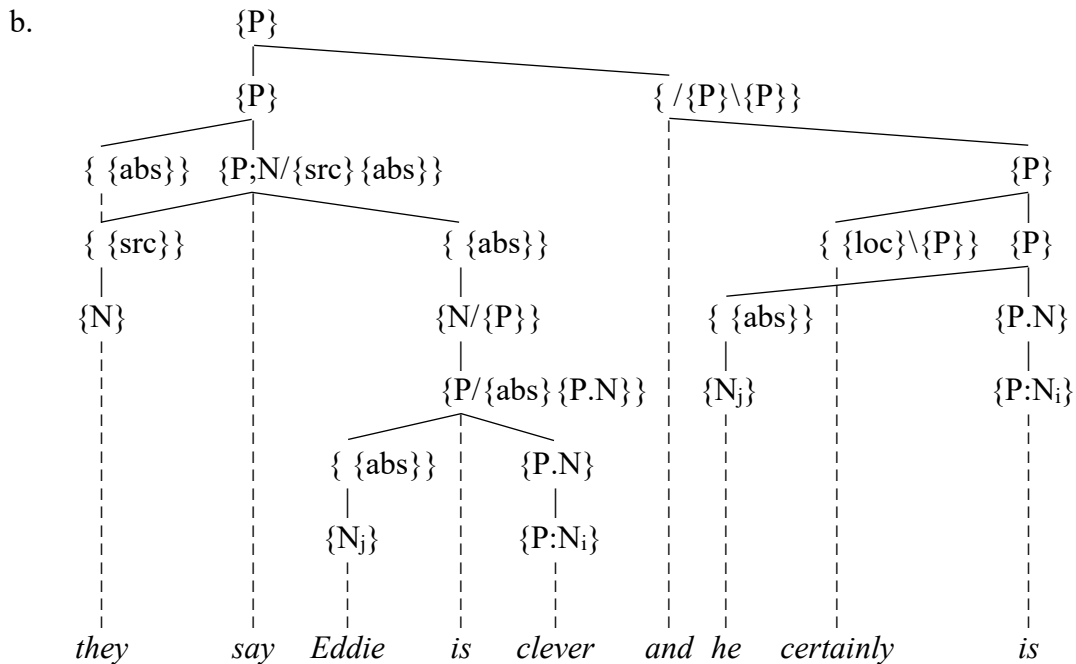
S.R. Anderson (2005) devotes his §3.4 to ‘Phonological clitics and cliticization in English’, illustrating ‘the case of contracted auxiliary verbs’ (p. 64). However, almost all the relevant examples discussed there are not auxiliaries, governing verbals, but copulæ, governing predicative non-verbals. He also illustrates various circumstances where the weak form is not acceptable, which he attributes to ‘prosodic structure’, an unwarranted concept that appeals to such oxymoronic terms as ‘phonological words and phrases’. I suggest instead that these are instances where a distinct covert form complementing *is* is needed to carry a coindexing syntactic category.

Such a circumstance is illustrated in (9a) below (Anderson [2022b: ch. 41, (187b)]). The representation of the coordinating structure of this sentence in (9b) ignores mood and tense but includes a crucial argument coreference. As well as clause-final coindexing (and indeed coreference between the subjects of the subordinates), it also includes a ‘coordinating’ structure and the determiner that introduces subordinate finite clauses, overtly (typically *that*) or not (as here), while ignoring the mood

structure (as irrelevant in the present context). Also, as a suffix, the weak copula must immediately follow the subject, as shown in (2a) above.

Certainly modifies the existential $\{P\}$ to which the coindexed category is subjoined, whatever its content, which is available via coindexing, abbreviated in (9a) – and more fully in (9b). (see Anderson [2022a: ch. 17] for similar examples in [231–233]).

- (9) a. They say Eddie_k is clever_i and he_k certainly is $\{P:N\}_i$



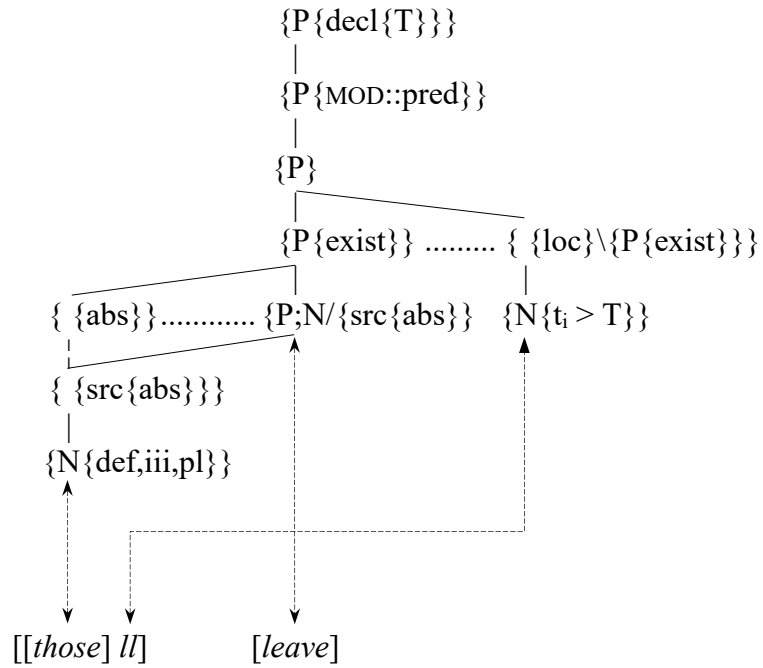
The weak form of passive/progressive *is* and the weak form of perfect *has* show a similar development, but the passive and perfect are distinguished by governing a perfect or passive form of participle (*He's been kidnapped*, *She's left*). Other weak forms of auxiliaries, i.e. such forms of otherwise independently-expressed $\{P\}$, grammaticalize in other ways. A number of them syllabify by acquiring a following schwa when preceding a consonant, as in the examples in (10a) vs. (10b).

- (10) a. *Those've left, Those'll leave*
 b. *They've left, They'll leave*
 c. *They'll've left, They'd've left*

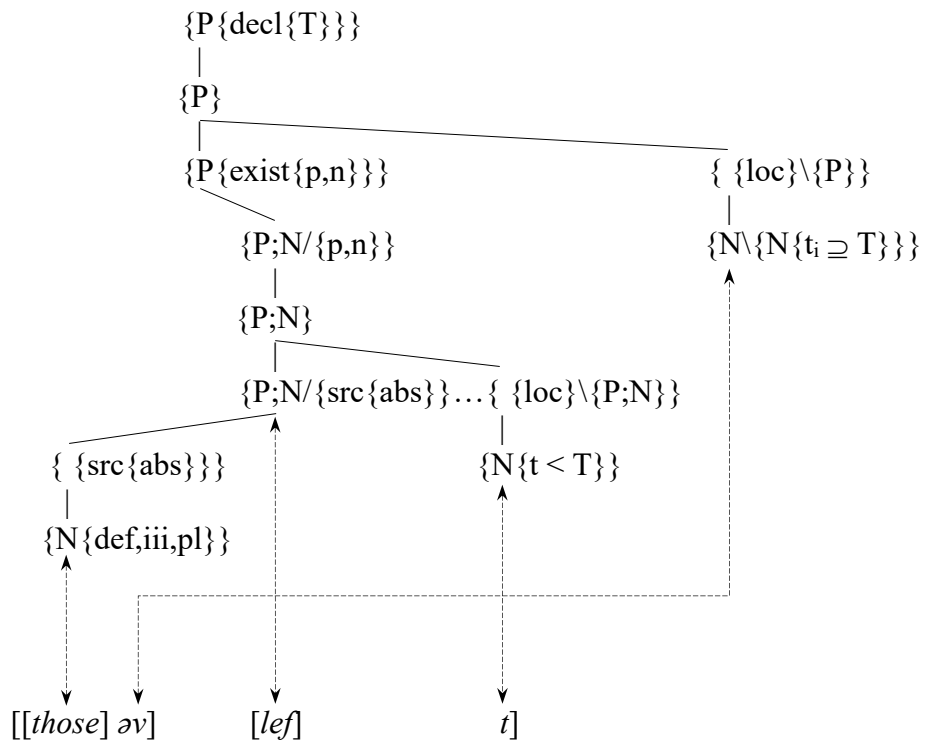
The examples in each of (10) is predictive or, under another interpretation, a volitional **modal** and that in (10a) is illustrated in (11). The first example in each of (10) is another periphrastic structure, the **perfect**. In these the existential $\{P\}$ and the dependent $\{P;N\}$ are stative ($\{p,n\}$, a simple combination of the minor features **p** and

n eventuating in a state, as is characteristic of perfect forms.); it involves a present state that is the result of a past event ($\{N\{t_i < T\}\}$).

(11)a.



b.



However, these weak forms are again syntactic inflections, and the perfect can indeed be inflected to an inflected modal, including the other modals in (12), though they do not conform to all of the syntactic morphophonological generalizations illustrated in (4-6).

- (12)a. *They'll've left, They'd've left*
 b. *Billy'd leave, Bill'əd leave*

Both of the weak forms in (10) exhibit syllabicity if the subject form ends in a consonant; compare a subject that ends in a vowel: *She'll refuse, They've refused*. The first of these has lost its {I}. We've seen, moreover, that the parallel of the weak 's of *has* and *is* with genitives, indeed with inflections in [S] and, in verbs, what we might represent as [D], extends to their participation in the generalization concerning the variation in the exponence of these inflections depending on the end of the stem. However, the weak modal/past habitual inflection [əd] (*would*) once more lacks its syllabic only after vowels: cf. [bɪlɪ d liv] vs. [bɪl əd liv], in a rough transcription of (12b) that separates the morphosyntactic elements. The past tense auxiliaries *was* and *were* retain their syntactic integrity though weakened: [wəz], [wər] (the latter for rhotic speakers only, unless a vowel follows), but the first-person singular loses it to its only host, to give *I'm*.

But, of course, not all weak auxiliaries conform to (6). And a more limited weakening with different morphosyntactic effects is to be found in the existential-locative syntactic-subject sentences whose unweakened forms are illustrated (13a-b).

- (13)a. *There is a solution*
 b. *There are solutions*

At the lexico-syntactic interface the locative here is preferred as the **positional subject** rather than the functor phrase that conforms to the subject-selection hierarchy, as in *A solution was found by the committee*. Nevertheless, number concord here is with the 'postposed' indefinite **morphosyntactic subject**.

Both *be*-forms may also be weakened, as illustrated by (14a-b), and the inflectional [r] in the latter syllabifies as [ər].

- (14)a. *There's a solution*
 b. *There're solutions*
 c. *There's solutions*
 d. *There is solutions*

But, though the usage in (14c) and particularly that in (14d), with no weakening, are discouraged by many prescriptivists, their occurrence is not uncommon. Here,

there is no concord with the existential in subject position, which does not inflect for number. We have competing systems with and without concord.

I have ignored here too social variations in the use of some of these weak forms. And I have not exhausted their behaviour: for instance, the capacity for interrogative ‘inversion’ of some weak-derived auxiliaries with simple subjects, such as that in *’m-I included?*. These obviously introduce further considerations. And I should note that there are weak colligations such as non-finite *gonna* and finite *wanna*, which amalgamate a weak final infinitive-*to* with the final syllable or segment of a verb form, which is itself weakened.

The data – better, *capta* – are taken from my experience as a native English speaker, rather than from a manufactured corpus, appeal to which would be particularly inappropriate when we are dealing with one manifestation of a present-day language, with minimal concern for sociolinguistic variation.

My main purpose in here has been to illustrate that the weak forms of English auxiliaries/copulæ that are implemented in certain (informal) situations have undergone a distinctive grammaticalization, specifically inflectionalisation. Some of these inflections have a phonic substance that is identical to and undergoes the same alternations as the denominal genitive inflections; and, like genitives, they are syntactic inflections, and are thus given their placement in structure in the lexicon/syntax interface rather than in the lexicon.

This exposition has been a modest continuation of the testing of the substance-based linguistic framework of Anderson (2022a, b). But it also illustrates that languages cannot be classified as either strictly inflectional or inflection-free, or are even as strictly becoming-inflectional or ceasing to be: English has a long history of inflection-loss, but we have just been looking at recent inflection-acquisition in some varieties of the language. And we have also found additional evidence in English for a distinction between traditional strictly morphological inflections that are a property of the lexicon and inflections that, though they are manifested as a suffix, the identity of the suffix-bearer is selected in the construction of the syntax.

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[†] References here to pages in publications that are reprinted in Joos (1957) are to the reprinted page-numbering. Martin Joos' volume was printed in the same year as Noam Chomsky submitted his Ph.D. dissertation, and thus the volume can be seen as an epitaph for an era in North American linguistics. Global ‘generativism’ itself, like the Roman empire, collapsed from within and has in turn been replaced by barbarian hordes whose members are devoted to ‘having a career’ within their horde, each of which, forms a social-media-linked ‘community’, or clique, including such generative remnants as the precarious ‘Byzantine empire’ in North America and such as the Eastern empirette embodied in the collection edited by Grohmann & Panagiotidis (2009).

In the writing of this piece I have benefitted from the contributions by Fran Colman and Roger Böhm.

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